

predictor of the model. To illustrate this, suppose we have n observations of a D -dimensional multivariate Gaussian distribution. To evaluate the correlations of this multivariate distribution in R-INLA, we can construct a component on the latent field $\mathbf{u}$, such that $\mathbf{u} \sim \text{MVN}(\mathbf{0}, \mathbf{Q})$, whose precision matrix is defined as $\mathbf{Q}_{ii} = \tau_{ii}$ and $\mathbf{Q}_{ij} = \rho_{ij} / \sqrt{\tau_{ii}\tau_{jj}}$. Each realization of this component produces a vector of D elements, so if this is replicated for each row of observations, we will have n replicas of this effect evaluated with the same hyperparameters $\{\boldsymbol{\tau}, \boldsymbol{\rho}\}$. This allows us to express the D -dimensional multivariate distribution in the likelihood with n components as a univariate distribution with $n \times D$ observations and fixed precision in the univariate Gaussian likelihood to a high value ($\log(\tau) \gg 1$). Therefore, in each term of the linear predictor we will have a realization from this effect u corresponding to a category $d \in \{1, \dots, D\}$ and an observation $i \in \{1, \dots, n\}$.

Another way to express this is to consider that we have multivariate observations organized in rows for each multivariate observation and in columns for the different components of the variable distributed according to a multivariate distribution. This can be expressed by defining the model for each multivariate observation in the following way:

$$\mathbf{y}_{1 \times D} = \boldsymbol{\mu}_{1 \times D} + \boldsymbol{\varepsilon} \quad (6.8)$$

where $\boldsymbol{\varepsilon} \sim \text{MVN}(\mathbf{0}_{1 \times D}, \boldsymbol{\Sigma}_{D \times D})$. If we incorporate this structure, $\boldsymbol{\varepsilon}$, into the latent field, then the model can be re-written as follows

$$y_i = \mu_i + \varepsilon_i \quad (6.9)$$

where $\varepsilon_i \sim \text{N}(0, \tau)$, and τ is fixed at a high value. The latent field must then incorporate a multivariate structure with correlations that relate the components of the multivariate structure we are re-expressing. If we rearrange the matrix structure into a vector such that the indices for the multivariate components are represented as $(1_1, \dots, n_1, \dots, 1_d, \dots, n_d, \dots, 1_D, \dots, n_D)$, then the multivariate component $\mathbf{u}$ must follow this indexing and can be incorporated into the latent field.

This demonstrates the flexibility to define various spatio-temporal structures for modelling compositional data, or more generally, multivariate data. It also highlights the combinatorial challenge of defining models within such a framework, where testing all possible combinations implies defining 2^D models for each component of the model, with D being the multivariate dimensionality. This issue suggests that constructing models in these contexts can benefit significantly from expert knowledge, which can help narrow down the feasible set of reasonable models. This approach

avoids the need for extensive model selection procedures among numerous sets, which can become intractable with even a few components. Therefore, implementing spatial models in the context of compositional data requires establishing criteria based on expert knowledge or other relevant factors to limit the set of models to be evaluated. This is especially important given the various possibilities for defining spatio-temporal components that we have mentioned in the previous section.

6.5 Dealing with 0's in land use data

In the first section, we presented various approaches to analyse land use data: beta models when focusing on a single category, and multivariate normal models for some transformation of logratios (additive logratios, centred logratios, isometric logratios) or Dirichlet models when dealing with multiple categories. These are all different approaches to CoDa analysis in general, and to land use data provided in proportions in particular. However, a significant challenge in dealing with compositional data across all these approaches is the presence of zeros and ones ([Martín-Fernández and Thió-Henestrosa, 2006](#); [Martín-Fernández et al., 2011](#); [Tsagris and Stewart, 2018](#)). Zeros can be problematic because they often indicate the absence of a component, complicating the application of standard statistical techniques that assume strictly positive data. Similarly, ones can signify the complete dominance of a single component (inducing zero values in the remaining components), which can distort correlations and overall analysis. Addressing these issues typically requires specific methods, such as zero-replacement strategies or transformations, to ensure that the compositional nature of the data is properly accounted for and that the analyses yield meaningful results.

In general, different types of zeros can be found in compositional data (CoDa): (i) rounded zeros or below-detection values, (ii) count zeros, and (iii) essential zeros. The first two types of zeros are associated with “false” zeros, where the structure of the experimental design, the sample size, or the sensitivity of the measuring instruments results in null values, even though the underlying phenomenon is not necessarily null. In contrast, the third type of zeros, also called structural or absolute zeros, assumes that the measurements indeed capture the real absence or null value of the phenomenon ([Martín-Fernández et al., 2011](#)). For the first two cases of rounded zeros and count zeros, replacement procedures or transformations have been proposed ([Rasmussen et al., 2020](#); [Lubbe et al., 2021](#)), while for essential zeros,

likelihood modifications such as zero-inflated models or hierarchical structures can be used (Aitchison and Kay, 2003; Tsagris and Stewart, 2018; Tang et al., 2022).

In line with Aitchison and Kay (2003), a joint hierarchical model similar to Hurdle models (Mullahy, 1986; Cameron and Trivedi, 2013; Martínez-Minaya et al., 2018), which are widely used in the environmental and ecological sciences, can be proposed to deal with null values in likelihoods that do not allow them to be evaluated. This approach involves conditional modelling of the components according to whether each particular component has a zero or non-zero value. In order to apply a structure similar to that of a Hurdle model, there must be a correspondence in the modelling of the composition and subcompositions with that of the null and non-null values of the subcompositions. This implies that we can use it in the case of dealing with data distributed according to a Dirichlet or when we use the centred logratios, since it allows us to make the joint analysis of the zeros with the process that generates the logratios for the CLR, or the shape parameters in the case of the Dirichlet.

Let's assume that we have a matrix of n observations and D compositions, $\mathbf{Y}_{n \times D}$ where each element Y_{ij} is the value for the i -th observation and j -th composition. Then, we can define an incidence matrix $\mathbf{I}_{n \times D}$, that for each element I_{ij} we have $I_{ij} = 0 \iff Y_{ij} = 0$ and $I_{ij} = 1 \iff Y_{ij} \neq 0$. Each column of the incidence matrix will be the values that we will model according to a Bernoulli distribution, and in principle this modelling will be independent between the different compositions. This assumption of strict independence is not entirely accurate, since all components cannot be null at the same time. However, it is also not clear how this restriction impacts the correlations that may exist between the components of the different Bernoulli distributions we are considering. Therefore, we propose a simpler approach by assuming independence, or a weak dependence that would not significantly affect the estimates. Consequently, considering independence for the incidence matrix, the model for each composition integrating the vectors of incidence $\mathbf{i}_d = \mathbf{I}_{1:n,d}$ can be written as:

$$\begin{aligned}
 \mathbf{i}_1 &\sim \text{Ber}(\boldsymbol{\pi}_1) \\
 \text{logit}(\boldsymbol{\pi}_1) &= \mathbf{X}\boldsymbol{\beta}_1 + \sum_{k=1}^K f_{k,1}(\mathbf{z}_k) + \mathbf{u}_{st,1} \\
 &\vdots \\
 \mathbf{i}_D &\sim \text{Ber}(\boldsymbol{\pi}_D) \\
 \text{logit}(\boldsymbol{\pi}_D) &= \mathbf{X}\boldsymbol{\beta}_D + \sum_{k=1}^K f_{k,D}(\mathbf{z}_k) + \mathbf{u}_{st,D}
 \end{aligned} \tag{6.10}$$

where β_d is the vector of linear coefficients, $\mathbf{f}_d$ are non-linear components and $\mathbf{u}_{st,d}$ the spatio-temporal components for the d -th component. While for the models for the values of the compositions, if we use the CLR we can write $\mathbf{Y}^* = \text{CLR}(\mathbf{Y})$, we have the following multivariate Gaussian model for the logratio of the compositions $\mathbf{Y}^* \sim \text{MVN}(\mathbf{0}, \Sigma)$. However, in this case, it is important to distinguish between rows where there is a component with a zero value, or where there is one component with a unit value and the rest are zeros, and rows where no values are zero, in order to analyse them analogously to how it is done with Hurdle models.

The multivariate model can be rewritten as D univariate Gaussian models in which the marginal precision is fixed at a very high value and the variance-covariance structure is evaluated by a random effect as described in the previous section in order to implement these models in R-INLA:

$$\begin{aligned}
 \mathbf{y}_1^* &\sim N(\boldsymbol{\mu}_1, \tau^*) \\
 \boldsymbol{\mu}_1 &= \mathbf{X}\boldsymbol{\beta}_1 + \sum_{k=1}^K f_{k,1}(\mathbf{z}_k) + \mathbf{u}_{st,1} + \mathbf{u} \\
 &\vdots \\
 \mathbf{y}_D^* &\sim N(\boldsymbol{\mu}_D, \tau^*) \\
 \boldsymbol{\mu}_D &= \mathbf{X}\boldsymbol{\beta}_D + \sum_{k=1}^K f_{k,D}(\mathbf{z}_k) + \mathbf{u}_{st,D} + \mathbf{u}
 \end{aligned} \tag{6.11}$$

where τ^* is a fixed precision with a high value ($\log(\tau^*) \gg 1$), $\mathbf{u}$ is the random effect that accounts for the variance-covariance structure, $\mathbf{u} \sim \text{MVN}(\mathbf{0}, \mathbf{Q})$, such that $\mathbf{Q} = \Sigma^{-1}$. In addition, for each observation row with a zero value component, the CLR transformation is re-evaluated to exclude those null values. This procedure is the same whether we use the Dirichlet distribution or the Beta distribution, which is a simplification of the Dirichlet when only two categories are available.

6.6 Downscaling models

Empirical studies involving land use change often use aggregate data for regions, countries, or other geographic scales (Chakir, 2009; Chakir and Le Gallo, 2013). However, some land use studies have had access to individual parcel-level data for analysing water quality, the extent of urban sprawl, carbon sequestration costs, and habitat fragmentation (You and Wood, 2006; Chakir, 2009). The spatial pattern of land use is of particular interest and has been useful in identifying factors that drive changes in land use at the disaggregated level (Chakir, 2009; An and Lee, 2020). In a more general sense, this issue of aggregated, disaggregated, and different scales of

data availability and analysis can be related to fusion models (Wang et al., 2018; Wang and Furrer, 2021; Villejo et al., 2023), data misalignment (Moraga et al., 2017), change of support (Gelfand, 2001; Bradley et al., 2016) and disaggregation modelling or downscaling (Nandi et al., 2023). In this general context, different issues arise concerning the availability at different scales of information concerning the response variable, as in data fusion models (Wang and Furrer, 2021), or availability at different scales of relative information, as in downscaler models (You and Wood, 2006; Berrocal et al., 2010). This implies extracting information from an aggregated scale to a disaggregated or continuous scale using an effect with a continuous spatial or spatio-temporal structure, structure that is commonly used in geostatistics to account for the data spatial dependence. This is possible to implement in INLA thanks to the definition of spatial models using the SPDE-FEM approach, which is integrated in the R-INLA package (Lindgren and Rue, 2015; Bakka et al., 2018).

6.6.1 SPDE-FEM

The SPDE-FEM approach (which stands for *Stochastic Partial Differential Equations* and *Finite Element Methods*) allows for expressing the continuous spatial structure by approximating a Gaussian Field with a Gaussian Markov Random Field. This makes the inferential process efficient by reducing computational costs. This approach was initially proposed in (Lindgren et al., 2011) by means of an approximate stochastic weak solution to the SPDE in Equation (6.12), proving that a stationary SPDE solution has a Matérn covariance function and allowing to calculate directly its precision matrix, bypassing the need for resource-intensive inverse processes. Thus, formally we get that the structured spatial effect ξ with precision τ is related to a non-structured Gaussian random effect (white noise) through a filtering operator $(\kappa - \Delta)^{\alpha/2}$:

$$\begin{aligned} (\kappa^2 - \Delta)^{\alpha/2} \tau \cdot \xi(\mathbf{s}) &= \mathcal{W}(\mathbf{s}), \\ \mathbf{s} \in \mathcal{D}, \alpha &= \nu + d/2, \kappa > 0, \nu > 0, \end{aligned} \tag{6.12}$$

where $\mathbf{s}$ are the locations; κ is a spatial scale parameter positive defined and related to ρ and ν through $\kappa = \sqrt{8\nu/\rho}$; α controls the smoothness of the performances, which by default is 2; τ regulates the variance, so it is also a parameter defined positive as $\tau^2 = \Gamma(\nu) [\Gamma(\alpha)(4\pi)^{d/2}\kappa^2\sigma^2]^{-1}$; and $\mathcal{D}$ is the spatial domain $\mathcal{D} \subset \mathbb{R}^d$, being d the Euclidean spatial dimension of such spatial domain. Additionally, $\Delta = \sum_i \partial^2 / \partial s_i^2$ is the Laplacian operator and $\mathcal{W}$ denotes a spatial stochastic Gaussian process with unit variance. Therefore, the solution for $\xi(\mathbf{s}_i)$ leads to a

Gaussian field with a covariance matrix defined by the Matérn function correlation $\mathcal{C}(h)$, where the covariance is the correlation multiplied by the marginal deviation:

$$\mathcal{C}(h) = \sigma^2 \cdot \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \cdot \frac{h}{\rho} \right)^\nu \mathcal{K}_\nu \left(\sqrt{2\nu} \cdot \frac{h}{\rho} \right). \quad (6.13)$$

In the above equation, σ is the marginal standard deviation, ρ is the spatial range, $\mathcal{K}_\nu$ is a modified Bessel function of the second kind of order ν , where ν is a parameter of smoothness defined as $\nu = \alpha - d/2$. In INLA, $\alpha = 2$ is taken by default, and in our cases, the dimension of the analysis space is the plane ($d = 2$), resulting in a constant smoothness parameter value $\nu = 1$. Furthermore, under the SPDE approximation, the spatial range ρ is considered as the distance at which the correlation value is close to 0.1 (Lindgren et al., 2011).

This approach allows defining a spatial effect whose precision structure is sparse and approximates the solution of a continuous Gaussian field with a variance-covariance structure belonging to the Matérn family; where one particular case of this family is the exponential covariance function when $\nu = 1/2$. This spatial effect is defined by a Gaussian Markov Random Field (GMRF), $\mathbf{u} \sim \text{GMRF}(\mathbf{0}, \mathbf{Q})$, whose precision matrix for the standard case of $d = 2$ and $\nu = 1$ can be expressed as $\mathbf{Q} = \tau^2(\kappa^4 \mathbf{C} + 2\kappa^2 \mathbf{G} + \mathbf{G}\mathbf{C}^{-1}\mathbf{G})$. Here, $\mathbf{C}$ and $\mathbf{G}$ are two matrices that can be computed through the structure of a constrained refined Delaunay triangulation $\mathcal{T}$ by Finite Element Methods. The GMRF realisations at the node locations $\mathbf{s}_n$ can be projected to any other point within the mesh domain, usually to the observation locations $\mathbf{s}_o$, through a linear basis approximation synthesised in the matrix product of the projection matrix $\mathbf{A}(\mathbf{s}_n, \mathbf{s}_o)$ with the GMRF realisation at the mesh nodes $\mathbf{u}(\mathbf{s}_n)$, $\mathbf{u}(\mathbf{s}_o) = \mathbf{A}(\mathbf{s}_n, \mathbf{s}_o)\mathbf{u}(\mathbf{s}_n)$. This allows the spatial effect $\mathbf{u}(\mathbf{s}_o)$ to be rewritten as $\mathbf{u}(\mathbf{s}_o) \sim \text{GMRF}(\mathbf{0}, \mathbf{A}^T \mathbf{Q} \mathbf{A})$, which is clearly equivalent to $\mathbf{u}(\mathbf{s}_o) = \mathbf{A}(\mathbf{s}_n, \mathbf{s}_o)\mathbf{u}(\mathbf{s}_n)$.

6.6.2 Downscaling model with SPDE-FEM

The definition of a spatial effect whose structure is continuous in space and space-time allows for spatial and spatio-temporal downscaling. Let's assume we have the following structure for the linear predictors of the model of our data:

$$\begin{aligned} \eta_{i,1} &= \mathbf{X}_i \boldsymbol{\beta}_1 + \sum_{k=1}^K f_{k,1}(z_i) + u_{st,1}(C_i, t_i) \\ &\vdots \\ \eta_{i,D} &= \mathbf{X}_i \boldsymbol{\beta}_D + \sum_{k=1}^K f_{k,D}(z_i) + u_{st,D}(C_i, t_i) \end{aligned} \quad (6.14)$$

the downscaling is defined for the spatial structure when, for each area C_i , the spatially structured effect is calculated as follows:

$$u_{st}(C_i, t_i) = \frac{\int_{\mathbf{s} \in C_i} u_{st}(\mathbf{s}, t_i) d\mathbf{s}}{|C_i|} \approx \frac{\sum_{i=1}^{n(\mathbf{s}_i \in C_i)} u_{st}(\mathbf{s}_i, t_i)}{n(\mathbf{s}_i \in C_i)} = \frac{\mathbf{A}(\mathbf{s}_n, s_i) \mathbf{u}_{st}(\mathbf{s}_n, t_i)}{n(\mathbf{s}_i \in C_i)} \quad (6.15)$$

where s_i are integration points, allowing the approximation of the integral in the region C_i by averaging the sum of the field at the integration points, and $n(\mathbf{s}_i \in C_i)$ is the number of integration points in C_i ; being the area of C_i determine as $|C_i| = \int_{\mathbf{s} \in C_i} d\mathbf{s}$. If a spatio-temporal downscaling is implemented, then the integration would be performed in both dimensions, spatial s and temporal t , using integration points for each corresponding volume $C_i \times t_i$. Additionally, this procedure can also be applied to establish sub-models for those covariates with a spatial structure that are given in areas. These sub-models would involve the modelling of such covariates by incorporating the integral of the covariate scaled by a linear regression coefficient, analogous to how it is done in error models (Muff et al., 2015, 2017) in cases without downscaling. The linear predictors in Equation (6.14) can be associated with either a Dirichlet distribution or a multivariate normal distribution related to some of the logratio transformations previously shown. If they correspond to a logratio transformation, it is necessary to re-write the linear predictors to incorporate the variance-covariance structure Σ of the multivariate normal distribution, as explained in previous sections.

The downscaling approach not only provides a way to obtain estimates on a continuous scale of the spatial effect but, due to its structure and implementation, also allows us to consider the same spatial effect for different spatial supports. This has several implications, as detailed in the initial references. One particular implication we would like to highlight is that when the spatial support varies over time—such as in the case of small areas for disease mapping or NUTS3—the use of a downscaled spatial or spatio-temporal effect ensures a coherent structure across the different supports, maintaining the spatial or spatio-temporal dependence of the data. Furthermore, this does not depend on the specific use of the SPDE-FEM approach, as it can be implemented with any other Gaussian field, such as spline and two-dimensional spline bases, or any other continuous field.

6.7 Big Data

Public smart card data, Wi-Fi access point data, wireless sensor networks, and data with spatial location information —such as social media data, mobile phone tracking, and other sensing information from Internet of Things devices— can provide useful ancillary data for LULC (Land Use and Land Cover) mapping. Compared to traditional geospatial data acquisition, these geospatial big data (GBD) are typically obtained at a lower cost and offer different coverages and better spatio-temporal resolutions (Liu et al., 2020; Zhang and Li, 2022). They contain abundant human activity information, which can compensate for the lack of socioeconomic data (Liu et al., 2020). By leveraging both Remote Sensing (RS) and GBD, it is possible to examine the physical and socioeconomic characteristics of the urban land system (Martí et al., 2019). Despite the great potential of integrating RS and GBD for enhanced insights into urban land use, significant challenges remain in storing, managing, analysing, and visualising these data due to differences in spatial data quality (e.g., semantics, timestamps, and scale), technical formats, and data structures (Li et al., 2016). Therefore, a large amount of land use information is accessible, particularly with global and low-scale land use data, which gives rise to a big data problem.

This entails dealing with very heterogeneous and high dimensional data, particularly when data are available at global scale (Stanimirova et al., 2023) or when LULC data at regional scales are available at fine grid (You and Wood, 2005; Chakir, 2009). The joint analysis of data that are at different scales and whose spatial structure is in different supports, as well as combining different sources of information, means that modelling structures can be particularly complex (Figueira et al., 2024, 2025). In order to deal with these large databases and complex modelling structures, sequential inference procedures can be implemented in INLA, which reduces the computational cost. In particular, a sequential consensus inferential procedure can be established and implemented in R-INLA (Figueira et al., 2025). This procedure, summarised in Figure 6.1, involves the marginal updating of fixed effects and hyperparameters over a given partition of the data. The partition of the data can be done along the various sources of information available, such as different likelihoods or groups of likelihoods, or by leveraging the structure of the latent field (spatial, temporal, or spatio-temporal). Once the sequential inference is performed, the random effects information is combined using a consensus approach according to their (i) marginal or (ii) multivariate distributions. This approach

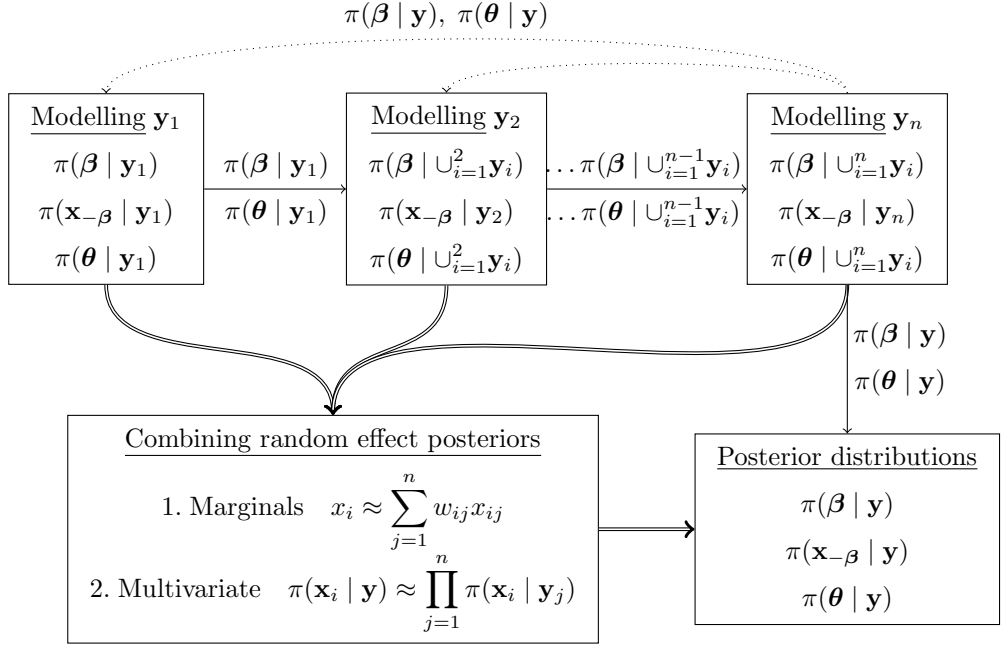


FIGURE 6.1: Scheme of the sequential consensus procedure. This approach assumes sequentially updating the fixed effects and hyperparameters, followed by performing a consensus for the random effects after these sequential updates (Figueira et al., 2025).

also accommodates complex models with shared components between different likelihoods.

This procedure reduces computational burden in terms of memory and CPU usage by solving the dataset or the original model through partitions. While it provides good estimates for the latent field, the evaluation of hyperparameters may differ from that obtained by analysing the full model without partitioning. This is because only the marginal information is updated, and hyperparameters can exhibit non-negligible correlations in the posterior distribution.

6.8 Examples

In this section, several examples are provided to illustrate the implementation of the various methodological approaches described in the previous sections. These range from handling zeros and ones using adaptations of Hurdle models—widely used in environmental and ecological sciences—to methods for downscaling and procedures for managing large databases or complex models.

The first example demonstrates how to handle compositional data with 0's and 1's, using simulated data. The second example illustrates the results of a downscaling model applied to real land use data from the European LAMASUS project. Finally, the third example uses a large simulated dataset to showcase the effectiveness of the sequential consensus procedure in analysing extensive datasets and accurately estimating the underlying process.

6.8.1 Dealing with 0's example

In this first example, we present two cases that illustrate how to handle the presence of zeros in CoDa. The first case uses data simulated from a Beta distribution to exemplify the simplest scenario, proving the Hurdle model for dealing with zeros. The second case also addresses CoDa, analysing 3 categories through a multivariate model of the CLR transformation.

In the Beta case, we simulate the data from the following hierarchical conditional model:

$$\begin{aligned} Z_i &\sim \text{Ber}(\pi_i), \\ \text{logit}(\pi_i) &= \beta_B + \alpha \cdot (\beta_0 + \beta_1 \cdot x_i + u_i), \\ Y_j | Z_j = 1 &\sim \text{Beta}(\mu_j, \phi), \\ \text{logit}(\mu_j) &= \beta_0 + \beta_1 \cdot x_j + u_{sj}, \end{aligned} \tag{6.16}$$

where Z_i are Bernoulli random variables explaining the process that generates the incidence vector (incidence matrix for CoDa), such that $z_i = 0 \iff y_i = 0$ and $z_i = 1 \iff y_i \neq 0$. Y_j are Beta random variables conditioned on $Z_j = 1$, which implies that $Y_j \neq 0$. The latent field comprises an intercept β_0 , a linear coefficient β_1 related to a covariate $\mathbf{x}$, and a spatial effect $\mathbf{u}_s$ following a Leroux structure. The spatial adjacency structure is built by partitioning mainland Spain with a Voronoi diagram, as shown in Figure 6.2. This latent field is shared in the linear predictor of

the Bernoulli process, scaled by α and including an intercept β_B that controls the overall proportion of 0's. Finally, ϕ stands for the precision of the Beta distribution.

Figure 6.2 shows the simulated spatial structure and the distribution of null and non-zero values, along with the inferential results of the beta model without incorporating the zeros and the Hurdle model incorporating the null values. It can be seen that, in general, the Hurdle model better captures the true values of the spatial effects where there are many zeros, as well as the posterior distributions of the model parameters. In the figure, we can see a slightly better identification of the β components, along with improved identification of the marginal precision τ and, in particular, the spatial dependence parameter λ of the Leroux distribution. To assess these improvements, we consider the DIC of the models and the RMSE for the spatial trend, as we have the true values of the spatial pattern from the simulation. RMSE is used to evaluate the improvement in identifying the spatial pattern according to the model. The DIC values for the models are -359 (Beta model) and -363 (Hurdle model), while the RMSE values are 0.56 for the Beta model and 0.51 for the Hurdle-like model.

The CoDa simulation is performed considering that only one ($\mathbf{Y}_1$) of the 3 categories ($\mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3$) can present zero values. This model serves as a conceptual example of how it can be implemented for CoDa, dealing with zeros and ones in several categories. Therefore, the model structure when no null values are present is as follows:

$$\begin{aligned} Z_{i1} &\sim \text{Ber}(\pi_i), \\ Y_{j1} \mid Z_{j1} = 1 &\sim \text{N}(\mu_{j1}, \tau^*), \\ Y_{j2} \mid Z_{j1} = 1 &\sim \text{N}(\mu_{j2}, \tau^*), \\ Y_{j3} \mid Z_{j1} = 1 &\sim \text{N}(\mu_{j3}, \tau^*), \end{aligned} \tag{6.17}$$

where the linear predictor for the CLRs are defined with the same structure as the one used for the Beta example with the addition of the $\mathbf{u}$ effect to take into account the correlation between the categories, $\eta_{id} = \beta_{0d} + \beta_{1d}x_i + u_{sid} + u_i$. However, when $Z_{i1} = 0$ the CLR change to the remaining two categories ($\mathbf{Y}_2, \mathbf{Y}_3$) using the same underlying process in the linear predictors as before.

Figure 6.3 shows the simulated spatial effects and the distribution of the zero and non-zero values for the first composition (*Comp. 1*). The figure also shows the inferred spatial effects (mean and stdev.) for the different compositions, along with the distributions for the fixed parameters and hyperparameters. In the figure, we

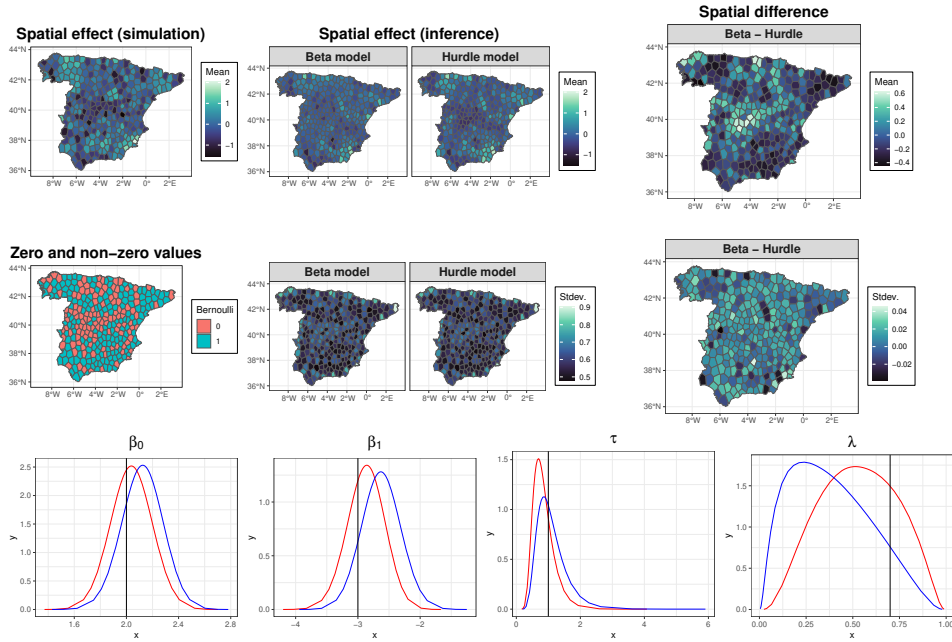


FIGURE 6.2: Simulated spatial effect, spatial distribution of zero and non-zero values, inferred spatial effect (mean and standard deviation), differences in the mean and standard deviation of the spatial structure between the Beta and Hurdle models, and posterior distributions of the fixed parameters and hyperparameters are presented. Distributions from the Hurdle model are shown in red, while those from the Beta model are shown in blue.

can see an improved identification of the β components, along with improved identification of the marginal precisions τ and slight changes in the spatial dependence parameters λ of the Leroux distributions for the spatial effects.

In both cases we can observe an improvement in the estimation of the model by incorporating the Hurdle-type structure to integrate the analysis of the zeros. To assess this improvement, the DIC is also considered for the two models, the Hurdle-like model and the standard model. As a measure for evaluating the improvement in identifying the spatial pattern according to the model, RMSE has been used, given that we have the true values of the spatial effect from the simulation. The DIC between the models is: -4615 (Hurdle-like) and -2754 (standard model), while the RMSE is $(0.50, 0.25, 0.20)$ for the Hurdle-like model and $(0.54, 0.34, 0.31)$ for the standard model, for spatial effects of component 1, 2, and 3, respectively.

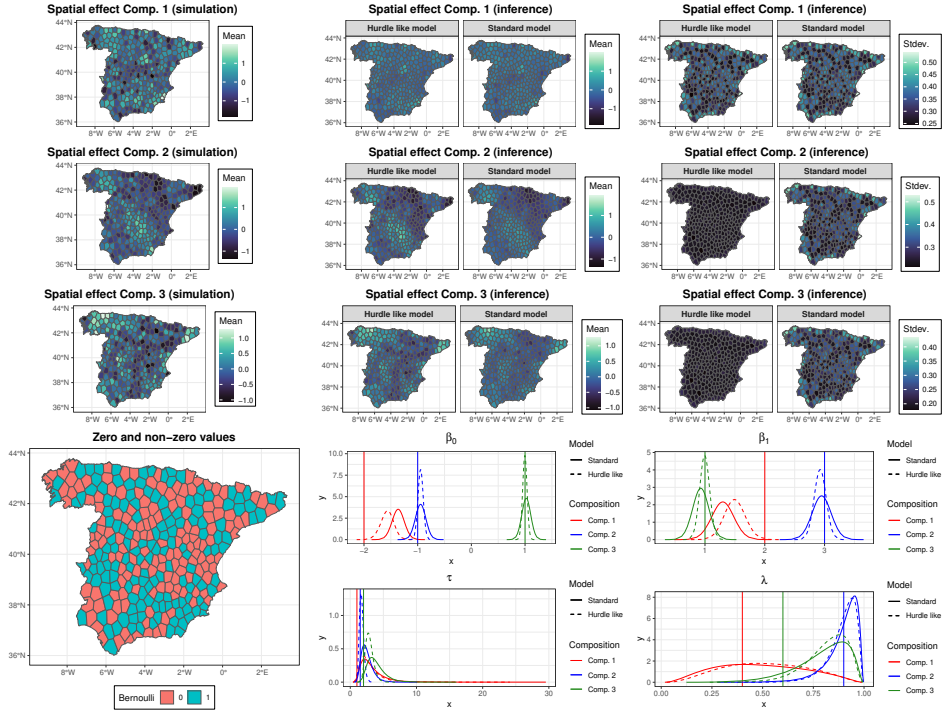


FIGURE 6.3: Simulated spatial effects, spatial distribution of the zero and non-zero values, and inferred spatial effects (mean and stdev.), along with the posterior distributions of the fixed parameters and hyperparameters.

6.8.2 Downscaling example: LAMASUS data

In this example, we present the implementation of the downscaling procedure for real land use data from the European LAMASUS project. In the context of the LAMASUS project, we aim to identify the components that determine or drive land use and land use change in Europe over the last decades. The data used in this example is spatially structured at the NUTS3 level between the years 2007 and 2018; where NUTS (Nomenclature of Territorial Units for Statistics) is the hierarchical system for dividing the economic territory of the European Union. The land use data analysed in this example is classified into five aggregated categories to avoid the presence of zero and unit values. These categories are: (i) cropland, (ii) grassland, (iii) forest, (iv) urban, and (v) other natural land. A wide range of explanatory variables are available to analyse the underlying processes that determine land

use, including socio-economic and demographic variables such as GDP, GVA, and employment, as well as environmental variables like aspect and elevation.

We begin by focusing on the proportion of agricultural land use, specifically the cropland category. To model exclusively the proportion of cultivated land use, we employ a downscaling model using a Beta distribution. Specifically, we use a spatio-temporal hierarchical Bayesian model, as outlined below:

$$\begin{aligned}
 Y_{it} &\sim \text{Beta}(\mu_{it}, \phi) \\
 \text{logit}(\mu_{it}) &= \beta X_{it} + \int_{\mathbf{s} \in C_i} \frac{u(\mathbf{s}) d\mathbf{s}}{|C_i|} + u_t,
 \end{aligned}
 \tag{6.18}$$

where Y_{it} represents the random variable for the proportion of cropland use in a specific region i and year t . The conditional distribution of Y_{it} follows a Beta distribution with mean μ_{it} and precision ϕ . The mean is linked through the logit function to the linear predictor. The covariates are represented by $\mathbf{X}_{it}$, and the vector of the linear regression coefficients are β . The downscaling spatial effect is $u(\mathbf{s})$, while the temporal effect is modelled as a first-order autoregressive process $u_t = \phi_t u_{t-1} + \epsilon(\tau_t)$, where ϕ_t and τ_t are the temporal autoregressive parameter and the marginal precision of the temporal autoregressive effect, respectively.

The selection of covariates to be included in the model was performed by exploring all possible combinations of models without spatial and temporal effects. The combination with the lowest Watanabe-Akaike Information Criterion (WAIC; [Watanabe 2013](#)) was selected. This means that for the 22 explanatory variables available, all possible inclusion and exclusion combinations were analyzed, resulting in a total of 2^{22} models. The model with the lowest WAIC was chosen.

Some results can be seen in Figure 6.4. As the figure shows, the temporal effect is small. As for the spatial effect, it captures the variability not explained by the covariates. We can observe that, in certain areas, the spatial effect has a greater impact on the overall average effect than in other regions, where its impact is less pronounced. Regarding the covariates, some notable ones include decoupled payments (in euros) normalized by the total used agricultural area (in hectares), and less favoured area subsidies (in euros) normalized by the total used agricultural area (in hectares). Additionally, other relevant variables are the area of the region, elevation of the terrain, GVA, and total output (in euros).

The analysis for the 5 aggregated land use categories is performed using a multivariate Gaussian model with the ALR transformation of the categories. To

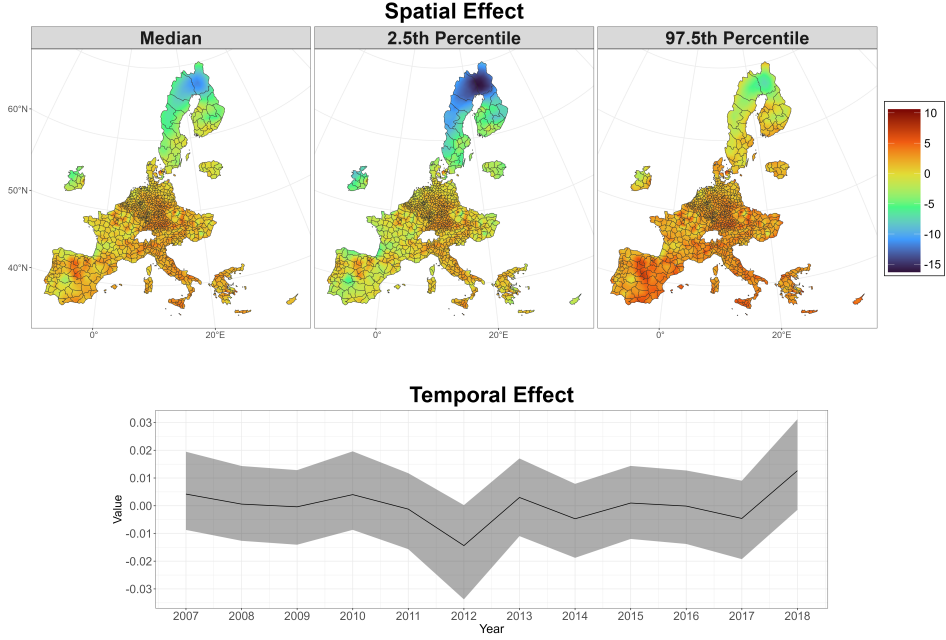


FIGURE 6.4: Cartographic representation of the continuous spatial effect and the temporal effect of the downscaling model.

select the covariates, we implemented a stepwise search algorithm driven by the WAIC of the models after a pre-selection where highly correlated ($\rho > 0.75$) variables were extracted. This algorithm involves the following steps: (i) a forward search among the covariates until no additional covariate can be included, then (ii) a backward step until no covariate can be deleted, and (iii) repeating steps (i) and (ii) until there is no change. The base model into which the algorithm is implemented encompasses a specific intercept, spatial, and temporal component for each ALR. The final model includes 9 explanatory covariates, and the model for the mean $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_4)$ of the multivariate Gaussian distribution $\text{MVN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is written as follows:

$$\begin{aligned}
 \mu_{i1} &= \beta_{01} + \mathbf{X}_i \boldsymbol{\beta}_1 + u_{si1} + u_{ti1} + u_i, \\
 \mu_{i2} &= \beta_{02} + \mathbf{X}_i \boldsymbol{\beta}_2 + u_{si2} + u_{ti2} + u_i, \\
 \mu_{i3} &= \beta_{03} + \mathbf{X}_i \boldsymbol{\beta}_3 + u_{si3} + u_{ti3} + u_i, \\
 \mu_{i4} &= \beta_{04} + \mathbf{X}_i \boldsymbol{\beta}_4 + u_{si4} + u_{ti4} + u_i,
 \end{aligned} \tag{6.19}$$

where u_{sid} is the downscaled spatial effect, $u_{sid} = \int_{\mathbf{s} \in C_i} u_{sd}(\mathbf{s}) d\mathbf{s} / |C_i|$, and the

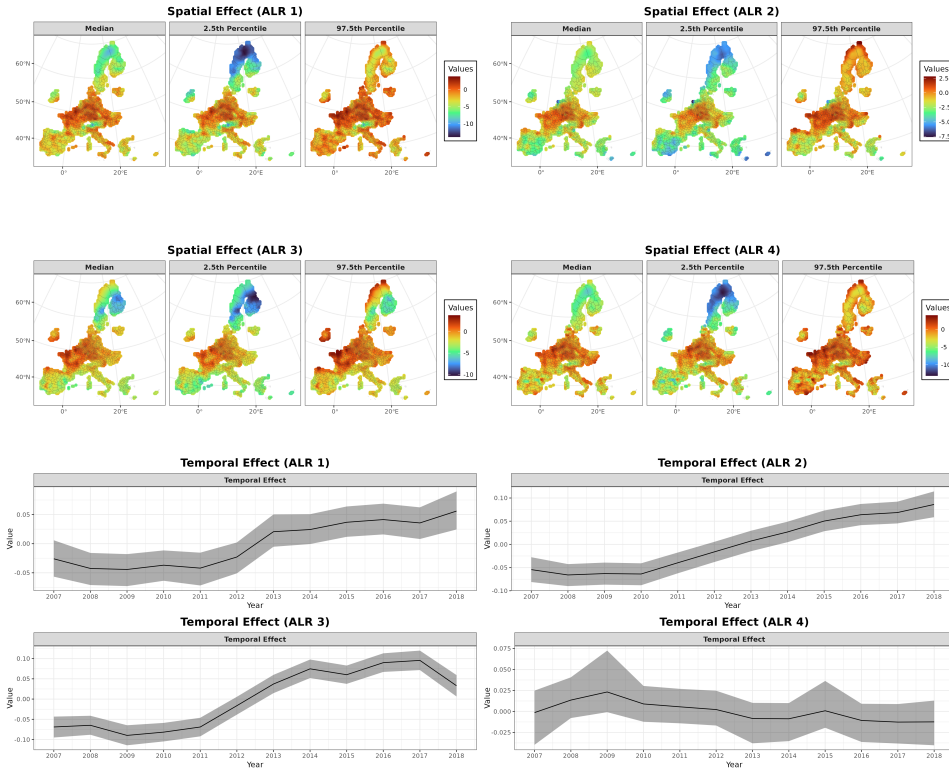


FIGURE 6.5: Cartographic representation of the continuous spatial effects and the temporal effects for the different ALRs of the downscaling model.

component $\mathbf{u}$ accounts for the correlation Σ^{-1} between the categories, as explained in the corresponding section.

Figure 6.5 shows the downscaled continuous spatial effects for the different ALRs, together with the temporal effects. In this case, we can observe the scaling effect for the different ALRs, which implies a common spatial dependency structure over the years. This is combined with the temporal effect for the log-ratios, where the trend is similar for the first three, while the fourth shows a temporal effect with greater uncertainty and is less pronounced.

6.8.3 Big Data example

In this final example, we present the implementation of the algorithm summarised in Figure 6.1 for evaluating large databases of compositional data. The purpose of the simulation is not merely to demonstrate the results of the algorithm implementation for handling large databases, but also to showcase its versatility and its application to spatio-temporal downscaling models. This enables the processing of structured data in areas with spatial support that varies over the years.

The simulation is conducted for three categories evaluated using the ALR transformation, so the multivariate Gaussian structure will have a dimension of 2. The spatial structure is simulated using a continuous Gaussian field aggregated over three different spatial supports, each structured into areas. Specifically, the aggregation is based on three distinct areal structures, with 200, 250, and 300 areas, as shown in Figure 6.6, making the downscaling model purely spatial. The Gaussian field is a separable spatio-temporal effect, defined by the following precision matrix: $\mathbf{Q}_{st} = \mathbf{Q}_s \otimes \mathbf{Q}_t$. The spatial precision matrix is defined using an SPDE in two dimensions, while the temporal part is determined by a first-order autoregressive structure with 600 (temporal) nodes. To construct the different spatial supports, the territory of mainland Spain has been partitioned using Voronoi diagrams.

The model used to simulate and infer the data is as follows:

$$\begin{aligned}\mu_{it1} &= \beta_{01} + \int_{\mathbf{s} \in C_i} u_1(\mathbf{s}, t) d\mathbf{s} + u_{it}, \\ \mu_{it2} &= \beta_{02} + \int_{\mathbf{s} \in C_i} u_2(\mathbf{s}, t) d\mathbf{s} + u_{it},\end{aligned}\tag{6.20}$$

where β_{0d} are the intercepts, $\mathbf{u}_d(\mathbf{s}, t)$ are the spatio-temporal effects aggregated over the different spatial supports for each ALR. Finally, the effect $\mathbf{u}$ accounts for the correlation structure between the different ALRs, allowing for the implementation of the multivariate Gaussian distribution in R-INLA.

Figure 6.6 shows the mesh for the spatial effect and the different spatial supports in which the data was aggregated over the various years. It also presents the aggregated spatial effect using simulated data, as well as the desegregated spatial effects obtained from the inferential process of the two ALRs for the first temporal node. The data set was divided along the temporal scale, grouping into 6 temporal nodes, as larger grouping would exceed the server's memory capacity, which had 253 GB of RAM and 93 cores. Additionally, the complete analysis by the sequential consensus procedure for the entire dataset took 314.28 minutes.

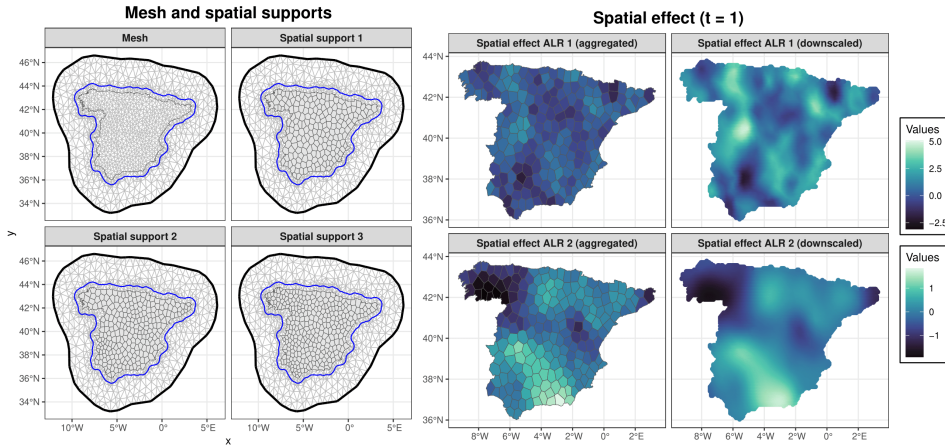


FIGURE 6.6: Mesh for spatial effects, spatial supports for data aggregation and outputs, comparing the aggregated simulated data with the downsampled spatial fields obtained from the inference for the first temporal node.

6.9 Conclusions

In the field of land use science, the application of spatial or spatio-temporal models is essential for evaluating the key factors that modify and direct land use and land use changes. Assessing spatial patterns alongside temporal patterns provides a precise understanding of their distribution in space and their temporal evolution. To achieve this, approaches often use either large-scale aggregated data or small-scale disaggregated data. In the latter case, when small-scale data is available, two natural problems may arise: the presence of null values and computational complications when handling large databases. Conversely, when dealing with large-scale data, the issue of changing the spatial aggregation support over time may occur. This study presents methodologies to address the problem of null values, downscaling procedures for spatial or spatio-temporal structures, and big data challenges. These methodologies are illustrated with various examples to demonstrate their implementation.

The results presented in the examples show the improvement when the presence of zeros is integrated into the analysis, along with the ability to perform downscaling or disaggregation models using continuous Gaussian fields, specifically through the SPDE-FEM approach. Additionally, the implementation of a downscaling model on a large dataset across different aggregation supports was shown. This highlights

the possibility of encountering aggregated data in different spatial structures over the years, and how to deal with them in a Big Data context. This can be achieved by implementing the sequential consensus algorithm; otherwise, computational limitations would render the analysis unfeasible.

References

- Aitchison, J. (1982). “The Statistical Analysis of Compositional Data.” *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 44: 139–160.
- (1986). *The statistical analysis of compositional data*. London: Chapman & Hall.
- Aitchison, J. and Egozcue, J. J. (2005). “Compositional Data Analysis: Where Are We and Where Should We Be Heading?” *Mathematical Geology*, 37: 829–850.
- Aitchison, J. and Kay, J. (2003). “Possible solution of some essential zero problems in compositional data analysis.” In *Proceedings of CoDaWork’03, The 1st Compositional Data Analysis Workshop*. University of Girona, Girona (Spain).
- An, Y. and Lee, S. (2020). “A Spatial Disaggregation Model to Improve Long-Term Land Use Forecasting with Transport Models Based on Zonal Data.” *Applied Spatial Analysis and Policy*, 13: 187–208.
- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D., and Lindgren, F. (2018). “Spatial modeling with R-INLA: A review.” *WIREs Computational Statistics*, 10(6): e1443.
- Banerjee, S., Carlin, B. P., and Gelfand, A. E. (2015). *Hierarchical Modeling and Analysis for Spatial Data*. Chapman & Hall/CRC Monographs on Statistics & Applied Probability. New York: Chapman and Hall/CRC, 2ed. edition.
- Bareille, F. and Chakir, R. (2024). “Structural identification of weather impacts on crop yields: Disentangling agronomic from adaptation effects.” *American Journal of Agricultural Economics*, 106(3): 989–1019.
- Berild, M. O., Martino, S., Gómez-Rubio, V., and Rue, H. (2022). “Importance Sampling with the Integrated Nested Laplace Approximation.” *Journal of Computational and Graphical Statistics*, 31: 1225–1237.

- Berrocal, V. J., Gelfand, A. E., and Holland, D. M. (2010). “A Spatio-Temporal Downscaler for Output From Numerical Models.” *Journal of Agricultural, Biological, and Environmental Statistics*, 15: 176–197.
- Besag, J. (1974). “Spatial Interaction and the Statistical Analysis of Lattice Systems.” *Journal of the Royal Statistical Society. Series B (Methodological)*, 36: 192–236.
- Besag, J., York, J., and Mollié, A. (1991). “Bayesian image restoration, with two applications in spatial statistics.” *Annals of the Institute of Statistical Mathematics*, 43(1): 1–20.
- Bivand, R. S., Gómez-Rubio, V., and Rue, H. (2014). “Approximate Bayesian inference for spatial econometrics models.” *Spatial Statistics*, 9: 146–165.
- Bradley, J. R., Wikle, C. K., and Holan, S. H. (2016). “Bayesian Spatial Change of Support for Count-Valued Survey Data With Application to the American Community Survey.” *Journal of the American Statistical Association*, 111: 472–487.
- Cameron, A. C. and Trivedi, P. K. (2013). *Regression Analysis of Count Data*. Econometric Society Monographs. Cambridge (UK): Cambridge University Press, 2 edition.
- Chakir, R. (2009). “Spatial Downscaling of Agricultural Land-Use Data: An Econometric Approach Using Cross Entropy.” *Land Economics*, 85(2): 238–251.
- Chakir, R. and Le Gallo, J. (2013). “Predicting land use allocation in France: A spatial panel data analysis.” *Ecological Economics*, 92: 114–125. Land Use.
- Chakir, R. and Le Gallo, J. (2021). *Spatial Autocorrelation in Econometric Land Use Models: An Overview*, 339–362. Cham: Springer International Publishing.
- Chakir, R. and Lungarska, A. (2017). “Agricultural rent in land-use models: comparison of frequently used proxies.” *Spatial Economic Analysis*, 12(2-3): 279–303.
- Chakir, R. and Parent, O. (2009). “Determinants of land use changes: A spatial multinomial probit approach.” *Papers in Regional Science*, 88(2): 327–345.
- Clayton, D. (1996). “Generalized linear mixed models.” In Gilks, W. R., Richardson, S., and Spiegelhalter, D. J. (eds.), *Markov Chain Monte Carlo in Practice*, chapter 16, 275–301. London: Chapman and Hall.

- Connor, R. J. and Mosimann, J. E. (1969). “Concepts of Independence for Proportions with a Generalization of the Dirichlet Distribution.” *Journal of the American Statistical Association*, 64: 194–206.
- Egozcue, J. J. and Pawlowsky-Glahn, V. (2006). “Simplicial geometry for compositional data.” *Geological Society, London, Special Publications*, 264: 145–159.
- Egozcue, J. J., Pawlowsky-Glahn, V., Mateu-Figueras, G., and Barceló-Vidal, C. (2003). “Isometric Logratio Transformations for Compositional Data Analysis.” *Mathematical Geology*, 35: 279–300.
- Elhorst, J. P. (2014). *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*. SpringerBriefs in Regional Science. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Figueira, M., Barber, X., Conesa, D., López-Quílez, A., Martínez-Minaya, J., Paradinas, I., and Pennino, M. G. (2024). “Bayesian feedback in the framework of ecological sciences.” *Ecological Informatics*, 84: 102858.
- Figueira, M., Conesa, D., López-Quílez, A., and Paradinas, I. (2025). “A computationally efficient procedure for combining ecological datasets by means of sequential consensus inference.” *Environmental and Ecological Statistics*, 32(2): 495–521.
- Fišerová, E. and Hron, K. (2011). “On the Interpretation of Orthonormal Coordinates for Compositional Data.” *Mathematical Geosciences*, 43: 455–468.
- Gelfand, A. E. (2001). “On the change of support problem for spatio-temporal data.” *Biostatistics*, 2: 31–45.
- Greenacre, M. and Grunsky, E. (2019). “The isometric logratio transformation in compositional data analysis: a practical evaluation. Proceedings of the 8th International Workshop on Compositional Data Analysis (CoDaWork2019): Terrassa, 3-8 June, 2019.” *Universitat Politècnica de Catalunya-BarcelonaTECH, 2019, p. 4–10*, 74–81.
- Greenacre, M., Grunsky, E., Bacon-Shone, J., Erb, I., and Quinn, T. (2023). “Aitchison’s Compositional Data Analysis 40 Years on: A Reappraisal.” *Statistical Science*, 38(3): 386 – 410.
- Gómez-Rubio, V. (2020). *Bayesian inference with INLA*. New York: Chapman and Hall/CRC Press.

- Gómez-Rubio, V. and Rue, H. (2018). “Markov chain Monte Carlo with the Integrated Nested Laplace Approximation.” *Statistics and Computing*, 28.
- Gómez-Rubio, V., Bivand, R. S., and Rue, H. (2021). “Estimating Spatial Econometrics Models with Integrated Nested Laplace Approximation.” *Mathematics*, 9(17).
- Hersperger, A. M. and Bürgi, M. (2009). “Going beyond landscape change description: Quantifying the importance of driving forces of landscape change in a Central Europe case study.” *Land Use Policy*, 26(3): 640–648.
- Hijazi, R. H. and Jernigan, R. W. (2009). “Modeling Compositional Data Using Dirichlet Regression Models.” *Journal of Applied Probability & Statistics*, 4: 77–91.
- James LeSage, R. K. P. (2009). *Introduction to Spatial Econometrics*. New York: Chapman and Hall/CRC, 1 edition.
- Knorr-Held, L. (2000). “Bayesian modelling of inseparable space-time variation in disease risk.” *Statistics in Medicine*, 19(17-18): 2555–2567.
- Krisztin, T., Piribauer, P., and Wögerer, M. (2022). “A spatial multinomial logit model for analysing urban expansion.” *Spatial Economic Analysis*, 17: 223–244.
- Leroux, B. G., Lei, X., and Breslow, N. (2000). “Estimation of Disease Rates in Small Areas: A new Mixed Model for Spatial Dependence.” In Halloran, M. E. and Berry, D. (eds.), *Statistical Models in Epidemiology, the Environment, and Clinical Trials*, 179–191. New York, NY: Springer New York.
- LeSage, J. P. (2008). “An Introduction to Spatial Econometrics.” *Revue d’économie industrielle*, 123(4): 19–44.
- Li, S., Dragicevic, S., Castro, F. A., Sester, M., Winter, S., Coltekin, A., Pettit, C., Jiang, B., Haworth, J., Stein, A., and Cheng, T. (2016). “Geospatial big data handling theory and methods: A review and research challenges.” *ISPRS Journal of Photogrammetry and Remote Sensing*, 115: 119–133.
- Lindgren, F. and Rue, H. (2015). “Bayesian Spatial Modelling with R-INLA.” *Journal of Statistical Software*, 63(19): 1–25.
- Lindgren, F., Rue, H., and Lindström, J. (2011). “An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential

- equation approach.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(4): 423–498.
- Lindgren, F., Bakka, H., Bolin, D., Krainski, E., and Rue, H. (2024). “A diffusion-based spatio-temporal extension of Gaussian Matérn fields.” *SORT-Statistics and Operations Research Transactions*, 48(1): 3–66.
- Liu, W., Wu, W., Thakuriah, P., and Wang, J. (2020). “The geography of human activity and land use: A big data approach.” *Cities*, 97: 102523.
- Lubbe, S., Filzmoser, P., and Templ, M. (2021). “Comparison of zero replacement strategies for compositional data with large numbers of zeros.” *Chemometrics and Intelligent Laboratory Systems*, 210: 104248.
- MacNab, Y. C. (2022). “Bayesian disease mapping: Past, present, and future.” *Spatial Statistics*, 50: 100593. Special Issue: The Impact of Spatial Statistics.
- (2023). “Revisiting Gaussian Markov random fields and Bayesian disease mapping.” *Statistical Methods in Medical Research*, 32: 207–225.
- Martinez-Beneito, M. A. and Botella-Rocamora, P. (2019). *Disease Mapping: From Foundations to Multidimensional Modeling*. New York: Chapman and Hall/CRC, 1 edition.
- Martinez-Beneito, M. A., Botella-Rocamora, P., and Banerjee, S. (2017). “Towards a Multidimensional Approach to Bayesian Disease Mapping.” *Bayesian Analysis*, 12(1): 239 – 259.
- Martí, P., Serrano-Estrada, L., and Nolasco-Cirugeda, A. (2019). “Social Media data: Challenges, opportunities and limitations in urban studies.” *Computers, Environment and Urban Systems*, 74: 161–174.
- Martín-Fernández, J. A. and Thió-Henestrosa, S. (2006). “Rounded zeros: some practical aspects for compositional data.” *Geological Society, London, Special Publications*, 264(1): 191–201.
- Martín-Fernández, J. A., Palarea-Albaladejo, J., and Olea, R. A. (2011). *Dealing with Zeros*, chapter 4, 43–58. John Wiley & Sons, Ltd.
- Martínez-Minaya, J. and Rue, H. (2024). “A flexible Bayesian tool for CoDa mixed models: logistic-normal distribution with Dirichlet covariance.” *Statistics and Computing*, 34.

- Martínez-Minaya, J., Cameletti, M., Conesa, D., and Pennino, M. G. (2018). “Species distribution modeling: a statistical review with focus in spatio-temporal issues.”
- Martínez-Minaya, J., Lindgren, F., López-Quílez, A., Simpson, D., and Conesa, D. (2023). “The Integrated Nested Laplace Approximation for Fitting Dirichlet Regression Models.” *Journal of Computational and Graphical Statistics*, 1–19.
- Moindjié, I.-A., Pinsard, C., Accatino, F., and Chakir, R. (2022). “Interactions between ecosystem services and land use in France: A spatial statistical analysis.” *Frontiers in Environmental Science*, 10.
- Moraga, P., Cramb, S. M., Mengersen, K. L., and Pagano, M. (2017). “A geostatistical model for combined analysis of point-level and area-level data using INLA and SPDE.” *Spatial Statistics*, 21: 27–41.
- Muff, S., Riebler, A., Held, L., Rue, H., and Saner, P. (2015). “Bayesian analysis of measurement error models using integrated nested Laplace approximations.” *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 64: 231–252.
- Muff, S., Ott, M., Braun, J., and Held, L. (2017). “Bayesian two-component measurement error modelling for survival analysis using INLA—A case study on cardiovascular disease mortality in Switzerland.” *Computational Statistics & Data Analysis*, 113: 177–193.
- Mullahy, J. (1986). “Specification and testing of some modified count data models.” *Journal of Econometrics*, 33(3): 341–365.
- Nandi, A. K., Lucas, T. C. D., Arambepola, R., Gething, P., and Weiss, D. J. (2023). “disaggregation : An R Package for Bayesian Spatial Disaggregation Modeling.” *Journal of Statistical Software*, 106.
- Pirzamanbein, B., Poska, A., and Lindström, J. (2020). “Bayesian Reconstruction of Past Land Cover From Pollen Data: Model Robustness and Sensitivity to Auxiliary Variables.” *Earth and Space Science*, 7(1): e2018EA00057. E2018EA00057 10.1029/2018EA00057.
- Prates, M. O., Azevedo, D. R. M., MacNab, Y. C., and Willig, M. R. (2022). “Non-Separable Spatio-Temporal Models via Transformed Multivariate Gaussian Markov Random Fields.” *Journal of the Royal Statistical Society Series C: Applied Statistics*, 71: 1116–1136.

- Rasmussen, C. L., Palarea-Albaladejo, J., Johansson, M. S., Crowley, P., Stevens, M. L., Gupta, N., Karstad, K., and Holtermann, A. (2020). “Zero problems with compositional data of physical behaviors: a comparison of three zero replacement methods.” *International Journal of Behavioral Nutrition and Physical Activity*, 17(1): 126.
- Stanimirova, R., Tarrio, K., Turlej, K., McAvoy, K., Stonebrook, S., Hu, K.-T., Arévalo, P., Bullock, E. L., Zhang, Y., Woodcock, C. E., Olofsson, P., Zhu, Z., Barber, C. P., Souza, C. M., Chen, S., Wang, J. A., Mensah, F., Calderón-Loor, M., Hadjikakou, M., Bryan, B. A., Graesser, J., Beyene, D. L., Mutasha, B., Siame, S., Siampale, A., and Friedl, M. A. (2023). “A global land cover training dataset from 1984 to 2020.” *Scientific Data*, 10(1): 879.
- Tang, M., Wu, Q., Yang, S., and Tian, G. (2022). “Dirichlet composition distribution for compositional data with zero components: An application to fluorescence in situ hybridization (FISH) detection of chromosome.” *Biometrical Journal*, 64: 714–732.
- Thomas-Agnan, C., Laurent, T., Ruiz-Gazen, A., Nguyen, T. H. A., Chakir, R., and Lungarska, A. (2021). *Spatial Simultaneous Autoregressive Models for Compositional Data: Application to Land Use*, 225–249. Cham: Springer International Publishing.
- Tsagris, M. and Stewart, C. (2018). “A Dirichlet Regression Model for Compositional Data with Zeros.” *Lobachevskii Journal of Mathematics*, 39: 398–412.
- van Vliet, J., de Groot, H. L., Rietveld, P., and Verburg, P. H. (2015). “Manifestations and underlying drivers of agricultural land use change in Europe.” *Landscape and Urban Planning*, 133: 24–36.
- Ver Hoef, J. M., Hanks, E. M., and Hooten, M. B. (2018). “On the relationship between conditional (CAR) and simultaneous (SAR) autoregressive models.” *Spatial Statistics*, 25: 68–85.
- Villejo, S. J., Illian, J. B., and Swallow, B. (2023). “Data fusion in a two-stage spatio-temporal model using the INLA-SPDE approach.” *Spatial Statistics*, 54: 100744.
- Wang, C. and Furrer, R. (2021). “Combining heterogeneous spatial datasets with process-based spatial fusion models: A unifying framework.” *Computational Statistics & Data Analysis*, 161: 107240.

- Wang, C., Puhan, M. A., and Furrer, R. (2018). “Generalized spatial fusion model framework for joint analysis of point and areal data.” *Spatial Statistics*, 23: 72–90.
- Watanabe, S. (2013). “A widely applicable Bayesian information criterion.” *J. Mach. Learn. Res.*, 14(1): 867–897.
- You, L. and Wood, S. (2005). “Assessing the spatial distribution of crop areas using a cross-entropy method.” *International Journal of Applied Earth Observation and Geoinformation*, 7: 310–323.
- (2006). “An entropy approach to spatial disaggregation of agricultural production.” *Agricultural Systems*, 90: 329–347.
- Zhang, C. and Li, X. (2022). “Land Use and Land Cover Mapping in the Era of Big Data.” *Land*, 11: 1692.

Integrating Expert Knowledge and Recursive Bayesian Inference: A Framework for Spatial and Spatio-Temporal Data Challenges

In this chapter, we present our paper *Integrating Expert Knowledge and Recursive Bayesian Inference: A Framework for Spatial and Spatio-Temporal Data Challenges* by Mario Figueira (University of Valencia), David Conesa (University of Valencia), Antonio López-Quílez (University of Valencia) and Håvard Rue (KAUST), which has been submitted to a journal and it is published in arXiv ([arXiv:2506.00221](https://arxiv.org/abs/2506.00221)). The chapter contains at the end the references used in this work.

Abstract

Expert elicitation is a critical approach for addressing data scarcity across various disciplines. But moreover, it can also complement big data analytics by mitigating the limitations of observational data, such as incompleteness and reliability issues, thereby enhancing model estimates through the integration of disparate or conflicting data sources. The paper also outlines various strategies for integrating prior information within the Integrated Nested Laplace Approximation method and proposes a recursive approach that allows for the analysis of new data as it arrives. This paper presents a comprehensive approach to expert elicitation, with a particular emphasis on spatial and spatio-temporal contexts. Specifically, it introduces a typology of expert-based model implementations that addresses different change of support scenarios between observational and expert data. Detailed examples

illustrating clear and replicable procedures for implementing expert elicitation and recursive inference are also presented.

Keywords

Bayesian hierarchical models, recursive inference, sequential learning, expert elicitation, INLA.

7.1 Introduction

Expert elicitation is a fundamental approach to address data scarcity in various applied fields, including ecology, environmental sciences, public policy, and risk management (Kynn, 2008; Kuhnert et al., 2010; Morgan, 2014; Zhang and Thai, 2016). Additionally, it complements “big data” analytics by filling gaps where observational data is incomplete or unreliable. Data combination is particularly valuable in integrated models (Ovaskainen and Soininen, 2011; Taylor-Rodríguez et al., 2017; Paradinas et al., 2023) and in fusion models (Wang et al., 2018; Chacón-Montalván et al., 2024), where expert knowledge can help reconcile conflicting or sparse data sources. It can involve incorporating new data, strengthening weakly informed areas, or leveraging expert judgment to address inconsistencies and uncertainties, as seen in expert elicitation methods and computer models (Kennedy and O’Hagan, 2001; O’Hagan, 2006; Dave Higdon and Rightley, 2008; Morgan, 2014; Zhang and Thai, 2016; Wang et al., 2018; Cadham et al., 2021).

The information provided through expert elicitation can be categorized into two types: (1) prior elicitation, which involves defining the prior distribution for some unknown parameters of the model (O’Hagan, 2006), and (2) the Supra-Bayesian approach, where experts provide information on more intuitive observable variables (Albert et al., 2012), such as the probability of the presence of a species in a given area, along with an uncertainty quantification for that estimate (Pearman-Gillman et al., 2020).

In this paper, we present a comprehensive approach to perform expert elicitation, with a particular focus on spatial and spatio-temporal scenarios, although not exclusively. Our proposal accounts for both types of expert elicitation previously

described, and it is implemented within the Bayesian inference framework, specifically leveraging the Integrated Nested Laplace Approximation (INLA, [Rue et al. \(2009\)](#)) approach for high computational performance. Furthermore, implementing our approach within the Bayesian framework allows us to incorporate it into a recursive Bayesian inference process, enabling a sequential model fitting as new data sources arrive. This is particularly advantageous due to the sequential nature of Bayes' theorem, as it allows new information to be incorporated into the inference sequence without requiring a complete model recomputation, as in online updating ([Schifano et al., 2016](#)) and Bayesian filtering ([Särkkä, 2013](#)).

In prior elicitation, experts define the specific shape of the distribution or characterize it by specifying certain moments ([Cadham et al., 2021](#)), such as the mean or variance. Additionally, prior information can also be obtained from another model, where the estimation of certain effects is assumed to be very similar to those in the target hierarchical Bayesian model. To incorporate this information into the target model, we can use alternative data sources (e.g., data from citizen science, institutions, or previous research) and fit auxiliary models, which yield estimates of the effects that can be incorporated as initial conditions in the target model. That is, posterior distributions from these auxiliary models can be used to inform prior distributions of the parameters and hyperparameters in the target model. In this paper, our proposal for prior elicitation encompasses different strategies exploiting the INLA approach to implement a sequential consensus strategy and a recursive framework for updating prior information as new data arrive.

The integration of expert data along with observational data can be achieved through the Supra-Bayesian approach ([Albert et al., 2012](#); [Billari et al., 2014](#)), analyzed within the framework of Bayesian Hierarchical Joint Models ([Albert et al., 2012](#); [Perälä et al., 2020](#)). In this context, experts may provide information at the same scale as the observational data or at a different scale, often due to spatial or temporal change of support or misalignment; this is the case in several real-world examples where both sources are misaligned ([Gobbi et al., 2012](#); [Aizpurua et al., 2015](#); [Pearman-Gillman et al., 2020](#); [Kaurila et al., 2022](#)). Our proposal for performing expert elicitation within the Supra-Bayesian approach addresses the challenge of change of support between observational and expert data. It enhances model estimates within a joint modeling structure that simultaneously integrates expert knowledge calibration and uncertainty analysis alongside observational data in change of support scenarios. Additionally, this information is not limited to expert judgment but can also come from simulators or computer models. These

deterministic models emulate the reference process and can be incorporated in the same way as expert-provided data. Accordingly, we also present a typology of expert-based model implementations, addressing different change of support scenarios between observational and expert data.

In summary, Section 2 provides a brief overview of the general framework for conducting expert elicitation, either through joint models or prior elicitation, along with the inferential approach used. Section 3 outlines the elicitation of model parameter distributions and the various ways it can be implemented, including the recursive Bayesian inference approach. Section 4 details expert elicitation via expert integration through joint models, discussing the different types of integration and the necessary methodological considerations. Section 5 presents concrete examples that illustrate the procedure in a clear, concise, and replicable manner. Finally, the conclusions discuss the scope of the proposed procedures and the potential limitations encountered in their implementation.

7.2 Expert elicitation framework

Elicitation models integrate expert knowledge from individuals or structured expert-based frameworks, including technicians, policymakers, and citizen science participants (O'Hagan, 2006). They can also incorporate information from deterministic or computer models developed in specific scientific fields, which mathematically encode the dynamics of a phenomenon under given conditions (Kennedy and O'Hagan, 2001). Since observational statistical models fundamentally differ from deterministic models, the latter often serve as a more effective source of expert information when developing new experimental models. In this context, leveraging deterministic models as “expert models” may be more efficient than relying solely on observational statistical models based on prior experiments. Another valuable source of information comes from previous experiments related to the same phenomenon under study.

A prior elicitation approach can be applied in such cases, allowing information from existing statistical models to inform the prior distributions of parameters and hyperparameters within a Bayesian framework. This process, often referred to as prior updating, can be visualized using a graphical scheme similar to the one proposed for integrating expert information or expert models into an observational statistical model, as the scheme illustrated in Figure 7.1. In this scheme, the prior distributions

of a model are updated using experiment $\mathbf{y}_0$ and model $\mathcal{M}_0 = \{\ell, g, \boldsymbol{\eta}^*, \mathbf{x}^*, \mathbf{A}^*, \boldsymbol{\theta}^*\}$ to refine the analysis for a new set of observations or experiment $\mathbf{y}_1$ assuming a model $\mathcal{M}_1 = \{\ell, g, \boldsymbol{\eta}, \mathbf{x}, \mathbf{A}, \boldsymbol{\theta}\}$. Here ℓ represents the likelihood, g the link function, $\boldsymbol{\eta}$ is the linear predictor $\boldsymbol{\eta} = \mathbf{A}\mathbf{x}$, $\{\mathbf{A}, \mathbf{A}^*\}$ are the design matrices, $\{\mathbf{x}, \mathbf{x}^*, \mathbf{u}\}$ stands for the latent field, where $\mathbf{u}$ are the non common effects between the models, and $\{\boldsymbol{\theta}, \boldsymbol{\theta}^*\}$ denote the hyperparameters. The updating process is defined with respect to the common parameters between both models and can also be informed by expert knowledge. A key aspect is the procedure used to integrate this prior information, for which multiple approaches exist (O'Hagan, 2006; O'Hagan and Oakley, 2010; O'Hagan, 2019).

Alternatively, expert information may pertain to observable variables rather than parameter distributions. A straightforward way to incorporate this information into observational data modeling is through calibrated auxiliary models. This involves joint models, where the observational model shares information with auxiliary models that incorporate expert-observable variables and are calibrated alongside uncertainty analysis. Figure 7.2 presents a scheme for combining expert information using the Supra-Bayesian approach. This is done by defining a joint model that integrates observational data with expert knowledge, embedding the linear predictor of the observational model ($\boldsymbol{\eta}$) into that of the auxiliary model $\boldsymbol{\eta}^*$. Here, $\{\mathbf{A}, \mathbf{A}^*\}$ denote the design matrices of covariates for the observational and expert models, respectively. $\{\mathbf{x}, \mathbf{x}^*\}$ represent the latent fields, while $\mathbf{u}$ corresponds to random variables (structured or unstructured) used to calibrate the auxiliary expert model. However, expert information sources can be classified based on their relationship to the observational data. This classification depends on whether the sources operate at the same scale or different scales, i.e., whether there is a change of support (misalignment or cross-reference of information scales) or not.

The approach proposed in this paper for integrating expert knowledge and the implementation of recursive inference to allow an updatable procedure of the model, has been implemented within the framework of the Integrated Nested Laplace Approximation (Rue et al., 2009; van Niekerk et al., 2023) in the R-INLA software (Martins et al., 2013). INLA is a deterministic approximation approach deeply rooted in GMRF theory (Rue and Held, 2005) for Bayesian inference (Rue et al., 2009; van Niekerk et al., 2023) and Latent Gaussian Models (LGM). INLA focuses on estimating the marginal posterior distributions of the model parameters, allowing the computation of marginal likelihoods and standard goodness-of-fit metrics such as Deviance Information Criterion (DIC) (Spiegelhalter et al., 2002), Watanabe-Akaike

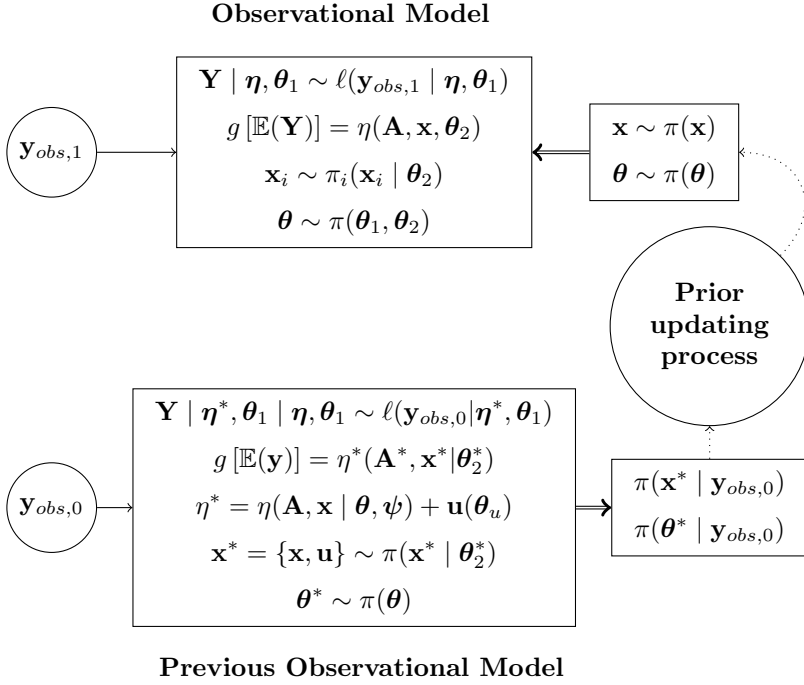


FIGURE 7.1: Implementation scheme of prior updating to integrate expert information along with observational data.

Information Criterion (WAIC) (Watanabe, 2013), Conditional Predictive Ordinates (CPO) (Pettit, 1990), and leave-group-out cross validation (Liu and Rue, 2024), all of which are essential for model evaluation and comparison. However, it could also be implemented with Markov Chain Monte Carlo (MCMC) methods.

7.3 Prior elicitation and recursive inference

Prior elicitation, as previously stated, can be performed by leveraging different information sources, including experts, citizen science, deterministic models, computational models, or data from previous experiments and surveys that inform new studies on the same phenomenon. However, this information may not always be available simultaneously with the data being analyzed or may become available during new surveys. In such cases, it is preferable to incorporate expert information into models for new data without needing to reprocess all previous information. It

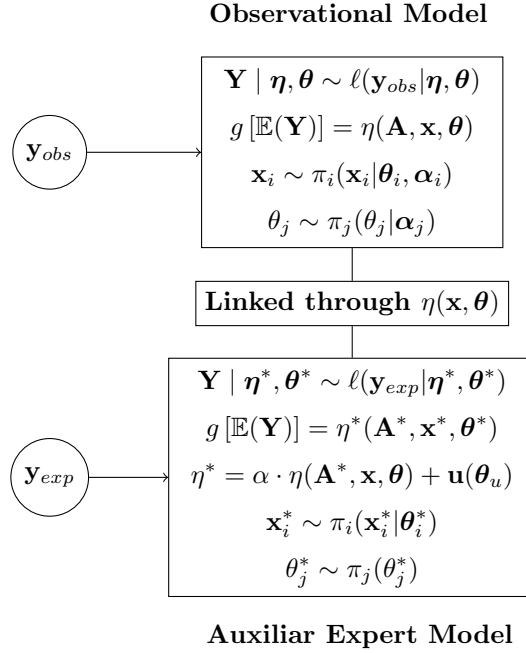


FIGURE 7.2: Implementation scheme of the Supra-Bayesian approach to integrate expert information along with observational data.

is possible to achieve this through recursive or sequential inference methods, such as online updating using a frequentist inferential approach (Schifano et al., 2016), sequential Bayesian inference (Scott et al., 2016; Hooten et al., 2021; Kessler et al., 2023), or a continual learning approach (Wang et al., 2024).

Sequential inference is approached from a wide range of perspectives and applied to various fields, such as online updating (Schifano et al., 2016), spatial statistics (Hooten et al., 2021), or distributed computing (Minsker et al., 2017; Guhaniyogi et al., 2022). In Bayesian statistics, many sequential inference methods rely on MCMC (Scott, 2017; Rendell et al., 2021) or in the properties of samplers for the MCMC, taking advantage of features from the Metropolis-Hastings sampler (Lunn et al., 2013; Hooten et al., 2021), the Weierstrass sampler (Wang and Dunson, 2014), and others. In this case, we build on the methodological foundations of INLA to implement a recursive inference procedure.

In this section, we introduce different strategies for integrating prior information within Bayesian inference, with a particular focus on implementation using the

INLA approach. These strategies can be applied using the INLA methodology through the R-INLA software (Martins et al., 2013), combined with other methods such as Monte Carlo (MC, Berild et al. (2022)) and MCMC (Gómez-Rubio and Rue, 2018), or implemented entirely using MCMC approaches. The section is divided into two parts. The first part discusses the updating of prior information and how, when combined with a consensus approach (Scott et al., 2016), it enables sequential consensus inference (Figueira et al., 2025). The second part introduces our proposed approach for performing recursive inference within the INLA framework, without relying on the assumptions and approximations of the previously discussed sequential consensus approach.

7.3.1 Prior updating

In a general sense, prior updating refers to the process of refining a prior distribution based on new knowledge or relevant information. This is a fundamental concept in Bayesian statistics, as Bayes' theorem itself is built upon the principle of updating prior beliefs to obtain a posterior distribution. In this section, we present different strategies for updating model information: direct prior updating (*prior updating by moments* and *full prior updating*) as well as an algorithm that integrates sequential prior updating with consensus to update the entire model's information.

7.3.1.1 Prior updating by moments

In this strategy, we focus on updating the moments that define the kernel of the prior distribution, such as the mean and variance in a Gaussian kernel. This approach can be easily implemented in both MCMC and INLA for univariate distributions. Since INLA relies on latent Gaussian models (LGM) and the fixed parameters (the β coefficients) are expected to follow a Gaussian distribution, we can update their Gaussian prior by modifying its characteristic moments, as discussed in Figueira et al. (2024). For hyperparameters in INLA, we can either update the existing kernel of the prior distribution or introduce a new one, refining it to better align with the posterior distribution from a previous model. However, for the linear coefficients β , we remain constrained to Gaussian priors in the INLA framework.

This can be achieved by determining the parameters ξ of $\pi(x \mid \xi)$ based on the moments (e.g., mean and variance) of the marginal posterior distribution $\pi(x \mid \mathbf{y})$.

Alternatively, ξ can be estimated by minimizing the Kullback-Leibler divergence (KLD) between the two distributions, i.e., solving $\min_{\xi} \text{KLD}(\pi(x | \xi) \| \pi(x | \mathbf{y}))$.

7.3.1.2 Full prior updating

Full prior updating enables the definition of arbitrary continuous distributions, including joint prior distributions for both parameters and hyperparameters. The goal is to define joint prior distributions $\pi(\mathbf{x} | \xi)$ based on the posterior distributions $\pi(\mathbf{x} | \mathbf{y})$. Within the R-INLA framework, three key features facilitate hyperparameter updating: the table and expression functionalities (Gómez-Rubio, 2020), along with the `rprior` function, which streamlines the creation of custom priors. This functionality allows us to define a distribution by specifying a set of values for the random variable along with their corresponding density values, $\{x_i, \pi(x_i)\}$. It is important to note that these density values do not always directly correspond to the parameter itself but may instead relate to a transformation of the parameter. Therefore, careful consideration of the appropriate transformation, accounting for internal parameterization, is essential. However, constructing joint priors and assigning non-Gaussian priors to latent field parameters presents certain challenges.

To address these challenges, integrating MCMC methods with INLA provides a viable solution (Gómez-Rubio and Rue, 2018). For instance, a Metropolis-Hastings algorithm can be implemented to jointly update a set ν of latent field parameters ($\mathbf{x}$) or hyperparameters (θ), where $\xi = \{\mathbf{x}, \theta\}$. This is achieved by defining a suitable proposal distribution $q(\cdot | \cdot)$ and computing the acceptance probability:

$$\alpha = \min \left\{ 1, \frac{\pi(\mathbf{y} | \xi^*) \pi(\xi^*) q(\xi^{(j)} | \xi^*)}{\pi(\xi^{(j)} | \mathbf{y}) \pi(\xi^{(j)}) q(\xi^* | \xi^{(j)})} \right\},$$

where $\pi(\mathbf{y} | \xi^{(j)})$ and $\pi(\mathbf{y} | \xi^*)$ are the conditional marginal likelihoods for $\xi^{(j)}$ and ξ^* , respectively. Alternatively, Importance Sampling or Adaptive Multiple Importance Sampling (Berild et al., 2022) can be used to perform full prior updating.

7.3.1.3 Sequential Consensus inference

The combination of prior updating procedures, particularly prior updating by moments, along with a consensus approach, allows us to define sequential algorithms

for conducting Bayesian inference with INLA. The sequential consensus inference approach (Figueira et al., 2025) focuses on sequentially updating the marginal distributions of the fixed effects of the latent field β and hyperparameters θ according to prior updating by moments, and reaching a consensus on the latent field related to the random effects, either marginal or multivariate. Specifically, the following approximation is assumed for the marginal posterior distribution of the fixed effects:

$$\pi(\beta_i \mid \mathbf{y}_2, \mathbf{y}_1) \approx \pi(\mathbf{y}_2 \mid \beta_i) \pi(\beta_i \mid \mathbf{y}_1), \quad (7.1)$$

and a similar approximation is assumed for the hyperparameters. Once the sequence is completed, a marginal consensus of the random effects is proposed, either by the weighted sum of the n parts of the sequence for the random effects of the latent field:

$$x_i \approx \sum_{j=1}^n w_{ij} \cdot x_{ij}, \quad (7.2)$$

where $w_{ij} = \tau_{ij} / \sum_{i=1}^n \tau_{ij}$ represents the optimal weights. Alternatively, a multivariate consensus of the nodes of the latent field can be employed:

$$\pi(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2) \approx \pi(\mathbf{x} \mid \mathbf{y}_1) \cdot \pi(\mathbf{x} \mid \mathbf{y}_2), \quad (7.3)$$

where, assuming both $\pi(\mathbf{x} \mid \mathbf{y}_1)$ and $\pi(\mathbf{x} \mid \mathbf{y}_2)$ are multivariate Gaussian distributions.

The precision and mean of the consensus distribution of the posteriors are:

$$\begin{aligned} \mathbf{Q} &= \mathbf{Q}_1 + \mathbf{Q}_2, \\ \boldsymbol{\mu} &= (\mathbf{Q}_1 + \mathbf{Q}_2)^{-1} (\mathbf{Q}_1 \cdot \boldsymbol{\mu}_1 + \mathbf{Q}_2 \cdot \boldsymbol{\mu}_2). \end{aligned} \quad (7.4)$$

The precision matrices $\mathbf{Q}$ and vector means $\boldsymbol{\mu}$ for performing the multivariate consensus would be those obtained in the modal configuration for the hyperparameters at the different steps of the sequence. For the marginal consensus, the random effects are distributed according to the marginal distributions of the nodes of the latent field related to those random effects.

7.3.2 Recursive Bayesian inference

INLA focuses in the posterior marginal distributions for the nodes of the latent field ($\mathbf{x}$) and for the hyperparameters ($\theta = \{\theta_1, \theta_2\}$). However, to achieve this it

needs to approximate the posterior distribution of the hyperparameters and the posterior distribution of the latent field given a set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ in the hyperparameter space (Rue et al., 2009). The approximation used to compute the posterior of the hyperparameters is $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$, defined as:

$$\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{\pi(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})}{\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})}, \quad (7.5)$$

where $\tilde{\pi}_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ is the Gaussian approximation for the posterior of the latent field, and the whole expression is analyzed in the modal configuration of posterior of the latent field under the Gaussian approximation.

To calculate the posterior distribution of the latent field, a Gaussian approximation is used, evaluated at the support points, and a low-rank correction is applied to the joint mean of the posterior distribution of the latent field under the Gaussian approximation (van Niekerk et al., 2023; van Niekerk and Rue, 2024). This correction can also be used to correct the marginal variance. The Gaussian approximation to the true $\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ is done by matching the mode and the curvature around the mode of $\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ using a second-order Taylor series expansion around the mode of $\boldsymbol{\mu}(\boldsymbol{\theta})$ (Rue et al., 2009; van Niekerk et al., 2023):

$$\begin{aligned} \pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \approx & \exp(\log \pi(\boldsymbol{\mu}(\boldsymbol{\theta}) \mid \mathbf{y}, \boldsymbol{\theta}) + \nabla \log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} \mathbf{x} \\ & + \frac{1}{2} \mathbf{x}^T \nabla^2 \log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} \mathbf{x}). \end{aligned} \quad (7.6)$$

This posterior approximation is obtained for each support point $\boldsymbol{\theta}^k$, and the mean is corrected through an iterative procedure of low-rank variational Bayes correction (van Niekerk and Rue, 2024). This yields a GMRF (Gaussian Markov Random Field) with a corrected mean for each support point $\boldsymbol{\theta}^k$:

$$\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}^k) = \text{GMRF}(\boldsymbol{\mu}^*(\boldsymbol{\theta}^k), \mathbf{Q}(\boldsymbol{\theta}^k)), \quad (7.7)$$

where $\boldsymbol{\mu}^*(\boldsymbol{\theta})$ is the corrected mean and $\mathbf{Q}(\boldsymbol{\theta})$ is the precision matrix. The GMRF of the latent field for each $\boldsymbol{\theta}^k$ is one of the outputs returned by the `inla` call in the R-INLA package, and it is be one of the key elements for implementing a recursive inference procedure with INLA.

Finally, two distinct strategies are used to obtain the marginals of the hyperparameters and the latent field. For the hyperparameters, interpolation is performed over the values obtained at the support points $\{\boldsymbol{\theta}^k\}$ to yield a marginal distribution

for each hyperparameter. Meanwhile, to obtain the marginal of the latent field, the following approximation is defined as a mixture of the marginal distributions at each support point:

$$\tilde{\pi}(x_i | \mathbf{y}) \approx \sum_{k=1}^K \pi_G(x_i | \boldsymbol{\theta}^k, \mathbf{y}) \tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y}) \Delta_k, \quad (7.8)$$

where $\pi_G(x_i | \boldsymbol{\theta}^k, \mathbf{y})$ is the marginal distribution given the GMRF with the corrected mean, and Δ_k are weights associated with each support point, related to their distribution.

Therefore, based on what we have explained about the INLA methodology, and in the fact that INLA also relies in the conditional independence of the data process given the latent field and hyperparameters $\pi(y_i, y_j | \mathbf{x}, \boldsymbol{\theta}) = \pi(y_i | \mathbf{x}, \boldsymbol{\theta}) \cdot \pi(y_j | \mathbf{x}, \boldsymbol{\theta})$, we can implement a recursive procedure to perform inference.

The recursive procedure assumes that we have a dataset that can be split—or is naturally partitioned—into N disjoint subsets $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_N\}$. The core structure for performing recursive inference involves analyzing the first partition, $\mathbf{y}_1$, as usual. From this we obtain the support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$, the joint posterior density of the hyperparameters evaluated at these points, $\tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y}_1)$, and the posterior GMRF of the latent field, $\pi_G(\mathbf{x} | \mathbf{y}_1, \boldsymbol{\theta}^k)$, for each support point. Using these outputs, we can reuse the support points $\{\boldsymbol{\theta}^k\}$ and the posterior GMRF of the latent field for each support point. This allows us to perform an analysis of the next subset by fixing the posterior GMRF and running parallel calls for each support point, resembling the following expression:

$$\begin{aligned} \pi(\mathbf{x} | \boldsymbol{\theta}^k, \mathbf{y}) &\propto \prod_{i=1}^N \pi(\mathbf{y}_i | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k), \\ &= \prod_{i=2}^N \pi(\mathbf{y}_i | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \mathbf{y}_1, \boldsymbol{\theta}^k), \end{aligned} \quad (7.9)$$

where for the i – *th* step we use as prior distribution for the latent Gaussian field the posterior from the previous one ($i - 1$). This allow us to compute the joint posterior distribution of the latent field.

The next step is to compute the posterior distribution for the hyperparameters throughout the sequence. In the first step, we have the support points and the density of the joint posterior distribution at these points. To update the density at the support points in subsequent steps, we can rely on the marginal likelihood of the model. Since we are fixing the computations at these support points, the

marginal likelihood in steps beyond the first is conditioned on them:

$$\begin{aligned}\pi(\boldsymbol{\theta}^k | \mathbf{y}) &\propto \pi(\mathbf{y} | \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k), \\ &= \pi(\boldsymbol{\theta}^k | \mathbf{y}_1) \prod_{i=2}^N \pi(\mathbf{y}_i | \boldsymbol{\theta}^k),\end{aligned}\tag{7.10}$$

where $\pi(\mathbf{y}_i | \boldsymbol{\theta}^k)$ is the marginal likelihood conditioned on the set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$.

Therefore, the posterior distribution $\tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y})$ computed through the recursive algorithm is:

$$\tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y}) \propto \left[\frac{\pi(\mathbf{y}_1 | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k)}{\pi_G(\mathbf{x} | \mathbf{y}_1, \boldsymbol{\theta}^k)} \right]_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta}^k)} \cdot \prod_{i=2}^N \pi(\mathbf{y}_i | \boldsymbol{\theta}^k),\tag{7.11}$$

and from this expression, the marginal likelihood can be approximated using a numerical integration scheme (e.g., central composite design, grid exploration, etc.):

$$\tilde{\pi}(\mathbf{y}) = \int \frac{\pi(\mathbf{y} | \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta})}{\pi_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})} d\boldsymbol{\theta} \approx \sum_{k=1}^K \frac{\pi(\mathbf{y} | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k)}{\pi_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta}^k)} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta}^k)}.\tag{7.12}$$

The final expression for the joint distribution of the hyperparameters when performing the recursive INLA procedure is:

$$\begin{aligned}\tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y}) &\propto \prod_{i=2}^N \left[\frac{\pi(\mathbf{y}_i | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \cup_{l=1}^{i-1} \mathbf{y}_l, \boldsymbol{\theta}^k)}{\pi_G^*(\mathbf{x} | \cup_{l=1}^i \mathbf{y}_l, \boldsymbol{\theta}^k)} \right]_{\mathbf{x}=\mathbf{x}_i^*(\boldsymbol{\theta}^k)} \\ &\quad \times \left[\frac{\pi(\mathbf{y}_1 | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k)}{\pi_G^*(\mathbf{x} | \mathbf{y}_1, \boldsymbol{\theta}^k)} \right]_{\mathbf{x}=\mathbf{x}_1^*(\boldsymbol{\theta}^k)} \\ &= \prod_{i=1}^N \left[\frac{\pi(\mathbf{y}_i | \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} | \cup_{l=1}^{i-1} \mathbf{y}_l, \boldsymbol{\theta}^k)}{\pi_G^*(\mathbf{x} | \cup_{l=1}^i \mathbf{y}_l, \boldsymbol{\theta}^k)} \right]_{\mathbf{x}=\mathbf{x}_i^*(\boldsymbol{\theta}^k)} \pi(\boldsymbol{\theta}^k)\end{aligned}\tag{7.13}$$

Once the joint posterior distributions of the hyperparameters and the GMRF of the latent field are obtained, with the mean rectified by the low-rank correction, the marginal distributions can be calculated. For the hyperparameters, this is done through interpolation and integration, while for the latent field, we use the same approximation employed by INLA, as described in Eq. 7.8, where we reuse the weights Δ_k obtained in the first step of the sequence.

In the proposed procedure for performing recursive inference, the first discrepancy that may arise, compared to simultaneous inference on all the data, comes from the identification of the support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$. The main issue stems from the potential shift in the mode of $\pi(\boldsymbol{\theta} \mid \mathbf{y})$ throughout the sequence, leading to a possible inadequate exploration of the posterior distribution. This could be corrected by assessing the shift in the mode. However, the non-rectifiable discrepancy lies in the fact that the computed posterior distribution density is not exactly the same as that obtained through the simultaneous analysis of all the data:

$$\tilde{\pi}(\boldsymbol{\theta}^k \mid \mathbf{y}) \propto \left\{ \begin{array}{l} \frac{\pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} \mid \boldsymbol{\theta}^k)}{\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta}^k)} \pi(\boldsymbol{\theta}^k), \\ \prod_{i=1}^N \left[\frac{\pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}^k) \pi(\mathbf{x} \mid \cup_{l=1}^{i-1} \mathbf{y}_l, \boldsymbol{\theta}^k)}{\pi_G^*(\mathbf{x} \mid \cup_{l=1}^i \mathbf{y}_l, \boldsymbol{\theta}^k)} \right]_{\mathbf{x}=\mathbf{x}_i^*(\boldsymbol{\theta}^k)} \pi(\boldsymbol{\theta}^k). \end{array} \right. \quad (7.14)$$

This discrepancy would transfer to the computation of the marginal distributions of the nodes of the GMRF in the latent field, being the only non-rectifiable discrepancy. However, it is estimable and, therefore, analysable.

7.4 Dealing with change of support in elicitation models

The fundamental distinction between expert data with and without a change of support lies in whether the scale at which expert data integrate with observational data remains consistent. This aspect directly impacts the Supra-Bayesian approach, as expert information in this context pertains to observable variables provided by the expert. Scenarios where data are available at different scales are commonly referred to as misalignment or change of support, sometimes described as cross-referencing information scales.

In many cases, expert information—regardless of its source—is provided on a scale that differs from that of the observational data. It is common for expert knowledge to be aggregated or presented at a different spatial, temporal, or categorical scale, and in many instances, combinations of these differences occur. For example, in ecology, numerous studies highlight cases where experts provide information at spatial and/or temporal scales distinct from those of field-collected data (Yamada et al., 2003; Gobbi et al., 2012; Aizpurua et al., 2015; Di Febbraro et al., 2018;

[Fitzgerald et al., 2021](#); [Kaurila et al., 2022](#)). Consequently, expert information may be available at a lower (or higher) resolution than observational data, requiring an integration approach that accounts for the specific type of change of support. The classification below provides a framework for assessing and integrating expert information based on its relationship to observational data: (i) spatial change of support, (ii) temporal change of support, (iii) spatio-temporal change of support, and (iv) categorical change of support.

The first three cases of expert information can be effectively integrated through a downscaling process using a Gaussian field with a Matérn structure ([Moraga et al., 2017](#); [Wang et al., 2018](#)). This can be applied in one dimension for temporal downscaling, in two dimensions for spatial downscaling, or as a combination of both for spatio-temporal downscaling. In the case of categorical change of support, where categorical levels are grouped or intersected, expert information should influence the aggregated category, either by informing the new grouping or by adjusting the parametric structure assessing the categorical level being aggregated.

An additional challenge arises when both observational and expert data are provided at an areal level but do not match. In such cases, downscaling can be applied to both datasets to facilitate their joint evaluation, or appropriate aggregation methods can be used when feasible. Furthermore, following the recommendations of [O’Hagan \(2006\)](#), it is crucial to integrate uncertainty quantification within the modeling framework to account for the experts’ confidence in their own knowledge.

In the following subsections, we examine the different cases and propose how to deal with change of support within the Supra-Bayesian approach: (i) no change of support, (ii) spatial change of support, (iii) temporal change of support, (iv) spatio-temporal change of support, and (v) categorical change of support.

7.4.1 No change of support

In the case with no change of support, both observational and expert data exist at the same scale, allowing for an integration as illustrated in [Figure 7.2](#). To integrate a single expert information source with observational data—or multiple expert-based sources—we propose the following mechanisms for constructing an integrated model that calibrates expert auxiliary models and assesses their associated uncertainty.

An integrated model with no change of support can be defined by a likelihood function ℓ for the observational data $y_{i,obs}$, within a given model structure $\mathcal{M} \equiv$

$\{\ell, \boldsymbol{\eta}, \mathbf{A}, \mathbf{x}, \boldsymbol{\theta}\}$. Here, $\boldsymbol{\eta}$ represents the linear predictor, $\mathbf{A}$ is the design matrix of covariates, $\mathbf{x}$ denotes the latent Gaussian field, and $\boldsymbol{\theta}$ consists of the hyperparameters $\boldsymbol{\theta} = \{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2\}$, where $\boldsymbol{\theta}_1$ corresponds to the likelihood parameters and $\boldsymbol{\theta}_2$ to the latent field parameters.

This observational data model is combined with expert data using the Supra-Bayesian approach (Albert et al., 2012; Billari et al., 2014; Perälä et al., 2020). In this framework, the expert data $y_{i,exp}$ is treated as additional observations. Since expert data is at the same scale as the observational data, it follows the same structure for the latent effects. Consequently, the integrated model can be expressed as:

$$\begin{aligned} y_{i,obs} &\sim \ell(\mu_{i,obs}, \boldsymbol{\theta}_1), \\ g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}\boldsymbol{\beta} + \sum_{k=1}^K f_k(z_{ik}), \\ y_{i,exp} &\sim \ell_e(\mu_{i,exp}, \boldsymbol{\theta}_1^{(e)}), \\ g_e(\mu_{i,exp}) &= \alpha \cdot \eta_{i,obs} + u_i = \alpha \left[\mathbf{A}\boldsymbol{\beta} + \sum_{k=1}^K f_k(z_{ik}) \right] + u_i, \end{aligned} \tag{7.15}$$

where $\mathbf{x} = \{\boldsymbol{\beta}, \mathbf{f}\}$ is the latent field composed by the fixed $\boldsymbol{\beta}$ and random effects $\mathbf{f}$. In this integrated model, the linear predictor of the observational model, $\boldsymbol{\eta}_{obs}$, is incorporated into the expert auxiliary model and scaled by a parameter α .

The scaling parameter α is expected to be close to 1 if the expert data is consistent with the observational data. However, α can also be defined as a spline function, allowing for an assessment of its deviation from 1 across different predictor values. The term u_i represents a random effect that captures the residual uncertainty in the expert model that is not explained by the scaled predictor from the observational model. Together, these components define a joint model that integrates an observational model with a calibrated auxiliary expert model.

In Eq. 7.15, it is crucial to analyze the hyperparameter associated with the variance (or precision) of the expert auxiliary model, as recommended by O'Hagan (2006). This analysis helps quantify the uncertainty introduced by the expert. If we define the precision of the likelihood for the expert data by $\tau_{i,exp}$, the expert may provide an estimate of the deviation or precision for each data point, $\{y_{i,exp}, \phi_{i,exp}\}$. The corresponding uncertainty can be represented as:

$$\tau_{i,exp} = \psi \cdot \phi_{i,exp},$$

where $\phi_{i,exp}$ is the expert-provided deviation or precision for each data point $y_{i,exp}$, and ψ is a parameter to be estimated.

The elicitation process can incorporate information from multiple expert sources, including expert-informed models, computational models, or specialized professionals. In such cases, it is beneficial to explore potential correlations among these experts, as highlighted by [Albert et al. \(2012\)](#). Within the INLA framework, these correlations can be conceptualized as a multivariate Gaussian error term.

Extending the previous model, if we have multiple experts $m \in \{1, \dots, M\}$, each with their own scaling parameter α_m , the model can be expressed as:

$$\begin{aligned}
 Y_{i,obs} &\sim \ell(\mu_{i,obs}, \boldsymbol{\theta}_1), \\
 g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}\boldsymbol{\beta} + \sum_{k=1}^K f_k(z_{ik}), \\
 Y_{i,exp1} &\sim \ell_e(\mu_{i,exp1}, \boldsymbol{\theta}_1^{(e1)}), \\
 g_e(\mu_{i,exp1}) &= \alpha_1 \cdot \eta_{i,obs} + u_{i1} + u_{iC} = \alpha_1 \left[\mathbf{A}\boldsymbol{\beta} + \sum_{k=1}^K f_k(z_{ik}) \right] + u_{i1} + u_{iC}, \\
 &\vdots \\
 Y_{i,expM} &\sim \ell_e(\mu_{i,expM}, \boldsymbol{\theta}_1^{(eM)}), \\
 g_e(\mu_{i,expM}) &= \alpha_M \cdot \eta_{i,obs} + u_{iM} + u_{iC} = \alpha_M \left[\mathbf{A}\boldsymbol{\beta} + \sum_{k=1}^K f_k(z_{ik}) \right] + u_{iM} + u_{iC},
 \end{aligned} \tag{7.16}$$

where each expert m has its own scaling parameter α_m . As above mentioned, the scaling structure can be made more flexible by defining it as a spline, allowing the scaling value to vary depending on the linear predictor. The term u_{iC} captures the correlation among expert linear predictors, assuming a Multivariate Normal (MVN) distribution for the experts:

$$\mathbf{u}_C \sim \text{MVN}(\mathbf{0}, \Sigma_C),$$

where the diagonal elements of the covariance matrix are defined as $\Sigma_{ii} = 1/\tau_i$, and the off-diagonal elements ($i \neq j$) are $\Sigma_{ij} = \rho_{ij}/\sqrt{\tau_i\tau_j}$.

It is important to note that real case studies often involve more complex models, where expert assessments (or expert data) may follow a different likelihood function than the one used for observational data. A common example in ecological studies—such as habitat suitability or fisheries management—occurs when observational data are presence/absence-based (Bernoulli distribution), while expert information relates to the probability of presence (Beta distribution). In such cases, when data distributions differ, the modeled quantity must remain compatible across likelihoods. For instance, the probability in a Bernoulli model must align with the probability in a Beta model. In some scenarios, linear predictors within the likelihood functions

must be transformed to ensure that the shared information between them is correctly defined.

7.4.2 Spatial change of support

Spatial change of support implies that expert information does not match the spatial structure of the observational data. This can be assessed based on whether the observational data is continuously structured or organized into areas (or networks).

The different scenarios that we may encounter for spatial change of support are: (i) geostatistical-area change of support, where observations are continuous geostatistical data, $s_i \in \mathcal{D} \subseteq \mathbb{R}^2$, and expert information is structured into areas, $C_i \subset \mathcal{D}$, and (ii) area-area change of support, where both observational and expert data is structured in areas, $B_i \in \mathbf{B}$ and $C_i \in \mathbf{C}$ respectively, but these spatial partitions are different ($\mathbf{B} \neq \mathbf{C}$).

In the case of geostatistical-area change of support we propose to implement the development around the fusion or melding models framework (Gelfand, 2001; Berrocal et al., 2012; Moraga et al., 2017; Wang et al., 2018; Wang and Furrer, 2021; Villejo et al., 2023; Chacón-Montalván et al., 2024; Zhong et al., 2025). The core idea is to use a continuous Gaussian field $\mathbf{u}(\mathbf{s}) \sim \text{GF}(\mathbf{0}, \mathbf{Q})$ in the study area $\mathcal{D}$, similar to a standard geostatistical model. However, in the model for expert information, we integrate the Gaussian field within each corresponding area to obtain the mean aggregated effect or the total aggregated effect. This involves setting a set of integration points (s_k) to approximate the integral via numerical integration. Moreover, we can compute the mean aggregated effect as $\iint_{s \in C_i} u(s)/|C_i|ds \approx \sum_{s_k \in C_i} u(s_k)/n : (s \in C_i)$, where we consider that each integration point has associated the same area, or we can compute the total aggregation $\iint_{s \in C_i} u(s)$.

A model within the Supra-Bayesian approach to deal with this change of support in expert elicitation can be defined as in the previous section. This integrated model consists of a likelihood ℓ for the observations $\mathbf{y}_{obs}$ in a continuous study region $\mathcal{D}$, and a set of expert data $\mathbf{y}_{expert}$ structured in a (non-intersecting) partition of the study region $\mathcal{D} = \cup_{k \in K} C_k$. For this case, the integration model we propose, assuming that the observational and expert data share the same core underlying process $\eta_{i,obs}$, while incorporating an additional component u_j to account for the

uncertainty in expert assessments, can be expressed as follows:

$$\begin{aligned}
Y_{i,obs} &\sim \ell(\mu_{i,obs}, \tau_{obs}), \\
g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}\mathbf{x} + \mathbf{A}_u(s)\mathbf{u}(s|\rho, \sigma), \\
Y_{j,exp} &\sim \ell_e(\mu_{j,exp}, \tau_{exp}), \\
g_e(\mu_{j,exp}) &= \alpha \cdot \iint_{s \in C_j} \frac{\eta_{j,obs}(s)}{|C_j|} ds + u_j, \\
&= \alpha \left[\sum_{s_k \in C_j} \frac{\mathbf{A}(s_k)\boldsymbol{\beta}}{n(s_k \in C_j)} + \sum_{s_k \in C_j} \frac{\mathbf{A}_u(s_k)\mathbf{u}(s_k|\rho, \sigma)}{n(s_k \in C_j)} \right] + u_j,
\end{aligned} \tag{7.17}$$

where we are integrating the effect associated with the observational predictor in each region where expert information is available. If the values of the covariates are not available at the required integration points, sub-models can be constructed for each covariate and incorporated into the observational predictor, similar to an Error model (Muff et al., 2015), or in a two-stage process (Villejo et al., 2023). Generalisation to non-Gaussian data and multiple experts with different aggregations is trivial given the previous formulation.

In the case of area-area change of support, we propose the same approach but perform the integration of the continuous field in both the observation predictor and when it is shared in the model associated with expert information. Following the structure of the previous example, we can present it as

$$\begin{aligned}
Y_{obs}(B_i) &\sim \ell(\mu_{obs}(B_i), \boldsymbol{\theta}_1), \\
g(\mu_{obs}(B_i)) &= \sum_{s_k \in B_i} \frac{\mathbf{A}(s_k)\boldsymbol{\beta}}{n(s_k \in B_i)} + \sum_{s_k \in B_i} \frac{\mathbf{A}_u(s_k)\mathbf{u}(s_k|\rho, \sigma)}{n(s_k \in B_i)}, \\
Y_{exp}(C_j) &\sim \ell_e(\mu_{exp}(C_j), \boldsymbol{\theta}_1^{(e)}), \\
g_e(\mu_{exp}(C_j)) &= \alpha \cdot \sum_{s_k \in C_j} \frac{\eta_{j,obs}(s)}{|C_j|} + u_j, \\
&= \alpha \left[\sum_{s_k \in C_j} \frac{\mathbf{A}(s_k)\boldsymbol{\beta}}{n(s_k \in C_j)} + \sum_{s_k \in C_j} \frac{\mathbf{A}_u(s_k)\mathbf{u}(s_k|\rho, \sigma)}{n(s_k \in C_j)} \right] + u_j,
\end{aligned} \tag{7.18}$$

where the observational and expert data are related to B_i and C_j areas respectively, and the continuous Gaussian field allow us to combine both datasets, which can also be applied to covariates in case we lack their information.

However, in the case of area-area change of support, it is also possible to explore alternative processes beyond defining a continuous Gaussian field. Specifically, when expert information is available at a coarser spatial scale than the observational data, an alternative downscaling approach can be considered without explicitly defining a continuous Gaussian field. This approach is particularly useful when the areas associated with observational data are entirely nested within a group of larger areas corresponding to expert information, such that $C_j = \bigcup_k B_k$. In this scenario, the

predictor associated with each polygon C_j can be expressed as:

$$\eta_j \propto \sum_{k: B_k \subset C_j} \frac{|B_k|}{|C_j|} \eta_k, \quad (7.19)$$

where the predictor is progressively constructed from finer-scale observational data and aggregated upwards. In the limit, this structure aligns with cases where information is not directly cross-referenced between scales, i.e. without change of support or misalignment. This component represents the shared predictor, and when expert data is available at multiple scales, additional calibration mechanisms can be introduced. This approach serves as an alternative to continuous Gaussian field-based downscaling while preserving the ability to align expert-based models with observational data and to quantify the uncertainty in expert assessments that is not explained by the observational model.

7.4.3 Temporal change of support

Temporal change of support implies that the expert information does not align in its temporal structure with the observational data. This discrepancy can be evaluated by determining whether the data are continuously structured or organised into intervals. A similar issue arises in spatial analysis when space is defined in only one dimension, such as in distance sampling models (Yuan et al., 2017) or metric graphs (Bolin et al., 2024).

Therefore, the procedures we have proposed for the two-dimensional spatial case can also be applied in this setting. In a joint model for a continuous process over time, where we have different observations over a period $\mathcal{T}$ and an expert provides information at intervals $T_i = (t_{i1}, t_{i2})$ that may or may not completely cover the period $\mathcal{T}$, a structure that integrates both sources of information can be written as follows:

$$\begin{aligned} Y_{i,obs} &\sim \ell(\mu_{i,obs}, \boldsymbol{\theta}_1), \\ g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}(s)\boldsymbol{\beta} + u(s|\rho, \sigma), \\ Y_{j,exp} &\sim \ell_e(\mu_{j,exp}, \boldsymbol{\theta}_1^{(e)}), \\ g_e(\mu_{j,exp}) &= \alpha \cdot \int_{s \in T_j} \frac{\eta_{j,obs}(s)}{|T_j|} ds + u_j, \\ &= \alpha \left[\sum_{s_k \in T_j} \frac{\mathbf{A}(s_k)\boldsymbol{\beta}}{n(s_k \in T_j)} + \sum_{s_k \in T_j} \frac{\mathbf{A}_u(s_k)u(s_k|\rho, \sigma)}{n(s_k \in T_j)} \right] + u_j, \end{aligned} \quad (7.20)$$

where s and s_k stand for temporal or one-dimension spatial locations and u_j is an additional component to account for the uncertainty in expert assessments. This equation is a reformulation of Equation (7.17) for the one-dimensional case, which is commonly encountered in temporal analysis. Moreover, it does not impose any regularity on the expert intervals—they can be irregular and do not necessarily represent an average over the interval. Instead, by setting the normalization factor to 1, we obtain the cumulative quantity.

7.4.4 Spatio-temporal change of support

An extension to the spatio-temporal case can be achieved by combining our previous two proposals (spatial and temporal). For spatio-temporal change of support, we propose the same procedures as for spatial and temporal change of support. The key difference is that, in the spatio-temporal case, the integration occurs in three dimensions, requiring the approximation of volume integration. In this setting, the response variable is continuous in both space and time, and an expert provides information within specific volumes of the spatio-temporal region. Consequently, a possible model that combines observational data $Y_{i,obs}$ and expert data $Y_{j,exp}$ given a spatio-temporal aggregation into voxels (V_j) of expert information, can be written as the following one:

$$\begin{aligned}
 Y_{i,obs} &\sim \ell(\mu_{i,obs}, \theta_1), \\
 g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}(s)\beta + u(s), \\
 Y_{j,exp} &\sim \ell_e(\mu_{j,exp}, \theta_1^{(e)}), \\
 g_e(\mu_{j,exp}) &= \alpha \cdot \iiint_{s \in V_j} \frac{\eta_{j,obs}(s)}{|V_j|} ds + u_j, \\
 &= \alpha \left[\sum_{s_k \in V_j} \frac{\mathbf{A}(s_k)\beta}{n(s_k \in V_j)} + \sum_{s_k \in V_j} \frac{u(s_k)}{n(s_k \in V_j)} \right] + u_j,
 \end{aligned} \tag{7.21}$$

where s and s_k represent spatio-temporal locations and u_j is an additional component to account for the uncertainty in expert assessments. This equation represents the extension of the previous two cases for spatial and temporal change of supports, but now accounting for the three dimensional nature of the data.

7.4.5 Categorical change of support

The latter situation arises when dealing with a categorical change of support. In this case, categorical change of support occurs when expert information is provided

using a smaller set of categories derived from an aggregation of a larger set in the observational data model; the reverse scenario is also possible. This means that the effects of categories can be combined if there is a clear correspondence between groups. For instance, consider an observational categorical variable with 5 categories, $\mathbf{u} = u_1, \dots, u_5$, and expert information consisting of 3 categories, $\mathbf{u}^* = \{u_1^*, \dots, u_3^*\}$, such that $u_2^* = \{u_2, u_3\}$ and $u_3^* = \{u_4, u_5\}$. In this case, the aggregation in the joint model assumes that in the expert data, the estimate for u_2^* and u_3^* is equivalent to the sum of the effects of the related categories: $u_2^* = u_2 + u_3$ and $u_3^* = u_4 + u_5$. In general, it can be stated as follows: if we have a categorical variable with K categories and expert information that provides information for the same variable in K^* categories, where $K > K^*$, such that there is a correspondence between disjoint groups $g_k = \{\mathbf{u}_k\}$ of levels of the observational variable with respect to the expert information, where $g_k \in \{u_1, \dots, u_K\}$ and $g_i \cap g_j = \emptyset \iff i \neq j$. Then, the aggregation can be written as follows:

$$u_k^* = \sum_{u_i \in g_k} u_i. \quad (7.22)$$

This approach relies on the underlying assumption that, across the grouped factor levels in the higher structure—i.e., at the lower resolution scale—the grouping from the higher resolution scale is balanced. However, a more general and flexible formulation accounts for unbalanced aggregation. In this case, Equation (7.22) is modified as follows:

$$u_k^* = \sum_{u_i \in g_k} w_{ik} u_i. \quad (7.23)$$

where w_i represents the weight associated with each factor level u_i in the aggregation $u_i \in g_k$, reflecting its actual proportion in the grouping, given by $n(u_i \in g_k) / \sum_{i=1}^K n(u_i \in g_k)$. The weights satisfy the constraints $w_{ik} \geq 0$ and $\sum_{i=1}^K w_{ik} = 1$.

Therefore, its implementation, following the model structures previously used, can be expressed as:

$$\begin{aligned} Y_{i,obs} &\sim \ell(\mu_{i,obs}, \boldsymbol{\theta}_1), \\ g(\mu_{i,obs}) &= \eta_{i,obs} = \mathbf{A}\boldsymbol{\beta} + u(z_i), \\ Y_{j,exp} &\sim \ell(\mu_{j,exp}, \boldsymbol{\theta}_1^{(e)}), \\ g_e(\mu_{j,exp}) &= \alpha \cdot (\mathbf{A}_e \boldsymbol{\beta} + u^*(g_j)) + u_j, \\ &= \alpha \cdot (\mathbf{A}_e \boldsymbol{\beta} + \sum_{z_i \in g_j} u(z_i)) + u_j, \end{aligned} \quad (7.24)$$

where $\{\mathbf{A}, \mathbf{A}_e\}$ are the design matrices for the observed and expert data, respectively, $u^*(g_j) = \sum_{z_i \in g_k} u(z_i)$ represents the change of support across different category levels, following Equation (7.22), and u_j is an additional component to account for the uncertainty in expert assessments.

A limiting case of this aggregation occurs when it is applied to the entire categorical variable. In this scenario, the effects of categories not considered in the expert information should not be shared between the observational model and the expert model. This is because categorical variables often assume a sum-to-zero constraint, $\sum_k^K u_k = 0$, where K is the total number of categories. Consequently, the aggregation of this effect in the auxiliary expert model is also equal to zero. Therefore, only the aggregated information across the category levels is considered. Since we have $\sum_k u_k = 0$, the integration over the entire category results in zero in the auxiliary expert model.

7.5 Examples

In this section, three examples are presented to demonstrate the implementation of expert elicitation and sequential inference using INLA. The first two examples focus on the implementation of expert elicitation: the first deals with spatial change of support, while the second addresses categorical change of support. The third example presents a case of sequential inference with a spatial-temporal structure, comparing the results of performing the procedure sequentially against fitting the model to all the data simultaneously.

7.5.1 Expert elicitation with spatial change of support

In this first example, we will consider a case where limited data is combined with expert knowledge, following the Supra-Bayesian approach (Albert et al., 2012). This simulated data example aims to replicate a structure for incorporating expert information similar to that presented in (Pearman-Gillman et al., 2020), where the expert could provide information related to the probability of presence in survey areas. It also refers to cases where expert elicitation may be based on the expert's knowledge of the species' spatial distribution. As such, this knowledge could be integrated using a strategy of this kind (Yamada et al., 2003; Gobbi et al., 2012; Aizpurua et al., 2015).

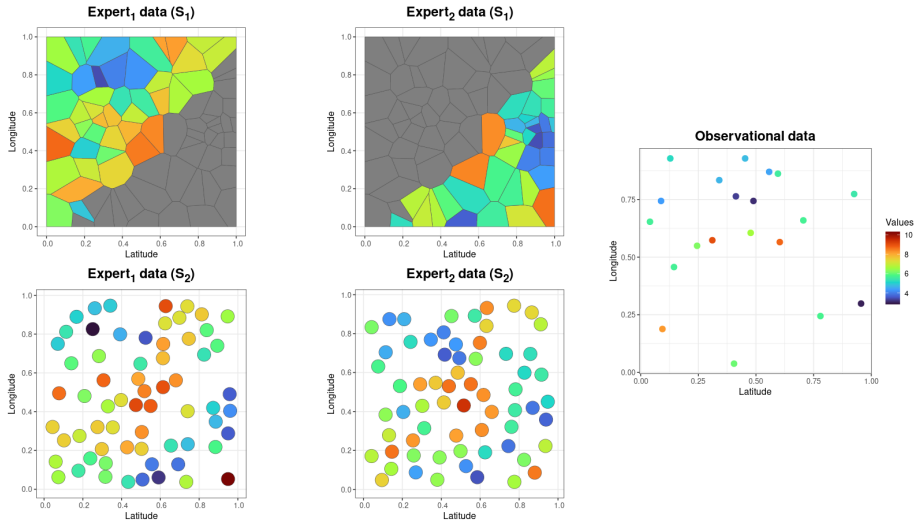


FIGURE 7.3: Observational data and information from two experts according to two different spatial structures.

In this specific case, we assume that there are two experts providing information according to a certain spatial structure. This can occur either by providing a predetermined structure that covers the study area (S_1), as in (Kaurila et al., 2022), or by allowing the expert to design specific areas (S_2) where they can provide information. In Fig. 7.3, we present the two ways in which experts could provide information (S_1 or S_2) along with the available observations in a scenario with limited spatial data. Therefore, the model used to simulate and infer the data is one that incorporates the spatial change of support between the expert and observational information, along with an assessment of the correlation between the experts:

$$\begin{aligned}
 y_{i,obs} &\sim \mathcal{N}(\mu_i, \tau), \\
 \mu_i &= \beta_0 + u_s(s_i), \\
 \mathbf{y}_{j,exp} &\sim \mathcal{MVN}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}), \\
 \boldsymbol{\mu}_j &= \boldsymbol{\beta}_{0,exp} + \int_{C_j} \frac{\beta_0 + u_s(s)}{|C_j|} ds,
 \end{aligned} \tag{7.25}$$

where we share the entire latent field between the observational model and the auxiliary expert model. The matrix $\boldsymbol{\Sigma}$ represents the variance-covariance between the experts, capturing both the marginal uncertainty of the experts and the correlation between them. In Fig. 7.4, the results of the spatial field are presented, either modeled using only the observational data or integrating expert information using

the aforementioned model, according to the two spatial structures in which the experts provided their information.

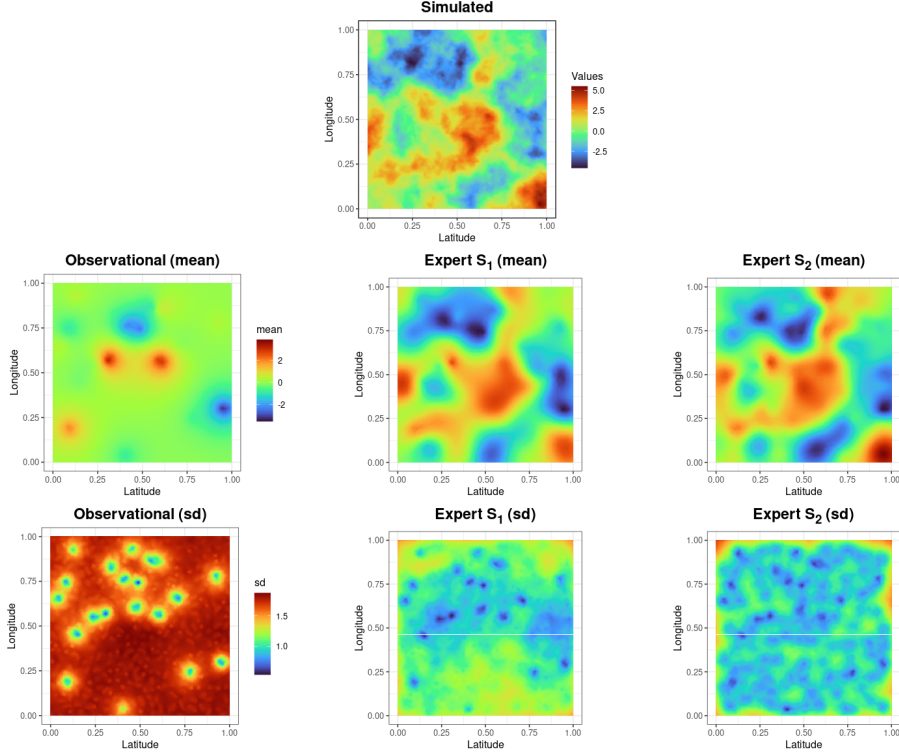


FIGURE 7.4: Simulated spatial effect, along with the mean and standard deviation of the posterior distribution of the spatial effect, analyzed either using only the observational data or by integrating expert information according to the two structures (S_1 or S_2) for expert information.

7.5.2 Expert elicitation with categorical change of support

In this example, it was assumed that the observational data is given with a categorical variable of 5 levels, while an expert would provide information structured by aggregating the first three levels of the categorical variable in the observational model, with the other two levels remaining unchanged. That is, the categorical variable in the observational model would be $\mathbf{u}_{obs} = \{u_1, u_2, u_3, u_4, u_5\}$, and for the expert, it would be $\mathbf{u}_{exp} = \{u_1^*, u_2^*, u_3^*\}$, such that $u_1^* = u_1 + u_2 + u_3$. Therefore, the

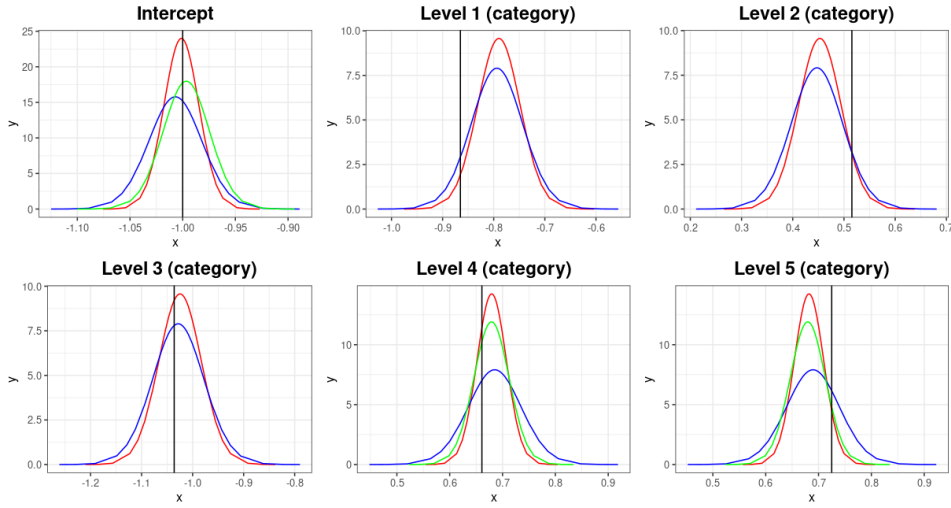


FIGURE 7.5: Posterior distributions for the intercept and the levels of the categorical variable. The vertical black line indicates the true value, the red line corresponds to the model that integrates both expert and observational information, the blue line is associated exclusively with the observational part, and the green line represents the expert's contribution (which is why it is not present for the first three categories).

model used is the following:

$$\begin{aligned}
 y_{i,obs} &\sim \mathcal{N}(\mu_i, \tau_{obs}), \\
 \mu_{i,obs} &= \beta_0 + u_i, \\
 y_{j,exp} &\sim \mathcal{N}(\mu_j, \tau_{exp}), \\
 \mu_{j,exp} &= \beta_0 + u_i^*,
 \end{aligned} \tag{7.26}$$

where the information about the intercept and the effect of category u_i would be shared, considering the previously mentioned aggregation in the expert model. In Fig. 7.5, the results for the intercept and the posterior distributions for each category level are presented.

7.5.3 Recursive inference for spatio-temporal data

In this last example, a recursive inference procedure is performed for a spatio-temporal model. The structure of the model to simulate and infer is as follows:

$$\begin{aligned}\mathbf{y} &\sim \mathcal{N}(\boldsymbol{\mu}, \tau \mathbf{I}), \\ \boldsymbol{\mu} &= \beta_0 + \mathbf{u}_s + \mathbf{u}_t,\end{aligned}\tag{7.27}$$

where β_0 is the intercept, $\mathbf{u}_s$ stands for the spatial effect, and $\mathbf{u}_t$ represents the temporal effect. The data have been simulated with 60 temporal nodes and 300 spatial points for each temporal node.

In the recursive inference procedure, we proposed splitting the data temporally, as if they were arriving at different moments for analysis. The data were divided into groups of 20 consecutive temporal nodes. Therefore, there are three subsets $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ for recursive analysis, corresponding to the temporal division of the data: $u_t^{(1)} = \{u_{1t}, u_{20t}\}$, $u_t^{(2)} = \{u_{21t}, u_{40t}\}$ and $u_t^{(3)} = \{u_{41t}, u_{60t}\}$. In Fig. 7.6, the results of the inferential process for the temporal structure are presented. Since we have split the data according to the spatial structure, evaluating this allows us to see more clearly the effect of the inferential procedure and the updates with the incorporation of each subset into which the original data have been split. Finally, in Fig. 7.7, the comparison of the mean and standard deviation of the posterior distribution of the spatial effect is shown between analyzing all the data simultaneously (standard INLA approach) or analyzing them recursively (recursive INLA), as previously explained.

7.6 Conclusions

In this work, several ways to incorporate prior information have been synthesised, either through prior elicitation or the Supra-Bayesian approach, focusing on its implementation in R-INLA. Regarding the Supra-Bayesian approach, it has been indicated that there may be a change in the support between the observation model and the auxiliary expert model, with respect to the support associated with the observational data. It has also been explained how to conduct prior elicitation, presenting various strategies to incorporate information into prior distributions in INLA or by combining INLA with other approaches, along with a proposal

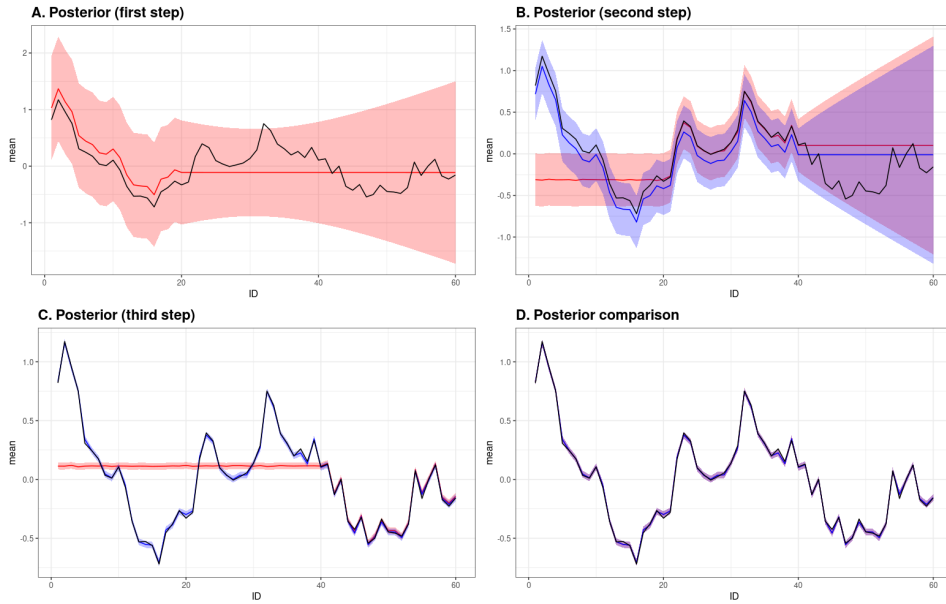


FIGURE 7.6: Mean values along with the 0.025 and 0.975 quantiles of the posterior distribution associated with the temporal effect as additional partitions are incorporated into the analysis. In plots A, B, and C, the black line represents the true temporal effect, while the red line shows the update introduced by the partition being added at each step. The blue line represents the posterior mean and quantiles of the temporal effect as the sequential updates progress (except at the first step, where it coincides with the red line and is thus omitted). Plot D displays the posterior mean and quantiles obtained using the recursive INLA approach (in blue), the standard INLA approach (in red), and the true simulated temporal effect (in black).

for conducting a recursive inferential procedure leveraging the methodology and underlying assumptions of the INLA framework.

In the recursive inferential approach, leveraging the underlying assumptions of INLA, we have highlighted that the determination of support points for the hyperparameters is likely to be different from those determined using the entire dataset. Another key difference is that the computation of the posterior distribution of the hyperparameters in the recursive approach differed from the one obtained when using the whole dataset at once. However, the results shown in the spatio-temporal example indicate that these differences were negligible.

In the various examples, the improvement in the posterior distributions of the spatial structure has been demonstrated in the presence of spatial change of support

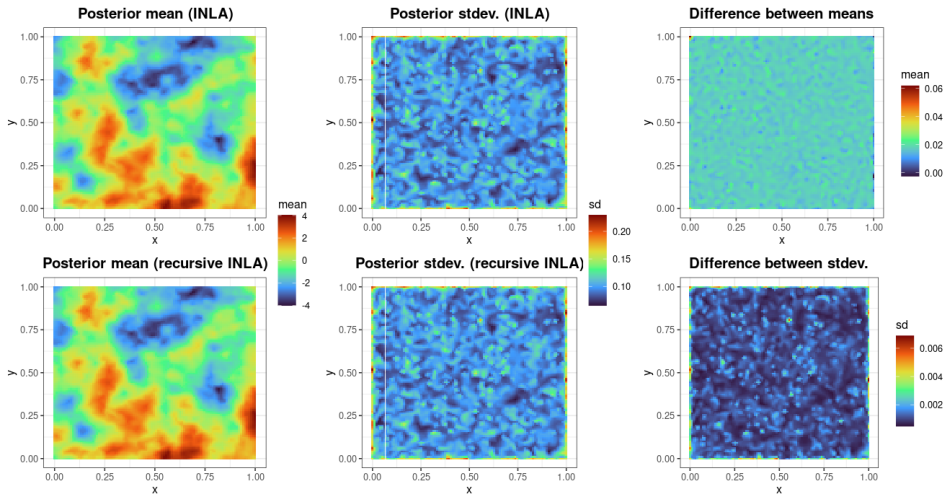


FIGURE 7.7: Mean and standard deviation of the spatial effect analyzed using both the recursive INLA and the standard INLA approaches, along with the differences in means and standard deviations between the two methods.

between the observations and the expert information, as well as in the case of categorical change of support. Finally, an example implementing the sequential inference procedure with INLA has been shown, where the data were temporally divided, as if accessed at different moments, and the result was compared with the model analyzing all data simultaneously. In this case, the effect of the updates as the sequence progressed was shown, with results that were nearly identical between the sequential procedure and the simultaneous analysis of all data.

References

- Aizpurua, O., Cantú-Salazar, L., San Martín, G., Biver, G., Brotons, L., and Titeux, N. (2015). “Reconciling expert judgement and habitat suitability models as tools for guiding sampling of threatened species.” *Journal of Applied Ecology*, 52(6): 1608–1616.
- Albert, I., Donnet, S., Guihenneuc-Jouyaux, C., Low-Choy, S., Mengersen, K., and Rousseau, J. (2012). “Combining Expert Opinions in Prior Elicitation.” *Bayesian Analysis*, 7(3): 503 – 532.

- Berild, M. O., Martino, S., Gómez-Rubio, V., and Rue, H. (2022). “Importance Sampling with the Integrated Nested Laplace Approximation.” *Journal of Computational and Graphical Statistics*, 31(4): 1225–1237.
- Berrocal, V. J., Gelfand, A. E., and Holland, D. M. (2012). “Space-Time Data fusion Under Error in Computer Model Output: An Application to Modeling Air Quality.” *Biometrics*, 68(3): 837–848.
- Billari, F. C., Graziani, R., and Melilli, E. (2014). “Stochastic Population Forecasting Based on Combinations of Expert Evaluations Within the Bayesian Paradigm.” *Demography*, 51(5): 1933–1954.
- Bolin, D., Simas, A. B., and Wallin, J. (2024). “Gaussian Whittle–Matérn fields on metric graphs.” *Bernoulli*, 30(2): 1611 – 1639.
- Cadham, C. J., Knoll, M., Sánchez-Romero, L. M., Cummings, K. M., Douglas, C. E., Liber, A., Mendez, D., Meza, R., Mistry, R., Sertkaya, A., Travis, N., and Levy, D. T. (2021). “The Use of Expert Elicitation among Computational Modeling Studies in Health Research: A Systematic Review.” *Medical Decision Making*, 42(5): 684–703.
- Chacón-Montalván, E. A., Atkinson, P. M., Nemeth, C., Taylor, B. M., and Moraga, P. (2024). “Spatial Latent Gaussian Modelling with Change of Support.” *arXiv preprint arXiv:2403.08514*.
- Dave Higdon, B. W., James Gattiker and Rightley, M. (2008). “Computer Model Calibration Using High-Dimensional Output.” *Journal of the American Statistical Association*, 103(482): 570–583.
- Di Febbraro, M., Sallustio, L., Vizzarri, M., De Rosa, D., De Lisio, L., Loy, A., Eichelberger, B., and Marchetti, M. (2018). “Expert-based and correlative models to map habitat quality: Which gives better support to conservation planning?” *Global Ecology and Conservation*, 16: e00513.
- Figueira, M., Barber, X., Conesa, D., López-Quílez, A., Martínez-Minaya, J., Paradinas, I., and Pennino, M. G. (2024). “Bayesian feedback in the framework of ecological sciences.” *Ecological Informatics*, 84: 102858.
- Figueira, M., Conesa, D., López-Quílez, A., and Paradinas, I. (2025). “A computationally efficient procedure for combining ecological datasets by means of sequential consensus inference.” *Environmental and Ecological Statistics*, 32(2): 495–521.

- Fitzgerald, D. B., Smith, D. R., Culver, D. C., Feller, D., Fong, D. W., Hajenga, J., Niemiller, M. L., Nolfi, D. C., Orndorff, W. D., Douglas, B., Maloney, K. O., and Young, J. A. (2021). “Using expert knowledge to support Endangered Species Act decision-making for data-deficient species.” *Conserv Biol*, 35(5): 1627–1638.
- Gelfand, A. E. (2001). “On the change of support problem for spatio-temporal data.” *Biostatistics*, 2(1): 31–45.
- Gobbi, M., Riservato, E., Bragalanti, N., and Lencioni, V. (2012). “An expert-based approach to invertebrate conservation: Identification of priority areas in central-eastern Alps.” *Journal for Nature Conservation*, 20(5): 274–279.
- Gómez-Rubio, V. (2020). *Bayesian Inference with INLA*. Chapman & Hall/CRC Press.
- Gómez-Rubio, V. and Rue, H. (2018). “Markov chain Monte Carlo with the Integrated Nested Laplace Approximation.” *Statistics and Computing*, 28(5): 1033–1051.
- Guhaniyogi, R., Li, C., Savitsky, T. D., and Srivastava, S. (2022). “Distributed Bayesian Varying Coefficient Modeling Using a Gaussian Process Prior.” *Journal of Machine Learning Research*, 23(84): 1–59.
- Hooten, M. B., Johnson, D. S., and Brost, B. M. (2021). “Making Recursive Bayesian Inference Accessible.” *The American Statistician*, 75: 185–194.
- Kaurila, K., Kuningas, S., Lappalainen, A., and Vanhatalo, J. (2022). “Species Distribution Modeling with Expert Elicitation and Bayesian Calibration.” *arXiv preprint arXiv:2206.08817*.
- Kennedy, M. C. and O’Hagan, A. (2001). “Bayesian calibration of computer models.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(3): 425–464.
- Kessler, S., Cobb, A., Rudner, T. G. J., Zohren, S., and Roberts, S. J. (2023). “On Sequential Bayesian Inference for Continual Learning.” *Entropy*, 25(6).
- Kuhnert, P. M., Martin, T. G., and Griffiths, S. P. (2010). “A guide to eliciting and using expert knowledge in Bayesian ecological models.” *Ecology Letters*, 13(7): 900–914.
- Kynn, M. (2008). “The ‘Heuristics and Biases’ Bias in Expert Elicitation.” *Journal of the Royal Statistical Society Series A: Statistics in Society*, 171: 239–264.

- Liu, Z. and Rue, H. (2024). “Leave-group-out cross-validation for latent Gaussian models.”
- Lunn, D., Barrett, J., Sweeting, M., and Thompson, S. (2013). “Fully Bayesian hierarchical modelling in two stages, with application to meta-analysis.” *J R Stat Soc Ser C Appl Stat*, 62(4): 551–572.
- Martins, T. G., Simpson, D., Lindgren, F., and Rue, H. (2013). “Bayesian computing with INLA: New features.” *Computational Statistics & Data Analysis*, 67: 68–83.
- Minsker, S., Srivastava, S., Lin, L., and Dunson, D. B. (2017). “Robust and Scalable Bayes via a Median of Subset Posterior Measures.” *Journal of Machine Learning Research*, 18: 1–40.
- Moraga, P., Cramb, S. M., Mengersen, K. L., and Pagano, M. (2017). “A geostatistical model for combined analysis of point-level and area-level data using INLA and SPDE.” *Spatial Statistics*, 21: 27–41.
- Morgan, M. G. (2014). “Use (and abuse) of expert elicitation in support of decision making for public policy.” *Proceedings of the National Academy of Sciences*, 111(20): 7176–7184.
- Muff, S., Riebler, A., Held, L., Rue, H., and Saner, P. (2015). “Bayesian analysis of measurement error models using integrated nested Laplace approximations.” *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 64(2): 231–252.
- O’Hagan, A. and Oakley, J. (2010). “The Sheffield Elicitation Framework (SHELF, v2. 0). An “Off the Shelf” package for eliciting probability distributions. Department of Probability and Statistics. The University of Sheffield. UK.”
- O’Hagan, A. (2006). *Uncertain judgements : eliciting experts’ probabilities*. Wiley.
- Ovaskainen, O. and Soininen, J. (2011). “Making more out of sparse data: hierarchical modeling of species communities.” *Ecology*, 92(2): 289–295.
- O’Hagan, A. (2019). “Expert Knowledge Elicitation: Subjective but Scientific.” *The American Statistician*, 73: 69–81.
- Paradinas, I., Illian, J. B., Alonso-Fernández, A., Pennino, M. G., and Smout, S. (2023). “Combining fishery data through integrated species distribution models.” *ICES Journal of Marine Science*, fsad069.

- Pearman-Gillman, S. B., Katz, J. E., Mickey, R. M., Murdoch, J. D., and Donovan, T. M. (2020). “Predicting wildlife distribution patterns in New England USA with expert elicitation techniques.” *Global Ecology and Conservation*, 21: e00853.
- Perälä, T., Vanhatalo, J., and Chrysafi, A. (2020). “Calibrating Expert Assessments Using Hierarchical Gaussian Process Models.” *Bayesian Analysis*, 15(4): 1251 – 1280.
- Pettit, L. I. (1990). “The Conditional Predictive Ordinate for the Normal Distribution.” *Journal of the Royal Statistical Society: Series B (Methodological)*, 52(1): 175–184.
- Rendell, L. J., Johansen, A. M., Lee, A., and Whiteley, N. (2021). “Global Consensus Monte Carlo.” *Journal of Computational and Graphical Statistics*, 30: 249–259.
- Rue, H. and Held, L. (2005). *Gaussian Markov Random Fields*. Chapman and Hall/CRC.
- Rue, H., Martino, S., and Chopin, N. (2009). “Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(2): 319–392.
- Schifano, E. D., Wu, J., Wang, C., Yan, J., and Chen, M.-H. (2016). “Online Updating of Statistical Inference in the Big Data Setting.” *Technometrics*, 58(3): 393–403.
- Scott, S. L. (2017). “Comparing consensus Monte Carlo strategies for distributed Bayesian computation.” *Brazilian Journal of Probability and Statistics*, 31(4): 668 – 685.
- Scott, S. L., Blocker, A. W., Bonassi, F. V., Chipman, H. A., George, E. I., and McCulloch, R. E. (2016). “Bayes and big data: the consensus Monte Carlo algorithm.” *International Journal of Management Science and Engineering Management*, 11: 78–88.
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., and Van Der Linde, A. (2002). “Bayesian measures of model complexity and fit.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(4): 583–639.
- Särkkä, S. (2013). *Bayesian Filtering and Smoothing*. Institute of Mathematical Statistics Textbooks. Cambridge University Press.

- Taylor-Rodríguez, D., Kaufeld, K., Schliep, E. M., Clark, J. S., and Gelfand, A. E. (2017). “Joint Species Distribution Modeling: Dimension Reduction Using Dirichlet Processes.” *Bayesian Analysis*, 12(4).
- van Niekerk, J. and Rue, H. (2024). “Low-rank Variational Bayes correction to the Laplace method.” *Journal of Machine Learning Research*, 25(62): 1–25.
- van Niekerk, J., Krainski, E., Rustand, D., and Rue, H. (2023). “A new avenue for Bayesian inference with INLA.” *Computational Statistics & Data Analysis*, 181: 107692.
- Villejo, S. J., Illian, J. B., and Swallow, B. (2023). “Data fusion in a two-stage spatio-temporal model using the INLA-SPDE approach.” *Spatial Statistics*, 54: 100744.
- Wang, C. and Furrer, R. (2021). “Combining heterogeneous spatial datasets with process-based spatial fusion models: A unifying framework.” *Computational Statistics & Data Analysis*, 161: 107240.
- Wang, C., Puhani, M. A., and Furrer, R. (2018). “Generalized spatial fusion model framework for joint analysis of point and areal data.” *Spatial Statistics*, 23: 72–90.
- Wang, L., Zhang, X., Su, H., and Zhu, J. (2024). “A Comprehensive Survey of Continual Learning: Theory, Method and Application.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(08): 5362–5383.
- Wang, X. and Dunson, D. B. (2014). “Parallelizing MCMC via Weierstrass Sampler.”
- Watanabe, S. (2013). “A widely applicable Bayesian information criterion.” *J. Mach. Learn. Res.*, 14(1): 867–897.
- Yamada, K., Elith, J., McCarthy, M., and Zerger, A. (2003). “Eliciting and integrating expert knowledge for wildlife habitat modelling.” *Ecological Modelling*, 165(2): 251–264.
- Yuan, Y., Bachl, F. E., Lindgren, F., Borchers, D. L., Illian, J. B., Buckland, S. T., Rue, H., and Gerrodette, T. (2017). “Point process models for spatio-temporal distance sampling data from a large-scale survey of blue whales.” *The Annals of Applied Statistics*, 11(4): 2270 – 2297.
- Zhang, G. and Thai, V. V. (2016). “Expert elicitation and Bayesian Network modeling for shipping accidents: A literature review.” *Safety Science*, 87: 53–62.

Zhong, R., Amaral, A. V. R., and Moraga, P. (2025). “Spatial data fusion adjusting for preferential sampling using integrated nested Laplace approximation and stochastic partial differential equation.” *Journal of the Royal Statistical Society. Series A: Statistics in Society*, 188(1): 140 – 157.

Distributed and recursive Bayesian inference for Big Data and complex spatio-temporal models

In this chapter, we present our paper *Distributed and recursive Bayesian inference for Big Data and complex spatio-temporal models* by Mario Figueira (University of Valencia), David Conesa (University of Valencia), Antonio López-Quílez (University of Valencia) and Håvard Rue (KAUST). The content of this chapter is currently in preparation for future submission. The chapter contains at the end the references used in this work.

Abstract

The rapid growth of massive and complex datasets in domains such as econometrics, environmental sciences, risk management, and public policy has reshaped statistical modeling, while posing significant computational and methodological challenges. These challenges stem not only from data scale, but also from the sequential or streaming nature of modern applications, which demand efficient updates of inference without full reprocessing. To address these issues, distributed and recursive inference methodologies have emerged as powerful strategies. Distributed approaches, typically based on divide-and-conquer principles, partition large datasets and likelihoods into manageable components that can be analyzed in parallel and subsequently recombined, with applications ranging from spatio-temporal models to privacy-preserving and federated analysis. Recursive approaches, in turn, enable sequential updating of posterior distributions as new information becomes available, providing particular advantages in online learning, continuous monitoring, and meta-analysis.

In this paper, we introduce a novel, comprehensive Bayesian framework for distributed and recursive inference, grounded in the integrated nested Laplace approximations (INLA) methodology, and exemplify its implementation using the **R-INLA** software. Our contributions include automated strategies for partitioning both data and models, thereby reducing computational complexity while retaining accuracy relative to centralized full-data analysis. We illustrate the effectiveness of the proposed framework through case studies that demonstrate its applicability to large-scale, streaming, and privacy-sensitive environments. By integrating distributed, recursive, and federated paradigms, this work provides scalable, adaptive, and generalizable tools for modern Bayesian inference.

Keywords

Distributed inference, recursive inference, latent Gaussian models, partitioning strategies, INLA.

8.1 Introduction

The increasing availability of large and complex datasets in fields such as econometrics, environmental sciences, ecology, risk management, and public policy has transformed the way statistical models are built and applied (Varian, 2014; Farley et al., 2018; Sebestyén et al., 2021). The ability to measure, store, and access these extensive databases provides comprehensive insights into various phenomena while also posing significant challenges for statistical and computational theory and methodology, which motivates the development of distributed methods capable of harnessing the collective computational power of multiple machines or servers (Gao et al., 2022; Zhou et al., 2024). These challenges arise not only from the sheer volume of data but also from the complexity of models, such as spatial and spatio-temporal models. Modern data are often not only massive in size but also arrive sequentially or in streams, requiring models to be updated regularly as new information becomes available. Re-analyzing the entire dataset each time new data are collected quickly becomes computationally infeasible, particularly for high-dimensional or spatio-temporal models. This challenge motivates the development of recursive inference methods, which enable the posterior distribution of model parameters to be updated incrementally, incorporating new data without revisiting the full historical dataset.

Such approaches are crucial not only for real-time decision-making, continuous monitoring and learning, or long-term data accumulation (Hooten et al., 2021; Kessler et al., 2023), but also for meta-analysis (Lunn et al., 2013), where evidence from new studies is published over time and must be integrated into an existing synthesis without restarting the full inferential process. In all these contexts, scalability and adaptability are as important as statistical rigor.

The challenges previously addressed have given rise to the development of a broad spectrum of distributed and recursive inference methodologies (Huang and Gelman, 2005; Vehtari et al., 2020; Hooten et al., 2021; Gao et al., 2022; Zhou et al., 2024). Distributed approaches, typically based on divide-and-conquer strategies (Huang and Gelman, 2005; Vehtari et al., 2020; Orozco-Acosta et al., 2023), partition the likelihood into subsets of data that can be analyzed in parallel and subsequently recombined using methods such as consensus Monte Carlo (Scott et al., 2016; Rendell et al., 2021), Weierstrass samplers (Wang and Dunson, 2013), or Wasserstein barycenter aggregation of subposteriors (Srivastava et al., 2018), among others. Recursive inference, in contrast, emphasizes sequential updating of posterior distributions as new data become available, avoiding the need to reprocess the entire dataset. Methods such as streaming variational Bayes (Broderick et al., 2013), stochastic variational inference (Hoffman et al., 2013), and particle-based sequential updates (Carvalho et al., 2010) enable efficient learning in settings where data arrive continuously or in large batches. These algorithmic frameworks are especially valuable in real-time applications, including environmental monitoring, financial risk assessment, and epidemiological surveillance. Moreover, beyond computational efficiency, distributed and recursive methods address increasingly important constraints on data accessibility. In privacy-preserving environments—such as health care, business analytics, or policy evaluation—data cannot always be centralized (Sarwate and Chaudhuri, 2013; Smith et al., 2017; Altman et al., 2018). Federated inference frameworks extend distributed computation by enabling collaborative learning across decentralized data sources without direct exchange of raw information, instead relying on communication of model summaries, gradients, or approximate posterior components (McMahan et al., 2017; Smith et al., 2017; Singh et al., 2022). This integration of distributed, recursive, and federated paradigms offers not only scalable solutions to big-data problems but also principled mechanisms for updating posterior information adaptively and responsibly in dynamic, heterogeneous, and privacy-sensitive environments.

In this paper, we present a comprehensive framework for performing distributed

and recursive inference within the Bayesian paradigm. Our proposals build upon the *integrated nested Laplace approximation* (INLA, [Rue et al. \(2009\)](#); [Van Niekerk et al. \(2023\)](#)) methodology, taking advantage of the high computational performance provided by the R-INLA software ([Martins et al., 2013](#)). This efficiency arises from the integration of sparse matrix algebra techniques —particularly effective for Gaussian Markov Random Fields (GMRFs)— and advanced numerical methods for deterministic approximation of posterior distributions. In addition, we provide algorithms for automatic partitioning of data and models, thereby reducing the computational complexity of the inference task while minimizing divergence from the results obtained through full-data analysis.

In the context of distributed inference, we discuss both its conceptual foundations and practical strategies for implementation. These methods are not only effective for large-scale datasets and complex models, such as spatio-temporal structures, but also crucial in privacy-sensitive settings, where raw data cannot be shared. In such cases, joint inference is achieved through the exchange of summaries or aggregated information, preserving confidentiality while enabling rigorous analysis. Two complementary paradigms illustrate this perspective: a top-down approach, motivated by the need to scale inference to massive datasets through data and likelihood partitioning; and a bottom-up approach, where data are naturally split across sources, as in privacy-preserving computation or meta-analysis. Together, these views underscore the versatility of distributed inference as both a computational solution and a methodological framework in Bayesian analysis. Within the framework of recursive inference, we also provide a detailed discussion of its conceptual foundations and limitations, as well as practical implementation strategies. Recursive methods are motivated by scenarios where data arrive sequentially or are too large to be processed in a single step, requiring inference to be updated incrementally as new information becomes available. This setting is especially relevant in online learning, real-time monitoring, and streaming data applications, where recomputing the posterior from scratch is computationally prohibitive. By reusing information from previous inference steps, recursive approaches balance accuracy and efficiency, enabling near real-time updates while keeping computational costs manageable. However, these methods face challenges such as error accumulation and approximation quality, which call for careful algorithmic design. Overall, recursive inference offers a principled way to adapt Bayesian analysis to dynamic and evolving data environments. Lately, we provide a detailed explanation of the algorithms proposed for the automatic partitioning of data and models, highlighting their connection to the distributed and recursive approaches. We then illustrate their

performance through examples that demonstrate the implementation of distributed and recursive frameworks within these partitioning strategies.

In summary, Section 2 provides a brief overview of the INLA methodology. Section 3 presents our proposed distributed inference approach within the INLA framework, offering a detailed discussion of various strategies and their limitations. Section 4 describes the recursive inference methods we propose, examining implementation strategies and addressing key methodological aspects. Section 5 introduces several algorithms for the automatic partitioning of data and models. Section 6 discusses the implementation of distributed and recursive inference within the R-INLA software, concisely connecting theory and practice. Section 7 illustrates the application of these methods using one simulated case study and two real-world examples. Finally, the conclusions synthesize our proposals, highlighting their scope, applicability, and potential impact.

8.2 Integrated Nested Laplace Approximations

INLA is a deterministic approximate Bayesian inference approach grounded in the properties of the GMRF and its computational efficiency. This methodology was developed by [Rue et al. \(2009\)](#), implemented in R through the R-INLA package ([Martins et al., 2013](#)), and later extended to more flexible modeling structures ([Lindgren and Rue, 2015](#); [Bakka et al., 2018](#); [Gómez-Rubio, 2020](#)). This approach allows Bayesian inference to be performed on a wide range of structured additive models, known as Latent Gaussian Models (LGMs). While INLA primarily focuses on computing marginal posterior distributions, it effectively addresses the challenges of determining the joint posterior distribution of the latent field $\mathbf{x}$ and hyperparameters $\boldsymbol{\theta}$ through a series of nested approximations or a low-rank variational Bayes correction for the latent field ([Van Niekerk et al., 2023](#); [Van Niekerk and Rue, 2024](#)). This technique enables the computation of marginal likelihoods, standard goodness-of-fit metrics (DIC, WAIC, CPO), cross-validation checks, and posterior predictive distributions.

Since INLA is based on GMRF principles, it offers inherent computational efficiency when modeling data with conditional independence. For example, if we have n observations denoted as $\mathbf{y} = (y_1, \dots, y_n)$, the likelihood of our data, assuming conditional independence, is described by the following expression:

$$p(\mathbf{y}|\boldsymbol{\eta}, \boldsymbol{\theta}) = \prod_{i=1}^n p(y_i | \eta_i, \boldsymbol{\theta}), \quad (8.1)$$

where the linear predictor is linked to the mean of the likelihood through a link function $g(\mu_i) = \eta_i$, $\boldsymbol{\eta}$ is related to the latent field $\mathbf{x}$ through a projection matrix $\mathbf{A}$, such that $\boldsymbol{\eta} = \mathbf{A} \cdot \mathbf{x}$. The latent field $\mathbf{x}$ encompasses the vector of fixed effects $\boldsymbol{\beta}$ and the vector of random effects $\mathbf{f}$, i.e., $\mathbf{x} = (\boldsymbol{\beta}, \mathbf{f})$. The matrix $\mathbf{A}$ serves as the design matrix of the model, incorporating covariate values, factor structures and their levels, and weights associated with random effects (e.g., basis weights for splines). The hyperparameters are represented by $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$, where $\boldsymbol{\theta}_1$ refers to the likelihood hyperparameters and $\boldsymbol{\theta}_2$ to the latent field hyperparameters.

Assuming this conditional independence property and the structure of the latent field, the INLA methodology computes the posterior of the hyperparameters by examining the following functional relation in the modal configuration $\mathbf{x}^*(\boldsymbol{\theta})$ of the Gaussian approximation for the conditional posterior of the latent field $\tilde{\pi}_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})$:

$$\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}) \propto \frac{\pi(\mathbf{y}, \mathbf{x}, \boldsymbol{\theta})}{\tilde{\pi}_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})}, \quad (8.2)$$

the evaluation of this posterior distribution for the hyperparameters is performed at K different support points, $\{\boldsymbol{\theta}^k\}_{k=1}^K$, based on a grid exploration or on a central composite design (CCD). The Gaussian approximation $\tilde{\pi}_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})$ is calculated from a second-order Taylor expansion of the likelihood around $\boldsymbol{\mu}^*(\boldsymbol{\theta})$, the mode of $\pi(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})$ (Rue et al., 2009; Van Niekerk et al., 2023):

$$\begin{aligned} \pi_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta}) \approx & \exp(\log \pi(\boldsymbol{\mu}(\boldsymbol{\theta}) | \mathbf{y}, \boldsymbol{\theta}) + \nabla \log \pi(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}^*(\boldsymbol{\theta})} \mathbf{x} \\ & + \frac{1}{2} \mathbf{x}^\top \nabla^2 \log \pi(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}^*(\boldsymbol{\theta})} \mathbf{x}). \end{aligned} \quad (8.3)$$

This posterior approximation is obtained for each support point $\boldsymbol{\theta}^k$, and the mean is corrected through an iterative low-rank variational Bayes procedure (Van Niekerk and Rue, 2024). This yields a GMRF with a corrected mean for each support point $\boldsymbol{\theta}^k$:

$$\pi'_G(\mathbf{x} | \mathbf{y}, \boldsymbol{\theta}^k) = \text{GMRF}(\boldsymbol{\mu}'(\boldsymbol{\theta}^k), \mathbf{Q}(\boldsymbol{\theta}^k));, \quad (8.4)$$

where $\boldsymbol{\mu}'(\boldsymbol{\theta})$ is the corrected mean, defined as $\boldsymbol{\mu}'(\boldsymbol{\theta}) = \boldsymbol{\mu}(\boldsymbol{\theta}) + \mathbf{M}\boldsymbol{\lambda}$. Here, $\mathbf{M}$ is a matrix that propagates the correction from p pre-selected nodes to the rest of

the latent field, and $\boldsymbol{\lambda}$ represents the explicit corrections at those p nodes. The precision matrix is denoted by $\mathbf{Q}(\boldsymbol{\theta})$.

The GMRF of the latent field for each $\boldsymbol{\theta}^k$ is one of the outputs returned by the `inla` function in the R-INLA package and is a key component for implementing a recursive inference procedure with the R-INLA software.

After fixing the support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ the INLA approach computes the marginals of latent field using one of two approaches: (i) a nested Laplace approximation or a simplified Laplace approximation (Rue et al., 2009) or (ii) a low-rank Variational Bayes correction (Van Niekerk and Rue, 2024) for the mean and marginal variances of the already computed conditional Gaussian approximation $\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$. Finally, it performs a numerical integration over the hyperparameters to compute the latent field marginals:

$$\pi(x_i \mid \mathbf{y}) = \sum_{k=1}^K \pi(x_i \mid \mathbf{y}, \boldsymbol{\theta}^k) \pi(\boldsymbol{\theta}^k \mid \mathbf{y}) \Delta_k. \quad (8.5)$$

A key aspect of this approach is the relationship between latent Gaussian fields (LGFs) and GMRFs. The latent field $\mathbf{x}$ follows a multivariate normal prior distribution with mean $\boldsymbol{\mu}$, usually equal to $\mathbf{0}$, and precision matrix $\mathbf{Q}(\boldsymbol{\theta})$. This connection is crucial because, under the assumption that the LGF precision matrix is sparse, computational efficiency is significantly improved by enabling efficient sparse matrix operations Rue and Held (2005).

8.3 Distributed inference

The distributed inference approach proposed in this paper builds on the properties and assumptions of the INLA methodological framework (Rue et al., 2009; Van Niekerk et al., 2023), particularly leveraging the version of INLA that implements a low-rank correction of the latent field (Van Niekerk and Rue, 2024).

This section is dedicated to a detailed presentation of the distributed approach for obtaining the marginal distributions of the latent field, $\pi(x_i \mid \mathbf{y})$, and the hyperparameters, $\pi(\boldsymbol{\theta}_i \mid \mathbf{y})$, using the INLA methodology. This involves computing the conditional posterior distribution of the latent field, $\pi(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})$, which is required to derive both the joint marginal distribution of the hyperparameters,

$\pi(\boldsymbol{\theta} \mid \mathbf{y})$, and the marginal distributions of the individual nodes in the latent field, $\pi(x_i \mid \mathbf{y})$. The marginal distribution for each hyperparameter, $\pi(\theta_i \mid \mathbf{y})$, is obtained by marginalizing the joint marginal posterior distribution of the hyperparameters, $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$.

8.3.1 Conditional posterior distribution of the latent field

This distributed method relies on the conditional independence of the data given the latent field and the hyperparameters, as stated in Eq. (8.1). This assumption allows us to partition or treat the data associated with the model in a distributed manner. The structure becomes more explicit when we write the conditional posterior distribution of the latent field, $\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$, given the data and the hyperparameters. In the case of a Gaussian likelihood and a dataset $\mathbf{y}$ that can be split into n subsets (or is naturally available in such a form), the posterior distribution of the latent field can be written by applying the conditional independence along the subsets $\mathbf{y}_i$:

$$\begin{aligned} \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) &= \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x} \mid \boldsymbol{\theta}) \frac{\pi(\boldsymbol{\theta})}{\pi(\boldsymbol{\theta} \mid \mathbf{y}) \pi(\mathbf{y})} \\ &\propto \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x} \mid \boldsymbol{\theta}) \\ &\propto \left[\prod_{i=1}^n \pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) \right] \pi(\mathbf{x} \mid \boldsymbol{\theta}) \end{aligned} \quad (8.6)$$

where the conditional independence property is applied. From this step, it is possible to compute the conditional posterior distribution of the latent field in a distributed way.

In the Gaussian case, and assuming a linear predictor of the form $\boldsymbol{\eta} = \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is a projection matrix and $\mathbf{x}$ is a GMRF (typically with zero mean and precision matrix $\mathbf{Q}(\boldsymbol{\theta}_2)$), the expression can be made explicit as:

$$\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) = C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \mathbf{A}_i \mathbf{x})^\top \mathbf{W}(\boldsymbol{\theta}_1) (\mathbf{y}_i - \mathbf{A}_i \mathbf{x}) - \frac{1}{2} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \quad (8.7)$$

where $C(\boldsymbol{\theta}, \mathbf{y})$ is a constant that depends on the hyperparameter vector $\boldsymbol{\theta} = \{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2\}$ and incorporates contributions from the likelihood, the prior on the latent field $\pi(\mathbf{x} \mid \boldsymbol{\theta}_2)$, the hyperprior $\pi(\boldsymbol{\theta})$, the marginal likelihood $\pi(\mathbf{y})$, and the marginal posterior $\pi(\boldsymbol{\theta} \mid \mathbf{y})$. If the conditional independence $\pi(\mathbf{y}_i, \mathbf{y}_j \mid \boldsymbol{\eta}, \boldsymbol{\theta}) = \pi(\mathbf{y}_i \mid \boldsymbol{\eta}, \boldsymbol{\theta}) \pi(\mathbf{y}_j \mid \boldsymbol{\eta}, \boldsymbol{\theta})$ holds, then the matrix $\mathbf{W}(\boldsymbol{\theta}_1)$ is diagonal and can be written as $\mathbf{W}(\boldsymbol{\theta}_1) = \tau \mathbf{I}$, where τ is the marginal precision associated with the likelihood.

To make explicit the equivalence between the conditional posterior distribution from standard analysis and that from distributed inference, we note that the projection matrix $\mathbf{A}$ can be rewritten as:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_n \end{pmatrix} \Rightarrow \begin{cases} \boldsymbol{\eta} = \mathbf{A}\mathbf{x} = \begin{pmatrix} \frac{\boldsymbol{\eta}_1}{\vdots} \\ \boldsymbol{\eta}_n \end{pmatrix} = \begin{pmatrix} \frac{\mathbf{A}_1\mathbf{x}}{\vdots} \\ \mathbf{A}_n\mathbf{x} \end{pmatrix} \\ \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} = \boldsymbol{\eta}^\top \boldsymbol{\eta} = \sum_{i=1}^n \mathbf{x}^\top \mathbf{A}_i^\top \mathbf{A}_i \mathbf{x} \end{cases} . \quad (8.8)$$

Therefore, in the case of conditional independence, where $\mathbf{W}(\boldsymbol{\theta}_1) = \tau \mathbf{I}$, the log-posterior conditional distribution can be written as:

$$\begin{aligned} \log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} (\tau \cdot \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} - 2 \cdot \tau \cdot \mathbf{y}^\top \mathbf{A} \mathbf{x}) - \frac{1}{2} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \\ &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} \sum_{i=1}^n (\tau \cdot \mathbf{x}^\top \mathbf{A}_i^\top \mathbf{A}_i \mathbf{x} - 2 \cdot \tau \cdot \mathbf{y}_i^\top \mathbf{A}_i \mathbf{x}) - \frac{1}{2} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \\ &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} \sum_{i=1}^n \left(\tau \cdot \mathbf{x}^\top \mathbf{A}_i^\top \mathbf{A}_i \mathbf{x} - 2 \cdot \tau \cdot \mathbf{y}_i^\top \mathbf{A}_i \mathbf{x} + \frac{1}{n} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \right) \\ &= C^*(\boldsymbol{\theta}, \mathbf{y}) + \sum_{i=1}^n \left(\log \pi(\mathbf{y}_i \mid \mathbf{A}_i \mathbf{x}, \boldsymbol{\theta}) + \frac{1}{n} \log \pi(\mathbf{x} \mid \boldsymbol{\theta}_2) \right) \\ &= C'(\boldsymbol{\theta}, \mathbf{y}) + \sum_{i=1}^n \log \pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}) \end{aligned} \quad (8.9)$$

where $C(\boldsymbol{\theta}, \mathbf{y})$, $C^*(\boldsymbol{\theta}, \mathbf{y})$ and $C'(\boldsymbol{\theta}, \mathbf{y})$ are constants ensuring the equivalence of the different formulations of $\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$. These constants can be ignored since the posterior is Gaussian, and given the mean and the precision matrix, we can immediately recover the analytical form of the posterior distribution without the need to compute the constants. Furthermore, when computing the posterior distribution associated with each partition $\log \pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta})$ one can use a prior weighted identically across partitions—as done above—or a prior weighted according to the number of observations in each partition:

$$\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) = C^*(\boldsymbol{\theta}, \mathbf{y}) + \sum_{i=1}^n (\log \pi(\mathbf{y}_i \mid \mathbf{A}_i \mathbf{x}, \boldsymbol{\theta}) + w_i \log \pi(\mathbf{x} \mid \boldsymbol{\theta}_2)), \quad (8.10)$$

where the weights can be defined as $w_i = (|\mathbf{y}| - |\mathbf{y}_i|) / ((n-1) \cdot |\mathbf{y}|)$ such that partitions with fewer observations are assigned priors that are weighted more heavily, thus remaining closer to the original prior. Any alternative weighting scheme that satisfies

$w_i > 0$ ($\forall i$) and $\sum_{i=1}^n w_i = 1$ can, in principle, also be used.

An alternative to weighting the prior distribution is to use the unweighted prior and modify the posterior distribution as follows:

$$\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) = C(\boldsymbol{\theta}, \mathbf{y}) + \sum_{i=1}^n (\log \pi(\mathbf{y}_i \mid \mathbf{A}_i \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \boldsymbol{\theta}_2)) - (n-1) \cdot \log \pi(\mathbf{x} \mid \boldsymbol{\theta}_2), \quad (8.11)$$

which is equivalent to compute the posterior as

$$\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \propto \pi^*(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) / \pi(\mathbf{x} \mid \boldsymbol{\theta})^{n-1},$$

where $\pi^*(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ is the posterior computed in a distributed fashion, using the unweighted prior in each partition.

This allows the conditional posterior distribution of the latent field in each partition, $\pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta})$, to be computed independently, either in parallel on different machines or servers, or in parallel on a single machine. The latter may be particularly useful, as we will discuss later, when partitioning the data is done through the partitioning of the latent field, generally reducing the computational cost of computing each of these posterior distributions. The distributed conditional posterior for the case of a Gaussian likelihood and under conditional independence of the observations—as discussed earlier—is analytically identical to the posterior obtained without data partitioning. However, this identity no longer holds in the case of non-Gaussian likelihoods, and the distributed approach becomes an approximation.

In general, for non-Gaussian likelihoods, the conditional posterior for each partition is approximated via a second-order Taylor expansion of the log-likelihood around the mode of the true conditional posterior $\pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta})$. The Gaussian approximation uses the mode $\boldsymbol{\mu}_i(\boldsymbol{\theta})$ for the i -th partition, computed as the minimum of the negative log conditional posterior:

$$\boldsymbol{\mu}_i(\boldsymbol{\theta}) = \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}) \}$$

and the precision matrix of the Gaussian approximation at the mode is given by the Hessian of the negative log-conditional posterior evaluated at $\boldsymbol{\mu}_i(\boldsymbol{\theta})$:

$$\mathbf{Q}_i(\boldsymbol{\theta}) = -\mathbf{H}(\log \pi(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}))|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})} = \mathbf{Q}(\boldsymbol{\theta}) + \mathbf{A}_i^\top \mathbf{D}_i \mathbf{A}_i,$$

where $\mathbf{Q}(\boldsymbol{\theta})$ is the prior precision matrix for the latent field, $\mathbf{A}_i$ is the projection matrix for the linear predictor, and $\mathbf{D}_i$ is a diagonal matrix computed from the second derivatives of the likelihood.

Hence, the conditional posterior distribution is approximated by a Gaussian distribution $\pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta})$ for each partition, expressed in canonical form:

$$\pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}) \propto \exp \left[-\frac{1}{2} \mathbf{x}^\top (w_i \mathbf{Q}(\boldsymbol{\theta}) + \mathbf{A}_i^\top \mathbf{D}_i \mathbf{A}_i) \mathbf{x} - \mathbf{b}_i^\top \mathbf{A}_i \mathbf{x} \right] \quad (8.12)$$

where $(w_i \mathbf{Q}(\boldsymbol{\theta}) + \mathbf{A}_i^\top \mathbf{D}_i \mathbf{A}_i)$ is the precision matrix of the approximated posterior and the prior precision is weighted by w_i , as explained earlier. The mean is computed as $\mathbf{b} = w_i \mathbf{Q}^{-1}(\boldsymbol{\theta}) \boldsymbol{\mu}$. This gives a Gaussian approximation for the conditional posterior of the latent field for each partition $\mathbf{y}_i$:

$$\pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}_i(\boldsymbol{\theta}), \mathbf{Q}_i(\boldsymbol{\theta})).$$

Therefore, to obtain the Gaussian approximation for the full dataset, one could apply any of the strategies discussed for the Gaussian case. However, with non-Gaussian likelihoods, the identity between the full and distributed conditional posteriors no longer holds. The posterior for the complete dataset is:

$$\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \propto \exp \left[-\frac{1}{2} \mathbf{x}^\top (\mathbf{Q}(\boldsymbol{\theta}) + \mathbf{A}^\top \mathbf{D} \mathbf{A}) \mathbf{x} - \mathbf{b}^\top \mathbf{A} \mathbf{x} \right],$$

which implies that the components $\mathbf{D}$ and $\mathbf{b}$ will differ across partitions. Thus, the identity used in the Gaussian case no longer holds:

$$\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \approx \pi_G^*(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) = C(\boldsymbol{\theta}, \mathbf{y}) \cdot \prod_{i=1}^n \pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}), \quad (8.13)$$

although we may still use this product as an approximation.

In the non-Gaussian case, we rely on the identity used in the Gaussian case, but only as an approximation:

$$\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \approx \pi_G^*(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \approx C(\boldsymbol{\theta}, \mathbf{y}) \cdot \prod_{i=1}^n \pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta}), \quad (8.14)$$

where the quality of this approximation depends on the quality of each individual approximation $\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ across all partitions. In other words, the approximation

$\pi_G^*(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ accumulates the errors of the per-partition approximations, whereas the global approximation $\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ can achieve greater accuracy, as it uses all available data to better identify the mode and the curvature at the mode.

Rue and Martino (2007) show that the error in the approximation $\pi_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})$ arises primarily from mode location and skewness. Therefore, Van Niekerk et al. (2023) propose correcting the mean using a low-rank variational Bayes approach while keeping the variance unchanged.

8.3.2 Joint marginal posterior distribution of the hyperparameters

The next step, following the inferential structure of INLA presented in Section 2, is to obtain the marginal posterior distribution of the hyperparameters $\pi(\boldsymbol{\theta} \mid \mathbf{y})$, which is generally obtained approximately using the functional form evaluated at the modal configuration of the conditional posterior of the latent field, as shown in Eq. (8.2):

$$\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}) \propto \frac{\pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})},$$

where $\pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta})$ is the likelihood, $\pi(\mathbf{x} \mid \boldsymbol{\theta})$ is the prior distribution for the latent field, $\pi(\boldsymbol{\theta})$ is the prior for the hyperparameters, and $\pi_G(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ is the conditional posterior distribution previously discussed, which is exact in the case of a Gaussian likelihood and approximated by a Gaussian distribution for non-Gaussian likelihoods. However, regardless of whether this is exact or a Gaussian approximation, evaluating it at the modal configuration of the conditional posterior distribution of the latent field $\pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})$ leads to the following expression:

$$\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}) \propto \frac{\pi(\mathbf{y} \mid \boldsymbol{\theta})|\mathbf{Q}(\boldsymbol{\theta})|\exp\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta})\mathbf{x}\right)}{|\mathbf{Q}_{\mathbf{x}|\mathbf{y},\boldsymbol{\theta}}(\boldsymbol{\theta})|} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})} \pi(\boldsymbol{\theta}), \quad (8.15)$$

where $|\mathbf{Q}(\boldsymbol{\theta})|$ is the determinant of the precision matrix of the prior distribution of the latent field, and $|\mathbf{Q}_{\mathbf{x}|\mathbf{y},\boldsymbol{\theta}}(\boldsymbol{\theta})|$ is the determinant of the precision matrix of the conditional posterior distribution of the latent field.

An important issue at this step arises when constraints are imposed on elements of the latent field, $\mathbf{C}\mathbf{x} = \mathbf{e}$. When such constraints are introduced, the precision matrix of the posterior distribution can be computed using *conditioning by kriging*

(Rue et al., 2009), such that the precision matrix with constraints, denoted $\mathbf{Q}_c$, is given by:

$$\mathbf{Q}_c^{-1} = \mathbf{Q}^{-1} - \mathbf{Q}^{-1}\mathbf{C}^\top(\mathbf{C}\mathbf{Q}^{-1}\mathbf{C}^\top)^{-1}\mathbf{C}\mathbf{Q}^{-1}, \quad (8.16)$$

and the determinant is computed over this precision matrix that incorporates the constraints. In Fattah and Rue (2022), an alternative approach is proposed for the case of space-time models with interaction (Knorr-Held, 2000), to compute the covariance matrix $\Sigma_c = \mathbf{Q}_c^{-1}$ associated with the conditional posterior distribution of the latent field, where the constraints are embedded by design in the null space structure of the prior distribution of the space-time effect. This approach leverages the Sherman–Morrison–Woodbury identity (also known as the Woodbury formula).

To compute the marginal posterior distribution of the hyperparameters in the distributed framework proposed in this paper, we can consider different strategies. One option is to compute the marginal distribution as stated in Eq. (8.14). This strategy would require evaluating the marginal posterior of the hyperparameters on a single machine. Alternatively, we can propose computing the marginal posterior leveraging a distributed framework, denoted by $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y})$, where the posterior is computed separately for each dataset and then combined across the n different subsets $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$:

$$\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}) \propto \prod_{i=1}^n \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i) = \prod_{i=1}^n \left. \frac{\pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x} \mid \boldsymbol{\theta})^{w_i} \pi(\boldsymbol{\theta})^{w_i}}{\pi_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y}_i)} \right|_{\mathbf{x}=\mathbf{x}_i^*(\boldsymbol{\theta})}, \quad (8.17)$$

where the weights w_i satisfy $\sum_{i=1}^n w_i = 1$ and $w_i > 0$ ($\forall i$), as defined in the previous section regarding the conditional posterior distribution of the latent field. Furthermore, $\pi_G(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y}_i)$ evaluated at the mode $\mathbf{x} = \mathbf{x}_i^*(\boldsymbol{\theta})$ reduces to

$$\pi_G(\mathbf{x} = \mathbf{x}_i^*(\boldsymbol{\theta}) \mid \boldsymbol{\theta}, \mathbf{y}_i) \propto |\mathbf{Q}_{\mathbf{x}|\boldsymbol{\theta}, \mathbf{y}_i}^c(\boldsymbol{\theta})|,$$

where it is assumed that this posterior has been computed by incorporating the relevant constraints.

In summary, the two possible strategies to compute the joint marginal posterior of the hyperparameters are: (1) obtain the conditional posterior distribution via Eq.(8.14) and apply Eq.(8.16) to compute the marginal posterior via Eq.(8.15), or (2) use the distributed approach described in Eq.(8.17) and apply Eq. (8.16) separately in each partition.

8.3.3 Marginal posterior distributions of the latent field

One of the key elements in the INLA framework is to find the mode of the marginal posterior distribution of the hyperparameters using optimization techniques based on finite differences, since this distribution is not analytically differentiable. From this mode, it is then possible to perform an exploration strategy (e.g. grid, central composite design) of the marginal posterior and obtain both the marginal distributions of the nodes of the latent field and the marginal distributions of each hyperparameter.

To construct the integration scheme—particularly the Central Composite Design (CCD)—we can start from the distributed expression of the marginal posterior of the hyperparameters in Eq. (8.17); based on this expression, we can compute the marginal distributions of the nodes of the latent field, $\pi(x_i | \mathbf{y})$, as well as the marginal distributions of the individual hyperparameters, $\pi(\theta_i | \mathbf{y})$.

In the distributed approach, instead of estimating the mode of $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$ directly, we can construct a Gaussian approximation of the mode of $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$ based on the Gaussian approximations at the modes of the individual posteriors $\tilde{\pi}d(\boldsymbol{\theta} | \mathbf{y}_i)$. Thus, the Gaussian approximation of $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$ can be defined as:

$$\tilde{\pi}_G(\boldsymbol{\theta} | \mathbf{y}) = N(\boldsymbol{\mu}_{\boldsymbol{\theta}^*}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}^*}), \quad (8.18)$$

where $\boldsymbol{\mu}_{\boldsymbol{\theta}^*} = \boldsymbol{\theta}^*$ is the mode of $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$, and $\boldsymbol{\Sigma}_{\boldsymbol{\theta}^*}$ is the covariance matrix evaluated at the mode $\boldsymbol{\mu}_{\boldsymbol{\theta}^*}$ of the marginal posterior distribution.

The mode is obtained by minimizing $-\log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}))$ using finite difference methods:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}^*} = \boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \{-\log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}))\}$$

and the covariance matrix is computed as the inverse of the Hessian matrix of $-\log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}))$, that is, $\boldsymbol{\Sigma}_{\boldsymbol{\theta}^*}^{-1} = -\mathbf{H}_{\boldsymbol{\theta}^*}(\log \tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}))$, evaluated at the mode:

$$\mathbf{H}_{\boldsymbol{\theta}^*}(\log \tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})) = \left(\begin{array}{ccc} \frac{\partial^2}{\partial^2 \theta_1} \log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})) & \cdots & \frac{\partial^2}{\partial \theta_1 \partial \theta_m} \log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial \theta_m \partial \theta_1} \log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})) & \cdots & \frac{\partial^2}{\partial^2 \theta_m} \log(\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})) \end{array} \right) \bigg|_{\boldsymbol{\theta}^*},$$

where these second-order derivatives are also estimated by finite differences, since the expression for the marginal posterior of the hyperparameters, as defined in Eq. (8.2), is not analytically tractable.

This Gaussian approximation at the modal configuration of $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$ serves as the starting point for constructing a global approximation using the Gaussian approximations associated with the modes of the marginal posteriors from each partition, $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$:

$$\tilde{\pi}_G(\boldsymbol{\theta} \mid \mathbf{y}) \approx \prod_{i=1}^n \pi_G(\boldsymbol{\theta} \mid \mathbf{y}_i), \quad (8.19)$$

where the precision matrix associated with $\tilde{\pi}_G(\boldsymbol{\theta} \mid \mathbf{y})$ is computed as the sum of the precision matrices evaluated at the modal configuration of the marginal posterior of the hyperparameters for each partition. That is, the precision matrix is approximated as:

$$\mathbf{H}_{\boldsymbol{\theta}^*} \approx \sum_{i=1}^n \mathbf{H}_{\boldsymbol{\theta}_i^*}(\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)), \quad (8.20)$$

where $\mathbf{H}_{\boldsymbol{\theta}_i^*}(\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i))$ is the Hessian of $\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$ evaluated at the corresponding mode $\boldsymbol{\theta}_i^*$, which is specific to each data partition.

The mean for each partition is given by $\boldsymbol{\mu}_{\boldsymbol{\theta}_i^*} = \boldsymbol{\theta}_i^*$, as in previous cases where the Gaussian approximation has been applied. Therefore, the mean of the global Gaussian approximation $\tilde{\pi}_G(\boldsymbol{\theta} \mid \mathbf{y})$ is approximated by:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}^*} \approx \mathbf{H}_{\boldsymbol{\theta}^*}^{-1} \sum_{i=1}^n \mathbf{H}_{\boldsymbol{\theta}_i^*} \boldsymbol{\mu}_{\boldsymbol{\theta}_i^*}. \quad (8.21)$$

Consequently, the Gaussian approximation at the modal configuration of the hyperparameters is given by:

$$\tilde{\pi}_G(\boldsymbol{\theta} \mid \mathbf{y}) \approx \mathcal{N} \left(\left[\sum_{i=1}^n \mathbf{H}_{\boldsymbol{\theta}_i^*} \right]^{-1} \sum_{i=1}^n \mathbf{H}_{\boldsymbol{\theta}_i^*} \boldsymbol{\mu}_{\boldsymbol{\theta}_i^*}, \sum_{i=1}^n \mathbf{H}_{\boldsymbol{\theta}_i^*} \right) \quad (8.22)$$

expressed in terms of the mean and precision matrix. Based on this Gaussian approximation, we can construct the CCD and compute the weights Δ_k related to the integration as explained in Rue et al. (2009).

Once the integration scheme has been defined, with its associated support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$, we can compute the marginal distribution of the latent field parameters

using Eq. (8.5), which incorporates the conditional posterior of the latent field, the marginal of the hyperparameters, and the weights from the integration scheme defined in the distributed approach:

$$\tilde{\pi}_d(x_i | \mathbf{y}) = \sum_{i=1}^n \tilde{\pi}_G(x_i | \boldsymbol{\theta}^k, \mathbf{y}) \tilde{\pi}_d(\boldsymbol{\theta}^k | \mathbf{y}) \Delta_k \quad (8.23)$$

where $\tilde{\pi}_G(x_i | \boldsymbol{\theta}^k, \mathbf{y})$ denotes the marginal distribution of node x_i from the conditional posterior of the latent field.

Note that at this stage we can apply the low-rank correction of the mean via Variational Bayes (Van Niekerk and Rue, 2024) to the conditional posterior distribution $\tilde{\pi}_G(\mathbf{x} | \boldsymbol{\theta}, \mathbf{y})$, obtained after aggregating all information from the different data subsets. Alternatively, we could apply the low-rank correction to the conditional posterior distribution of each subset, i.e., $\tilde{\pi}_G(\mathbf{x} | \boldsymbol{\theta}, \mathbf{y}_i)$, before merging the information. It is also important to note that the constraints imposed on the latent field have already been incorporated into the conditional posterior distribution of the latent field during the step in which the marginal posterior of the hyperparameters is computed.

8.3.4 Marginal posterior distributions of the hyperparameters

The marginal posterior distributions of each hyperparameter, $\tilde{\pi}(\theta_i | \mathbf{y})$, can be computed by following different strategies, such as those presented in Rue et al. (2009) and Martins et al. (2013), which can be summarized into the following four approaches: (i) interpolation, (ii) asymmetric Gaussian interpolation, (iii) Laplace approximation, and (iv) the numerical integration-free algorithm.

The first strategy—interpolation—requires a grid exploration to obtain a sufficient number of evaluation points for interpolation. In particular, one can use the grid exploration described in Rue et al. (2009) and Martins et al. (2013), and, based on the set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ where the posterior density $\{\tilde{\pi}(\boldsymbol{\theta}^k | \mathbf{y})\}_{k=1}^K$ has been evaluated, construct an interpolant $I(\boldsymbol{\theta} | \mathbf{y})$ such that the marginalization over each dimension of the hyperparameters can be computed numerically:

$$\tilde{\pi}(\theta_j | \mathbf{y}) \approx \int_{\boldsymbol{\theta}_{-j} \in \boldsymbol{\Theta}_{-j}} I(\boldsymbol{\theta} | \mathbf{y}) d\boldsymbol{\theta}_{-j},$$

where the interpolant $I(\boldsymbol{\theta} \mid \mathbf{y})$ can be selected from among a wide variety of available interpolation methods—for example, inverse distance weighting interpolation (and its variants), nearest-neighbour interpolation, radial basis function interpolation, kernel interpolation, or barycentric interpolation. An alternative grid strategy for exploring the posterior distribution of the hyperparameters involves using Korobov lattices and low discrepancy sequences, as described in [Brown et al. \(2021\)](#).

The second strategy considers that, under a rescaling and rotation of the internal parameterization of the hyperparameters, given by $\boldsymbol{\theta}(\mathbf{z}) = \boldsymbol{\theta}^* + \mathbf{U}\boldsymbol{\Lambda}^{1/2}\mathbf{z}$ —where $\mathbf{U}$ and $\boldsymbol{\Lambda}$ are, respectively, the matrix of eigenvectors and the diagonal matrix of eigenvalues from the eigendecomposition of $\mathbf{H}_{\boldsymbol{\theta}^*}$ —we can employ an *asymmetric Gaussian approximation* over this reparameterization $\mathbf{z}(\boldsymbol{\theta})$. First, we define the function

$$f(\boldsymbol{\theta}) = \prod_{j=1}^J f_j(z_j(\boldsymbol{\theta})) \quad (8.24)$$

where each component function $f_j(z_j(\boldsymbol{\theta}))$ is defined as:

$$f_j(z_j(\boldsymbol{\theta})) \propto \begin{cases} \exp\left(-\frac{1}{2(\sigma_j^-)^2}z_j^2\right) & \text{if } z_j < 0 \\ \exp\left(-\frac{1}{2(\sigma_j^+)^2}z_j^2\right) & \text{if } z_j \geq 0 \end{cases}. \quad (8.25)$$

These functions $f_j(z_j(\boldsymbol{\theta}))$ can be interpreted as a Gaussian approximation capable of capturing the asymmetry of $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$. In each dimension of $\mathbf{z}(\boldsymbol{\theta}) = (z_1(\boldsymbol{\theta}), \dots, z_J(\boldsymbol{\theta}))$, a piecewise-defined function is used, with a Gaussian kernel on each side. Each side of the axis $z_j(\boldsymbol{\theta})$ has a different variance, allowing for a more flexible representation of the asymmetry of $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$ depending on the direction with respect to the coordinate center in the transformed space $\mathbf{z}(\boldsymbol{\theta})$. The scaling parameters (σ_j^-, σ_j^+) can be estimated during the optimization process used to find the mode of Eq.(8.2). Once these scaling parameters are obtained, the marginal posteriors $\tilde{\pi}(\theta_j \mid \mathbf{y})$ can be computed via numerical integration of Eq.(8.24).

Nevertheless, these numerical integration approaches require evaluating Eq.(8.2)—or an approximation thereof, as in the second strategy—a considerable number of times, making both strategies computationally expensive, particularly the first. An alternative to these two interpolation-based strategies would be to replace the grid-based exploration, which typically implies a sequential search strategy, with

direct sampling from the Gaussian approximation at the mode. This would allow for parallel evaluation of Eq.(8.2), thus significantly improving computational efficiency.

The third strategy consists in applying another Laplace approximation to compute the marginal distributions:

$$\tilde{\pi}(\theta_j | \mathbf{y}) = \frac{\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})}{\tilde{\pi}_G(\boldsymbol{\theta}_{-j} | \theta_j, \mathbf{y})} \Big|_{\boldsymbol{\theta}_{-j} = \boldsymbol{\theta}_{-j}^*}, \quad (8.26)$$

where $\boldsymbol{\theta}_{-j}$ is the modal configuration of $\tilde{\pi}(\boldsymbol{\theta}_{-j} | \theta_j, \mathbf{y})$. While this approach can yield accurate results, it requires finding the maximum of the $(J - 1)$ -dimensional function $\pi(\boldsymbol{\theta}_{-j} | \theta_j, \mathbf{y})$ for each value of θ_j , which does not scale efficiently with the dimensionality J of hyperparameter space.

The fourth strategy, the *numerical integration free algorithm*, is the default one in the R-INLA software, and consists in assuming the following approximation for the marginals

$$\tilde{\pi}(\theta_j | \mathbf{y}) \propto \begin{cases} \exp\left(-\frac{1}{2(\sigma_j^-)^2}\theta_j^2\right) & \text{if } \theta_j < 0 \\ \exp\left(-\frac{1}{2(\sigma_j^+)^2}\theta_j^2\right) & \text{if } \theta_j \geq 0 \end{cases}. \quad (8.27)$$

where, as in the asymmetric Gaussian interpolation, we need to compute the scaling parameters (σ_j^-, σ_j^+) , but now without interpolation. To accomplish this, we can leverage the following lemma from Rue et al. (2009):

$$-\frac{1}{2}(\theta_j, \mathbb{E}(\boldsymbol{\theta}_{-j} | \theta_j))\boldsymbol{\Sigma}^{-1} \begin{pmatrix} \theta_j \\ \mathbb{E}(\boldsymbol{\theta}_{-j} | \theta_j) \end{pmatrix} = -\frac{1}{2} \frac{\theta_j^2}{\Sigma_{jj}}, \quad (8.28)$$

which states that the values of the joint density of $\boldsymbol{\theta}$, considered as a function of θ_j and $\boldsymbol{\theta}_{-j}$, evaluated at the conditional mean $\mathbb{E}(\boldsymbol{\theta}_{-j} | \theta_j)$, behaves as the marginal distribution of θ_i . Using this result, we can explore both sides of θ_j for each hyperparameter and evaluate, at least, three points to estimate the curvature, i.e. the scaling parameters (σ_j^-, σ_j^+) .

This approach assumes that the joint marginal distribution of the hyperparameters behaves like its Gaussian approximation around the mode, which allows us to compute the conditional expectation and apply the previously stated lemma. This, in turn, implies that at each support point in the hyperparameter space, the Hessian with respect to the hyperparameters must be computed. In the distributed

setting, this requires calculating the Hessian separately for each partition and then applying Eq. (8.20) to obtain the overall Hessian at each support point.

8.4 Recursive inference

The recursive inference approach shares many conceptual steps with the previously described distributed framework. Building on the results already established, we now summarize the relevant sections from the distributed setting, adapting them to the recursive context. Our focus will be on the specific procedures and features that distinguish recursive inference, while referencing shared components only as needed. This allows us to concentrate more thoroughly on the unique aspects of recursive inference within the INLA methodology.

Therefore, as in the distributed framework, we will examine how to obtain the posterior distributions in the recursive approach: (i) the conditional posterior distribution of the latent field, $\pi(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{y})$; (ii) the joint marginal posterior distribution of the hyperparameters, $\pi(\boldsymbol{\theta} \mid \mathbf{y})$; (iii) the marginal posterior distribution of each latent field node, $\pi(x_i \mid \mathbf{y})$; and (iv) the marginal posterior distribution of each hyperparameter, $\pi(\theta_j \mid \mathbf{y})$.

8.4.1 Conditional posterior distribution of the latent field

The conditional posterior distribution of the latent field is a fundamental component for implementing recursive inference. As in Section 8.3.1 of distributed inference, we can distinguish between two scenarios: one with Gaussian likelihood and another with non-Gaussian likelihood. In the Gaussian case, there is an exact identity between the standard and recursive inferential approaches. In contrast, for non-Gaussian likelihoods, the recursive formulation yields an approximation.

In the Gaussian likelihood case, we can proceed analogously to Eq. (8.9), restructuring the expression to show that the prior of the latent field at the i -th step of the recursive inference is updated using the conditional posterior of the latent field from step $(i - 1)$:

$$\begin{aligned} \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta}) &= C(\boldsymbol{\theta}, \mathbf{y}) + \sum_{j=1}^i \log \pi(\mathbf{y}_j \mid \mathbf{A}_j \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \boldsymbol{\theta}_2) \\ &= C'(\boldsymbol{\theta}, \mathbf{y}) + \log \pi(\mathbf{y}_i \mid \mathbf{A}_i \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta}) \end{aligned} \quad (8.29)$$

where $C(\boldsymbol{\theta}, \mathbf{y})$ and $C'(\boldsymbol{\theta}, \mathbf{y})$ are normalizing constants ensuring equivalence.

Therefore, the explicit expression that establishes the equivalence between the recursive approach in Eq. (8.29) and the standard inferential approach for the full dataset $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, in the case of Gaussian likelihoods, is:

$$\begin{aligned}
 \log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2}(\boldsymbol{\tau} \cdot \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} - 2 \cdot \boldsymbol{\tau} \cdot \mathbf{y}^\top \mathbf{A} \mathbf{x}) - \frac{1}{2} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \\
 &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} \sum_{j=1}^n (\boldsymbol{\tau} \cdot \mathbf{x}^\top \mathbf{A}_j^\top \mathbf{A}_j \mathbf{x} - 2 \cdot \boldsymbol{\tau} \cdot \mathbf{y}_j^\top \mathbf{A}_j \mathbf{x}) - \frac{1}{2} \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \\
 &= C(\boldsymbol{\theta}, \mathbf{y}) - \frac{1}{2} \sum_{j=i}^n (\boldsymbol{\tau} \cdot \mathbf{x}^\top \mathbf{A}_j^\top \mathbf{A}_j \mathbf{x} - 2 \cdot \boldsymbol{\tau} \cdot \mathbf{y}_j^\top \mathbf{A}_j \mathbf{x}) \\
 &\quad - \frac{1}{2} \left(\sum_{m=1}^{i-1} (\boldsymbol{\tau} \cdot \mathbf{x}^\top \mathbf{A}_m^\top \mathbf{A}_m \mathbf{x} - 2 \cdot \boldsymbol{\tau} \cdot \mathbf{y}_m^\top \mathbf{A}_m \mathbf{x}) + \mathbf{x}^\top \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x} \right) \\
 &= C'(\boldsymbol{\theta}, \mathbf{y}) + \sum_{j=i}^n \log \pi(\mathbf{y}_j \mid \mathbf{A}_j \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta}) \\
 &= C^*(\boldsymbol{\theta}, \mathbf{y}) + \log \pi(\mathbf{y}_n \mid \mathbf{A}_n \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{n-1}, \boldsymbol{\theta})
 \end{aligned} \tag{8.30}$$

where $C(\boldsymbol{\theta}, \mathbf{y})$, $C'(\boldsymbol{\theta}, \mathbf{y})$ and $C^*(\boldsymbol{\theta}, \mathbf{y})$ are again constants ensuring equivalence. The conditional posterior distributions $\pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta})$ and $\pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{n-1}, \boldsymbol{\theta})$ serve as priors for the i -th and n -th steps, respectively, in the recursive framework. This remarks that the recursive conditional posterior is equal to the one obtained from the standard (non-recursive) approach and that it is invariant to the ordering in which observations are introduced recursively.

In general, for non-Gaussian likelihoods, the conditional posterior at the i -th step of the recursive approach is approximated via a second-order Taylor expansion of the log-likelihood around the mode of the true conditional posterior $\pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta})$. This Gaussian approximation—used also in the standard inferential approach based on the full dataset—computes the mode $\boldsymbol{\mu}(\boldsymbol{\theta})$ of the conditional latent field as the minimizer of the negative log conditional posterior:

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}) \} = \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi(\mathbf{x} \mid \boldsymbol{\theta}) \}.$$

This is equivalent to stating that $\mathbf{x} = \boldsymbol{\mu}(\boldsymbol{\theta})$ is the point at which the gradient of the negative log conditional posterior is equal to zero:

$$\nabla_{\mathbf{x}} (-\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta}))|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} = -\nabla_{\mathbf{x}} \log \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} - \nabla_{\mathbf{x}} \log \pi(\mathbf{x} \mid \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} = \mathbf{0},$$

so that, for a prior distribution $\pi(\mathbf{x} \mid \boldsymbol{\theta}_2) = \text{GMRF}(\mathbf{0}, \mathbf{Q}(\boldsymbol{\theta}_2))$, the mode satisfies:

$$-\nabla_{\mathbf{x}} \log \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} + (\mathbf{Q}(\boldsymbol{\theta}_2)\mathbf{x})|_{\mathbf{x}=\boldsymbol{\mu}(\boldsymbol{\theta})} = \mathbf{0}. \quad (8.31)$$

This equation does not admit a closed-form solution, except for the case of Gaussian likelihoods, and must be solved using numerical optimization techniques. These may include Newton-type methods—which require explicit evaluation of the Hessian at each iteration (and thus benefit from known analytic expressions of it to accelerate convergence)—or quasi-Newton methods such as L-BFGS, which avoid the explicit computation of the Hessian. Other numerical optimization techniques can also be used. In particular, the R-INLA software employs Newton-Raphson methods that are enhanced with modifications to both the gradient and Hessian to improve convergence and exploration. These enhancements are implemented through the *Smart Gradient* and *Smart Hessian* strategies proposed in [Fattah et al. \(2022\)](#).

The expression for the Hessian, which corresponds to the precision matrix of the Gaussian approximation, is:

$$\begin{aligned} \mathbf{Q}_{\mathbf{x}|\mathbf{y},\boldsymbol{\theta}}(\boldsymbol{\theta}) &= \nabla_{\mathbf{x}}^2(-\log \pi(\mathbf{x} \mid \mathbf{y}, \boldsymbol{\theta})) \\ &= \mathbf{Q}_{\mathbf{x}|\boldsymbol{\theta}_2}(\boldsymbol{\theta}_2) - \nabla_{\mathbf{x}}^2 \log \pi(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}) \\ &= \mathbf{Q}_{\mathbf{x}|\boldsymbol{\theta}_2}(\boldsymbol{\theta}_2) - \mathbf{A}^\top \nabla_{\boldsymbol{\eta}}^2 \log \pi(\mathbf{y} \mid \boldsymbol{\eta}, \boldsymbol{\theta}) \mathbf{A} \end{aligned} \quad (8.32)$$

where $\boldsymbol{\eta} = \mathbf{A}\mathbf{x}$ is the linear predictor. This formulation simplifies the expression for the curvature of the likelihood since, under conditional independence, the likelihood factorizes as $\pi(\mathbf{y} \mid \boldsymbol{\eta}, \boldsymbol{\theta}) = \prod_{i=1}^n \pi(\mathbf{y}_i \mid \boldsymbol{\eta}_i, \boldsymbol{\theta})$, leading to a diagonal structure for the matrix $\nabla_{\boldsymbol{\eta}}^2 \log \pi(\mathbf{y} \mid \boldsymbol{\eta}, \boldsymbol{\theta})$, denoted by $\mathbf{D}$. This is particularly relevant when evaluating the conditional posterior at the mode.

In the recursive framework, these steps are carried out sequentially over the data partitions $\mathbf{y} = \mathbf{y}_1, \dots, \mathbf{y}_n$. At each step, the conditional prior for $\mathbf{x}$, used in the analysis of $\mathbf{y}_i$, is given by the conditional posterior from the previous step, $\pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta})$. Accordingly, the conditional posterior at step i is

approximated as:

$$\begin{aligned}
\log \pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta}) &= \log \mathbf{C}(\boldsymbol{\theta}, \mathbf{y}_1, \dots, \mathbf{y}_i) + \\
&\quad \nabla_{\mathbf{x}} \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})} \mathbf{x} + \\
&\quad \frac{1}{2} \mathbf{x}^\top \nabla_{\mathbf{x}}^2 \log \pi(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})} \mathbf{x} \\
&= \log \mathbf{C}(\boldsymbol{\theta}, \mathbf{y}_1, \dots, \mathbf{y}_i) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i(\boldsymbol{\theta}))^\top \mathbf{Q}_i(\boldsymbol{\theta}) (\mathbf{x} - \boldsymbol{\mu}_i(\boldsymbol{\theta})).
\end{aligned} \tag{8.33}$$

In this equation, the precision matrix $\mathbf{Q}_i(\boldsymbol{\theta})$ is updated recursively as:

$$\mathbf{Q}_i(\boldsymbol{\theta}) = \mathbf{Q}_{i-1}(\boldsymbol{\theta}) + \mathbf{A}_i^\top \nabla_{\boldsymbol{\eta}}^2 \log \pi(\mathbf{y}_i \mid \boldsymbol{\eta}_i, \boldsymbol{\theta})|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})} \mathbf{A}_i = \mathbf{Q}_{i-1}(\boldsymbol{\theta}) + \mathbf{A}_i^\top \mathbf{D}_i(\boldsymbol{\theta}, \mathbf{y}_i) \mathbf{A}_i \tag{8.34}$$

which leads to the following explicit recurrence relation:

$$\mathbf{Q}_i(\boldsymbol{\theta}) = \mathbf{Q}(\boldsymbol{\theta}_2) + \sum_{j=1}^i \mathbf{A}_j^\top \mathbf{D}_j(\boldsymbol{\theta}, \mathbf{y}_j) \mathbf{A}_j. \tag{8.35}$$

The mean $\boldsymbol{\mu}_i(\boldsymbol{\theta})$, corresponding to the mode of the true conditional posterior, is also computed recursively:

$$\boldsymbol{\mu}_i(\boldsymbol{\theta}) = \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta}) \}. \tag{8.36}$$

which is equivalent to solving:

$$-\nabla_{\mathbf{x}} \log \pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) + \mathbf{Q}_{i-1}(\boldsymbol{\theta})(\mathbf{x} - \boldsymbol{\mu}_{i-1}(\boldsymbol{\theta})) = \mathbf{0}. \tag{8.37}$$

Alternatively, by defining the change of variable $\mathbf{x} = \mathbf{z} + \boldsymbol{\mu}_{i-1}(\boldsymbol{\theta})$, we obtain:

$$-\nabla_{\mathbf{z}} \log \pi(\mathbf{y}_i \mid \mathbf{z}, \boldsymbol{\theta}) + \mathbf{Q}_{i-1}(\boldsymbol{\theta}) \mathbf{z} = \mathbf{0}.$$

These expressions highlight that the recursive approach is, in general, not invariant under reordering of the data partitions. This is because the conditional posterior distribution computed for non-Gaussian likelihoods is approximated by a Gaussian approximation of the true conditional posterior. Although the mode of the Gaussian approximation coincides with that of the true posterior, the geometric information of the latter is not fully captured by the approximation. As a result, when computing the modal configuration in the recursive framework using Eq. (8.37),

the update does not preserve the same information and is not invariant, since the operation is not commutative.

This can be understood as follows. Suppose we have two datasets, $\mathbf{y}_1$ and $\mathbf{y}_2$. If we assume conditional independence given the latent field and the hyperparameters, $\pi(\mathbf{y}_1, \mathbf{y}_2 \mid \mathbf{x}, \boldsymbol{\theta}) = \pi(\mathbf{y}_1 \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{y}_2 \mid \mathbf{x}, \boldsymbol{\theta})$, then it follows that:

$$\pi(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta}) \propto \pi(\mathbf{y}_2 \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \mathbf{y}_1, \boldsymbol{\theta}) \propto \pi(\mathbf{y}_1 \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \mathbf{y}_2, \boldsymbol{\theta}),$$

which would imply that recursive inference is invariant to the order of the data. However, this invariance holds if and only if closed-form expressions for the posterior distributions are available—something that occurs exclusively under conjugacy of the model component distributions.

In the case of log-likelihoods that are not quadratic in $\mathbf{x}$ —that is, non-Gaussian likelihoods—we apply a Gaussian approximation, which means that we cannot fully capture the underlying geometry. The curvature, when evaluated at different modes, depends on the specific point in the latent field at which it is computed, something that does not occur with Gaussian log-likelihoods. That is, if we analyze the two datasets mentioned in the previous paragraph and consider the posterior under both possible processing orders, $\pi_{1 \rightarrow 2}(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta})$ and $\pi_{2 \rightarrow 1}(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta})$, we find that the geometric information preserved along each of these recursive paths differs.

Let us suppose that the modes under each ordering are:

$$\begin{aligned} \mathbf{x}_1^* &= \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y}_1 \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi(\mathbf{x} \mid \boldsymbol{\theta}) \}, \\ \mathbf{x}_{1 \rightarrow 2}^* &= \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y}_2 \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi_G(\mathbf{x} \mid \mathbf{y}_1, \boldsymbol{\theta}) \}, \\ \mathbf{x}_2^* &= \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y}_2 \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi(\mathbf{x} \mid \boldsymbol{\theta}) \}, \\ \mathbf{x}_{2 \rightarrow 1}^* &= \underset{\mathbf{x}}{\operatorname{argmin}} \{ -\log \pi(\mathbf{y}_1 \mid \mathbf{x}, \boldsymbol{\theta}) - \log \pi_G(\mathbf{x} \mid \mathbf{y}_2, \boldsymbol{\theta}) \}. \end{aligned}$$

These expressions can be reformulated using a function $\mathcal{L}(\mathbf{x})$ to explicitly represent the recursive order. We define $\mathcal{L}_i(\mathbf{x}) = \log \pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) + \log \pi(\mathbf{x} \mid \boldsymbol{\theta})$ and $\mathcal{L}_{i \rightarrow j}(\mathbf{x}) = \log \pi(\mathbf{y}_j \mid \mathbf{x}, \boldsymbol{\theta}) + \log \pi_G(\mathbf{x} \mid \mathbf{y}_i, \boldsymbol{\theta})$, allowing the modes to be expressed as:

$$\begin{aligned} \nabla_{\mathbf{x}} [\mathcal{L}_i(\mathbf{x})]_{\mathbf{x}=\mathbf{x}_i^*} &= \mathbf{A}_i^\top [\nabla_{\boldsymbol{\eta}} \log \pi(\mathbf{y}_i \mid \boldsymbol{\eta}, \boldsymbol{\theta})]_{\boldsymbol{\eta}=\mathbf{A}_i \mathbf{x}_i^*} + \mathbf{Q}(\boldsymbol{\theta}_2) \mathbf{x}_i^* = \mathbf{0}, \\ \nabla_{\mathbf{x}} [\mathcal{L}_{i \rightarrow j}(\mathbf{x})]_{\mathbf{x}=\mathbf{x}_{i \rightarrow j}^*} &= \mathbf{A}_j^\top [\nabla_{\boldsymbol{\eta}} \log \pi(\mathbf{y}_j \mid \boldsymbol{\eta}, \boldsymbol{\theta})]_{\boldsymbol{\eta}=\mathbf{A}_j \mathbf{x}_{i \rightarrow j}^*} + \mathbf{Q}_i(\boldsymbol{\theta}) \mathbf{x}_{i \rightarrow j}^* = \mathbf{0}. \end{aligned}$$

Meanwhile, the curvature—i.e., the precision matrix—depends on the order and is given by:

$$\begin{aligned}\mathbf{Q}_i(\boldsymbol{\theta}) &= \nabla_{\mathbf{x}}^2 [\mathcal{L}_i(\mathbf{x})]_{\mathbf{x}=\mathbf{x}_i^*} = \mathbf{A}_i^\top \nabla_{\boldsymbol{\eta}}^2 [\log \pi(\mathbf{y}_i \mid \boldsymbol{\eta}, \boldsymbol{\theta})]_{\boldsymbol{\eta}=\mathbf{A}_i \mathbf{x}_i^*} \mathbf{A}_i \\ \mathbf{Q}_{i \rightarrow j}(\boldsymbol{\theta}) &= \nabla_{\mathbf{x}}^2 [\mathcal{L}_{i \rightarrow j}(\mathbf{x})]_{\mathbf{x}=\mathbf{x}_{i \rightarrow j}^*} = \mathbf{A}_j^\top \nabla_{\boldsymbol{\eta}}^2 [\log \pi(\mathbf{y}_j \mid \boldsymbol{\eta}, \boldsymbol{\theta})]_{\boldsymbol{\eta}=\mathbf{A}_j \mathbf{x}_{i \rightarrow j}^*} \mathbf{A}_j + \mathbf{Q}_i(\boldsymbol{\theta})\end{aligned}$$

If the Gaussian approximation were invariant under a change in data ordering, then the modal configuration would necessarily remain the same, that is, $\mathbf{x}_{1 \rightarrow 2}^* = \mathbf{x}_{2 \rightarrow 1}^*$, and likewise, the curvature —i.e., the precision matrix— would also have to match: $\mathbf{Q}_{1 \rightarrow 2}(\boldsymbol{\theta}) = \mathbf{Q}_{2 \rightarrow 1}(\boldsymbol{\theta})$. These would also have to match the Gaussian approximation obtained from the joint analysis of both datasets, i.e., $\pi(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta}) = \text{GMRF}(\mathbf{x}_{12}^*, \mathbf{Q}_{12}(\boldsymbol{\theta}))$. In other words, the modal configuration should satisfy:

$$\mathbf{x}_{1 \rightarrow 2}^* = \mathbf{x}_{2 \rightarrow 1}^* = \mathbf{x}_{12}^*$$

and identically for the curvature:

$$\mathbf{Q}_{1 \rightarrow 2}(\boldsymbol{\theta}) = \mathbf{Q}_{2 \rightarrow 1}(\boldsymbol{\theta}) = \mathbf{Q}_{12}(\boldsymbol{\theta}).$$

If we assume that the modes coincide, i.e., $\mathbf{x}_{12}^* = \mathbf{x}_{1 \rightarrow 2}^* = \mathbf{x}_{2 \rightarrow 1}^*$, then it must hold that:

$$\begin{aligned}\mathbf{A}^\top \nabla_{\boldsymbol{\eta}}^2 [\log \ell_{12}(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A} \mathbf{x}_{12}^*} \mathbf{A} &= \mathbf{A}_2^\top \nabla_{\boldsymbol{\eta}}^2 [\log \ell_2(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_2 \mathbf{x}_{12}^*} \mathbf{A}_2 + \mathbf{A}_1^\top \nabla_{\boldsymbol{\eta}}^2 [\log \ell_1(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_1 \mathbf{x}_{12}^*} \mathbf{A}_1 \\ &= \mathbf{A}_1^\top \nabla_{\boldsymbol{\eta}}^2 [\log \ell_1(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_1 \mathbf{x}_{12}^*} \mathbf{A}_1 + \mathbf{A}_2^\top \nabla_{\boldsymbol{\eta}}^2 [\log \ell_2(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_2 \mathbf{x}_{12}^*} \mathbf{A}_2\end{aligned}$$

where $\ell_i(\mathbf{x}) = \pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta})$. However, this condition generally does not hold, as the modes $\mathbf{x}_1^*$, $\mathbf{x}_2^*$, and $\mathbf{x}_{12}^*$ are not equal in general. Therefore:

$$\begin{aligned}\nabla_{\boldsymbol{\eta}}^2 [\log \ell_1(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_1 \mathbf{x}_{12}^*} &\neq \nabla_{\boldsymbol{\eta}}^2 [\log \ell_1(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_1 \mathbf{x}_1^*}, \\ \nabla_{\boldsymbol{\eta}}^2 [\log \ell_2(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_2 \mathbf{x}_{12}^*} &\neq \nabla_{\boldsymbol{\eta}}^2 [\log \ell_2(\mathbf{x})]_{\boldsymbol{\eta}=\mathbf{A}_2 \mathbf{x}_2^*}.\end{aligned}$$

If the modes do not coincide, i.e., $\mathbf{x}_{12}^* \neq \mathbf{x}_{1 \rightarrow 2}^* \neq \mathbf{x}_{2 \rightarrow 1}^*$, then even if the curvature were identical, the approach would not be invariant under reordering—since the approximation to the conditional posterior of the latent field would differ in the mode location, and the procedure would therefore not be invariant under reordering of the data.

Nevertheless, although this confirms that invariance under data reordering is not generally guaranteed by the proposed methodology, it does not necessarily imply that

the resulting differences are substantial. In many practical scenarios, the divergence (or a suitable distance measure) between the approximations $\pi_{G,1 \rightarrow 2}(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta})$ and $\pi_{G,2 \rightarrow 1}(\mathbf{x} \mid \mathbf{y}_1, \mathbf{y}_2, \boldsymbol{\theta})$ may be sufficiently small, below a pre-specified threshold ε , to consider the procedure effectively invariant with respect to the ordering of the data partitions. This nearly invariance can be formally justified in asymptotic regimes where each partition $\mathbf{y}_i$ contains enough information relative to the dimension of the latent field. In particular, if the latent dimension grows slower than the information per partition, and the conditional prior is Gaussian and sufficiently regular, the Bernstein–von Mises theorem applies (Freedman, 1999; Vaart, 2007; Koers et al., 2025). Under these conditions, the conditional posterior distribution of the latent field converges to a Gaussian, and the recursive procedure becomes asymptotically invariant under permutations of the data order. However, this result does not hold universally and must be interpreted with care, as convergence to Gaussianity depends on both the model structure and the relative scaling between data size and parameter dimension. In high-dimensional or misspecified models, the classical Bernstein–von Mises approximation may fail, and order dependence may persist (Castillo and Nickl, 2014; Spokoiny, 2014; Bochkina, 2023).

8.4.2 Joint marginal posterior distribution of the hyperparameters

The next step is to compute the joint marginal posterior distribution of the hyperparameters, for which we follow the standard procedure by evaluating the functional expression for $\pi(\boldsymbol{\theta} \mid \mathbf{y})$ given in Eq. (8.2).

In the recursive approach, as in the distributed one, there are two alternatives for approximating the joint marginal posterior. One option is to directly use Eq.(8.2), computing:

$$\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y}) \propto \frac{\pi(\mathbf{y}_1, \dots, \mathbf{y}_n \mid \mathbf{x}, \boldsymbol{\theta})\pi(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_n, \boldsymbol{\theta})} \quad (8.38)$$

where $\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_n, \boldsymbol{\theta})$ denotes the conditional posterior distribution of the latent field computed through the recursive inference approach, as described in Eq. (8.33). In this case, model constraints Eq. (8.16) must be applied to the Gaussian approximation $\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_n, \boldsymbol{\theta})$.

An alternative approach for approximating the joint marginal posterior of the hyperparameters leverages the information already computed and defines a recursive

procedure to approximate $\pi(\boldsymbol{\theta} \mid \mathbf{y})$. In this case, the recursive expression is:

$$\tilde{\pi}_r(\boldsymbol{\theta} \mid \mathbf{y}) \propto \left[\prod_{i=1}^n \frac{\pi(\mathbf{y}_i \mid \mathbf{x}, \boldsymbol{\theta}) \pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta})}{\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta})} \right]_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})} \pi(\boldsymbol{\theta}) \quad (8.39)$$

where, for each conditional posterior distribution $\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta})$ appearing in the numerator and denominator, the constraints defined by Eq. (8.16) must also be applied.

8.4.3 Marginal posterior distribution of the latent field

The marginal posterior distribution of the latent field nodes is one of the key components of the INLA framework. To compute it, it is necessary to build an integration scheme for evaluating the marginal distributions using Eq. (8.5). A required step for constructing such a scheme is to find the mode of the marginal posterior distribution of the hyperparameters, typically through optimization techniques based on finite differences, since the distribution is not analytically differentiable. Once this mode is identified, one can carry out an exploration strategy (e.g., a grid or central composite design, CCD) of the hyperparameter space to obtain both the marginal distributions of the latent field nodes and the marginal posteriors of the hyperparameters.

In the corresponding section of the distributed framework, we discussed several ways to construct these integration schemes by leveraging the structure of the distributed inference itself. However, in the recursive setting, building an integration scheme directly from the recursive approach is unfeasible if an accurate localization of the mode is desired. While one could, in principle, use the mode estimated from the first data subset, in practice this is inadequate, as the mode will likely shift as new data are incorporated. Thus, the only viable option for a recursive design is to use the distributed strategy in a sequential fashion, i.e., applying it as new data subsets become available.

The construction of the integration scheme—particularly the CCD—can be based on the same strategy discussed for the distributed framework, using Eqs. (8.19) through (8.22). With these, for each dataset $\mathbf{y}_i$ corresponding to the i -th step of the recursive procedure, we can build a Gaussian approximation centered at the modal configuration of the hyperparameters for that specific dataset. That is, for each $\mathbf{y}_i$ we construct $\pi_G(\boldsymbol{\theta} \mid \mathbf{y}_i)$, and combine the different approximations to obtain

a modal approximation for the hyperparameters associated with the combined datasets $\{\mathbf{y}_1, \dots, \mathbf{y}_i\}$ as $\pi_G(\boldsymbol{\theta} \mid \mathbf{y}_1, \dots, \mathbf{y}_i) \approx \prod_{j=1}^i \pi_G(\boldsymbol{\theta} \mid \mathbf{y}_j)$, from which we can then build the appropriate integration scheme for that step of the recursive procedure.

Once the integration scheme is defined—for instance, a CCD scheme without loss of generality—with support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$, we can use either Eq. (8.38), using the conditional posterior of the latent field obtained via the recursive approach, or Eq. (8.39) to compute the marginal posterior distribution of the hyperparameters. After evaluating this posterior and having computed the conditional posterior of the latent field (also through the recursive method) at each support point, we can finally compute the marginal posterior distribution for each latent field node:

$$\tilde{\pi}_r(x_j \mid \mathbf{y}_1, \dots, \mathbf{y}_i) = \sum_{k=1}^K \tilde{\pi}_G(x_j \mid \mathbf{y}_1, \dots, \mathbf{y}_i, \boldsymbol{\theta}^k) \tilde{\pi}_r(\boldsymbol{\theta}^k \mid \mathbf{y}_1, \dots, \mathbf{y}_i) \Delta_k, \quad (8.40)$$

where $\tilde{\pi}_r(\boldsymbol{\theta}^k \mid \mathbf{y}_1, \dots, \mathbf{y}_i)$ is the marginal posterior of the hyperparameters at support point $\boldsymbol{\theta}^k$, and Δ_k are the integration weights. The conditional posterior of the latent field and its marginal at this stage incorporate both the constraints and the low-rank correction.

It is important to note that, due to changes in the integration support points, the information computed in the $(i-1)$ -th step of the recursive approach cannot, in general, be reused in step i . This is because both the conditional posterior of the latent field, $\pi_G(\mathbf{x} \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1}, \boldsymbol{\theta}^k)$, and the marginal posterior of the hyperparameters, $\tilde{\pi}_r(\boldsymbol{\theta}^k \mid \mathbf{y}_1, \dots, \mathbf{y}_{i-1})$, depend on the specific integration support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$, which may change as $\pi_G(\boldsymbol{\theta} \mid \mathbf{y}_1, \dots, \mathbf{y}_i)$ is recomputed as a product of the Gaussians $\pi_G(\boldsymbol{\theta} \mid \mathbf{y}_j)$.

However, what can be reused throughout the recursive process is the Gaussian approximation of the hyperparameter mode for each individual dataset, since this approximation is computed independently for each $\mathbf{y}_i$. Moreover, if the Gaussian approximation of the hyperparameter mode at step i is sufficiently close to that from step $(i-1)$, then the integration scheme, and all associated quantities (the conditional posterior of the latent field and the marginal posterior of the hyperparameters), can also be reused from the previous step.

8.4.4 Marginal posterior distribution of the hyperparameters

The marginal posterior distributions of each hyperparameter, $\tilde{\pi}(\theta_i | \mathbf{y})$, can be computed using one of four strategies developed for the distributed inference framework. These strategies can be grouped into two categories: those that rely on interpolation—namely, (i) interpolation and (ii) asymmetric Gaussian interpolation—and those that do not, such as (iii) Laplace approximation for each hyperparameter θ_j , and (iv) the numerical integration-free algorithm.

The first approach, interpolation, evaluates the joint posterior at a grid or a sufficiently dense set of points to properly explore the hyperparameter space and constructs an interpolant $I(\boldsymbol{\theta} | \mathbf{y})$ to approximate the marginal posteriors:

$$\tilde{\pi}(\theta_j | \mathbf{y}) = \int_{\boldsymbol{\theta}_{-j} \in \Theta_{-j}} I(\boldsymbol{\theta} | \mathbf{y}) d\boldsymbol{\theta}_{-j}.$$

where the interpolant $I(\boldsymbol{\theta} | \mathbf{y})$ can be chosen from various interpolation methods. Although accurate, this strategy is computationally expensive due to the large number of evaluations of $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$ required.

The second method, asymmetric Gaussian interpolation, improves upon the first by transforming the parameter space into a rotated and scaled system and using piecewise Gaussian kernels:

$$f_j(z_j(\boldsymbol{\theta})) \propto \begin{cases} \exp\left(-\frac{1}{2(\sigma_j^-)^2} z_j^2\right) & \text{if } z_j < 0 \\ \exp\left(-\frac{1}{2(\sigma_j^+)^2} z_j^2\right) & \text{if } z_j \geq 0 \end{cases},$$

which allows for better approximation of asymmetric posteriors. This approach still requires numerical integration but typically involves fewer evaluations than the standard interpolation method.

To mitigate the computational cost of interpolation-based methods, the third strategy applies a Laplace approximation to compute the marginal distribution of each hyperparameter θ_j , by conditioning on the mode of the remaining hyperparameters $\tilde{\pi}(\boldsymbol{\theta}_{-j} | \theta_j, \mathbf{y})$. While effective in low dimensions, this method becomes impractical in higher-dimensional settings due to the repeated optimization needed.

The fourth strategy, known as the numerical integration-free algorithm, is the default in R-INLA. It assumes an asymmetric Gaussian shape for each marginal and

estimates its curvature using a lemma from [Martins et al. \(2013\)](#), which relates conditional expectations to marginal behavior. This method requires minimal numerical evaluation and is particularly efficient in high-dimensional settings.

Among these, the most practical for recursive implementation are the (i) simple interpolation method and the (iv) numerical integration-free algorithm. The former is especially convenient when the posterior exploration is restricted to a low-dimensional subspace $\Theta_0 \subset \Theta$, which can be covered with a fixed, well-chosen set of support points $\{\theta^k\}_{k=1}^K$. On the other hand, the numerical integration-free algorithm offers the lowest computational cost among the alternatives, requiring only the Gaussian approximation at the mode and three evaluations along each side of every hyperparameter dimension.

8.5 Latent field and data splitting

In the previous two sections, we have presented in detail the distributed and recursive inference approaches, taking advantage of the INLA methodology. In both approaches, it is assumed that the data are divided into subsets, $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, but at no point have we specified how to divide the data in cases where all of them—or a large subset—are available from the beginning. Therefore, this section is devoted to detailing various strategies for automatically dividing the data, either by leveraging the structure of the latent field itself or by exploiting a substructure of the latent field, e.g., spatial, temporal, or spatio-temporal structures. Moreover, we will show how these partitions, derived from the latent field, also allow for a reduction of the latent field itself within each partition, thus decreasing the computational burden.

8.5.1 Reordering of sparse matrices and clustering approaches

There is a wide variety of strategies that can be followed to generate partitions. These strategies may be based on the same principles used for reordering sparse matrices or may employ different clustering techniques. We focus on these two broad families of methods—sparse matrix reordering and clustering—to develop algorithms for constructing partitions of the latent field and the data. Traditional

clustering methods will also be used to propose partitions of the data, from which partitions of the latent field can be derived.

The reordering of sparse matrices is essential for the efficient solution of linear systems and for matrix factorization, as it reduces the fill-in during factorization and thereby the computational cost (Yannakakis, 1981; Duff, 2017). In particular, the minimum fill-in problem seeks to reorder the rows and columns of a given sparse matrix such that the triangular factors (e.g., from Cholesky or LU decomposition) remain as sparse as possible. Several reordering techniques have been proposed to address this problem, including: Average Minimum Degree (AMD, Amestoy et al. 1996), Multiple Minimum Degree (MMD, Liu 1985), Nested Dissection (ND, George 1973; Gilbert and Tarjan 1986), and Bandwidth Reduction techniques such as the Reverse Cuthill-McKee (RCM) algorithm (Liu and Sherman, 1976), the Gibbs-Poole-Stockmeyer algorithm (Gibbs et al., 1976), and variations thereof (Luo, 1992). We focus particularly on nested dissection and bandwidth reduction methods, both in terms of their conceptual approach and their computational output, to propose partitions of the latent field based on the graph associated with a given random effect —e.g., spatial, temporal, or spatio-temporal structures. We may also rely on different clustering techniques to generate data partitions via a latent field partition, or vice versa, e.g., classical clustering methods such as k-means (Kaufman and Rousseeuw, 1990; Jain, 2010), k-medoids (Kaufman and Rousseeuw, 1990), or bisecting K-means (Kaufman and Rousseeuw, 1990), as well as other approaches such as hierarchical clustering (Murtagh and Contreras, 2012) and spectral clustering (Chung, 1997; Jianbo Shi and Malik, 2000; Von Luxburg, 2007). Moreover, partitioning can also be improved by integrating insights from community detection algorithms (Blondel et al., 2008; Fortunato, 2010; Fortunato and Hric, 2016), applied to the graph associated with the relevant component of the latent field, thereby leveraging network-based structures to inform the segmentation of the latent field.

8.5.1.1 Nested dissection

Nested dissection is a divide-and-conquer heuristic algorithm for graph partitioning that generates efficient elimination orderings by identifying and recursively removing subsets of nodes known as separators (George, 1973; Gilbert and Tarjan, 1986). A separator is a small set of vertices whose removal divides the graph into two or more disconnected, and ideally balanced, subgraphs. These subgraphs can then

be independently reordered and factorized, leading to a recursive decomposition of the problem. The result is a nested dissection tree, in which each separator explicitly defines the interface between smaller subproblems (Duff, 2017). This hierarchical structure is particularly advantageous for sparse Cholesky factorization, as it minimizes fill-in and naturally exposes parallelism: the disconnected subgraphs can be processed independently and in parallel, with synchronization only required at the separators (Monien and Schulze, 1997). Nested dissection is especially effective for structured meshes or grid-like graphs, such as those arising in finite element or finite difference discretizations of partial differential equations. Moreover, the quality of the separators has a significant impact on both computational cost and memory usage. Modern implementations often combine nested dissection with other heuristics or use graph coarsening techniques (e.g., in multilevel methods like METIS (Karypis and Kumar, 1998) or Scotch (Naumann and Schenk, 2012)) to improve partition quality and reduce preprocessing time. In the context of latent field modeling, nested dissection can be exploited not only to reduce computational complexity but also to construct meaningful domain-aware partitions, facilitating recursive or distributed inference in large-scale spatial or spatio-temporal models.

8.5.1.2 Bandwidth reduction

Bandwidth reduction algorithms aim to minimize the bandwidth of a sparse matrix, defined as the maximum distance from the diagonal at which nonzero entries occur (Duff, 2017). This is particularly beneficial for matrices associated with graph-based structures, as a reduced bandwidth can significantly improve computational efficiency in applications such as finite difference methods, graph Laplacians, and iterative or direct solvers. Two widely used algorithms in this context are: (i) the Reverse Cuthill-McKee (RCM, Liu and Sherman 1976) algorithm, a greedy method that reorders the matrix rows and columns (or graph vertices) to bring nonzero entries closer to the diagonal, often yielding excellent practical results with low implementation complexity; and (ii) the Gibbs-Poole-Stockmeyer (GPS, Gibbs et al. 1976) algorithm, a more sophisticated technique that constructs orderings based on level structures derived from breadth-first search, aiming to reduce both bandwidth and matrix profile. Notably, both RCM and GPS can be adapted to identify graph partitions, either by exploiting hierarchical level structures or defining graph cut-points. These insights can be directly leveraged for constructing latent field partitions, especially in models with spatial or spatio-temporal dependence represented through graph structures.

8.5.1.3 K-means and Bisecting K-means

K-means clustering is a widely used partitioning algorithm that iteratively assigns observations to one of k clusters so as to minimize the within-cluster sum of squared distances to cluster centroids, which are defined as the mean of each cluster (Kaufman and Rousseeuw, 1990; Bishop, 2019; Jain, 2010). While efficient, K-means is sensitive to outliers and requires the number of clusters to be predefined (Kaufman and Rousseeuw, 1990). Meanwhile, bisecting K-means, commonly used in text and hierarchical clustering, improves K-means by recursively applying it in a divisive (top-down) manner: starting with one cluster containing all data, it bisects the cluster that leads to the largest reduction in overall variance, thus producing a hierarchical binary tree of partitions (Savaresi and Boley, 2001). These two approaches are fast, intuitive, and well-suited for large datasets or pre-processing in multistage partitioning strategies. Therefore, they can be employed in conjunction with Voronoi diagrams to partition the latent field, either directly or based on the available observations, allowing for the generation of partitions in continuous spaces of one, two, or three dimensions.

8.5.1.4 Spectral Clustering

Spectral clustering leverages the eigenstructure of a similarity graph constructed from the data to detect complex, non-convex cluster shapes (Chung, 1997; Von Luxburg, 2007). It operates by computing a graph Laplacian from a similarity or adjacency matrix, and then embedding the data in a low-dimensional space using a subset of its eigenvectors (Chung, 1997; Bolla, 2013). Clustering is then performed (typically via K-means) in this spectral embedding space or by leveraging the Fiedler vector (Fiedler, 1973) to define a clustering algorithm. The method is rooted in spectral graph theory and offers strong theoretical foundations, including connections to normalized cuts and random walks (Bolla, 2013). Spectral clustering is particularly effective when the data lie on a manifold or when clusters are connected components in a graph, making it a powerful tool for analyzing high-dimensional or structured datasets, including spatial networks or discretized spatial continuous structures, and more complex spatio-temporal graphs. Therefore, this approach can be leveraged by constructing a graph from the precision matrix associated with the structured effect of the model (e.g., spatial, temporal, spatio-temporal, or other latent components). Once the sparsity pattern of the precision matrix is established, a normalized Laplacian can be derived, and spectral clustering methods can be applied via its

spectral decomposition. This framework accommodates both purely discrete models and those that bridge continuous and discrete domains—such as models based on splines, SPDE-FEM, or other interpolation-based representations—enabling the construction of meaningful partitions across a wide range of latent structures.

8.5.1.5 Hierarchical Clustering

Hierarchical clustering methods build a tree-like structure (dendrogram) that captures the nested grouping relationships among observations, either by successively merging or splitting clusters (Johnson, 1967; Kaufman and Rousseeuw, 1990; Xu and WunschII, 2005). The two main approaches are the agglomerative (bottom-up algorithms, Murtagh and Contreras 2012; Ezugwu et al. 2022), which start with each observation in its own singleton cluster and iteratively merge the closest pairs of clusters based on a defined linkage criterion (e.g., single, complete, or average linkage), and the divisive (top-down algorithms, Ezugwu et al. 2022), which start with all observations grouped into a single cluster and recursively split it into smaller clusters by optimizing a criterion such as the largest dissimilarity or the greatest reduction in variance. While agglomerative methods are more common due to their lower computational cost, divisive methods can be more effective in capturing global structure.

One of the main advantages of hierarchical clustering is that it does not require a pre-defined number of clusters, can handle clusters of varying shapes and densities, and provides a multiscale view of the data structure (Kaufman and Rousseeuw, 1990). It is especially useful when the data naturally exhibit nested or tree-like relationships and can be applied to dissimilarity matrices (Murtagh and Contreras, 2012), allowing for a broad range of distance metrics. Particularly, the divide hierarchical clustering approach, including bisecting K-means as a specific case, can be used to sequentially construct partitions by repeatedly applying some of the previously described methods.

8.5.2 Partitioning algorithms

This section presents several algorithms, building upon the matrix reordering and clustering techniques previously discussed, to perform the partitioning of the latent field and the data. These algorithms can be broadly classified into two categories: (i) partitioning of the latent field is guided by the structure of one of its components,

such as spatial, temporal, or other types of structure; and (ii) partitioning of the latent field is derived from an initial partition of the observations. The latter approach is particularly suitable in cases where the nodes of the latent field—on which the partitioning is based—are linked to the observations through a projection matrix, as in the case of SPDE-FEM, splines, or interpolation.

For all algorithms, we assume that the partitioning is performed exclusively based on a specific component of the model. That is, given a linear predictor defined as $\boldsymbol{\eta} = \mathbf{A}\mathbf{x}$, where $\mathbf{A} = (\mathbf{A}_1, \dots, \mathbf{A}_J)$ and $\mathbf{x}^\top = (\mathbf{x}_1^\top, \dots, \mathbf{x}_J^\top)^\top$, with each matrix $\mathbf{A}_i$ representing the projection matrix associated with the j -th component $\mathbf{x}_j$ of the latent field (e.g., spatial, temporal, spatio-temporal, or other effects), the partitioning is carried out by focusing solely on the structure of a particular component $\mathbf{x}_j$.

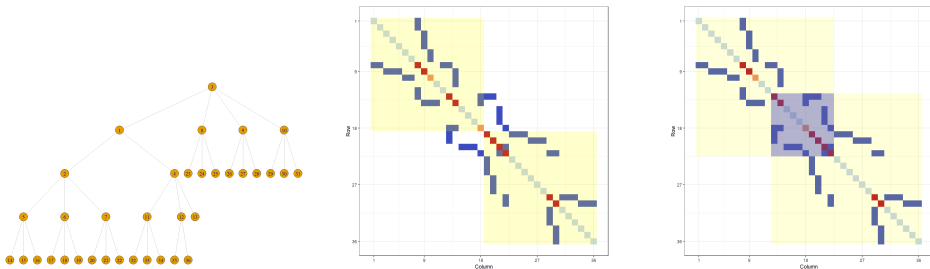
The purpose of partitioning extends beyond merely subdividing observations into manageable subsets for distributed or recursive inference; it also involves a structured decomposition of the latent field itself. This aspect is central to the design of the partitioning scheme and directly relates to Eqs. (8.9) and (8.30). Two principal strategies may be adopted: (i) a naïve approach that neglects correlations between adjacent nodes across partitions, or (ii) an extended formulation that incorporates such dependencies within each partition. The latter ensures that the global prior can be coherently reconstructed as the individual partitions are integrated in a distributed or recursive setting:

$$\pi(\mathbf{x} \mid \boldsymbol{\theta}) = \prod_{i=1}^n \pi(\mathbf{x}_i \mid \boldsymbol{\theta}),$$

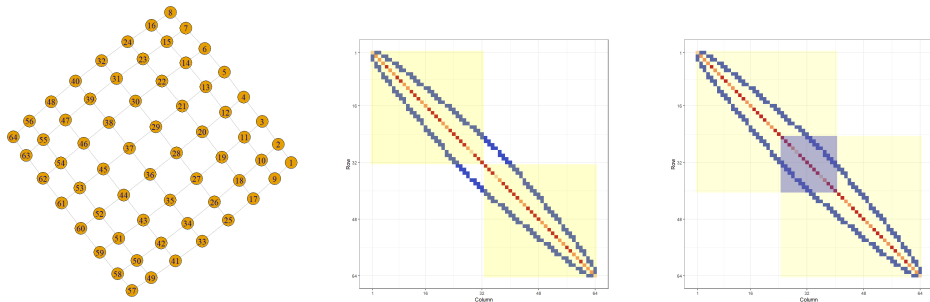
where the latent field $\mathbf{x} \mid \boldsymbol{\theta}$ is modeled as a GMRF with mean vector $\boldsymbol{\mu}$ (typically zero) and precision matrix $\mathbf{Q}$. Similarly, each partition-specific prior $\pi(\mathbf{x}_i \mid \boldsymbol{\theta})$ is a GMRF with mean $\boldsymbol{\mu}_i$ and precision matrix $\mathbf{Q}_i$.

In the particular case where $\boldsymbol{\mu}_i = \boldsymbol{\mu} = \mathbf{0}$, the global latent field remains a zero-mean GMRF with precision matrix $\mathbf{Q} = \sum_{i=1}^n \mathbf{Q}_i$, provided the components $\mathbf{Q}_i$ are suitably defined. Practically, the reconstruction of the precision matrix can be achieved either by enforcing strict block-diagonal structure in $\mathbf{Q}$, an approximated and naïve approach, thereby disregarding inter-block dependencies, or by allowing overlaps between partitions to preserve cross-boundary correlations, an exact approach. In the latter scenario, shared components of the matrices from the different partitions must be appropriately re-scaled to ensure exact recovery of the

original precision matrix. This concept is illustrated in Figure 8.1, which displays two example graphs and their corresponding Laplacian matrices. The left column shows the graph structures: a dendrogram (top-left) and a grid (bottom-left). The middle column presents the associated Laplacian matrices with disjoint blocks from the naïve approach highlighted in yellow, corresponding to each graph. The right column shows the same Laplacian matrices, now with the extended blocks from the naïve approach, where overlapping elements —rescaled by the number of partitions that shared those elements to preserve consistency— are highlighted in blue.



(A) Dendrogram example, with the graph (left), Laplacian matrix with partition highlighted in yellow (middle) and Laplacian matrix with the extended partition with overlapping highlighted in blue (right).



(B) Grid example, with the graph (left), Laplacian matrix with partition highlighted in yellow (middle) and Laplacian matrix with the extended partition with overlapping highlighted in blue (right).

FIGURE 8.1: Example with two graphs—a dendrogram and a grid—with the diagonal blocks in the Laplacian matrix and the extended blocks capturing the interactions between the original blocks, highlighting the common elements.

8.5.2.1 Algorithm 1: Partitioning by separators (nested dissection)

In this first algorithm, we build on the nested dissection approach, where identifying suitable separators is essential, based on the structure of the graph associated with the model component $\mathbf{x}_i$ used to define the partitioning. This component $\mathbf{x}_i$ is assigned a prior distribution modeled as a GMRF, $\mathbf{x}_i \sim \text{GMRF}(\boldsymbol{\mu}_i, \mathbf{Q}_i)$, where $\boldsymbol{\mu}_i$ is the mean—typically zero—and $\mathbf{Q}_i$ is the precision matrix.

The algorithm consists of five main steps: (i) identify graph separators, either from the precision matrix or the mesh structure; (ii) construct initial blocks for partitions and the separator set; (iii) assign observations and separator nodes to the corresponding partitions; (iv) build the block-diagonal structure; and (v) extend the blocks, and thus the partitions of the latent field component. Step (i) can be applied recursively, yielding a hierarchical partitioning algorithm based on nested dissection, where separators are identified at successive levels.

The first step of the algorithm focuses on constructing the graph, induced by the precision matrix $\mathbf{Q}$ or by the structure of a mesh (SPDE-FEM), and identifying separators—nodes that divide the graph into two or more subgraphs (Lipton and Tarjan, 1980; Alon et al., 1994; Goodrich, 1995; Har-Peled, 2011). These subgraphs can then be recursively partitioned by computing new separators within each subgraph, continuing this process until the desired number of partitions is achieved. This hierarchical procedure gives rise to a nested dissection design. In other words, the partitioning can be carried out either in a single step, by applying the separator rule once, or through a recursive, hierarchical process. In the latter case, the resulting structure can be represented as a tree, where each level corresponds to a stage in the application of the partitioning rule.

Once partitions and hierarchical separators are defined, observations and nodes can be consistently assigned to partitions, allowing construction of the diagonal and extended blocks of the precision matrix. In the discrete case, separator nodes are integrated into partitions using the dilation operation from mathematical morphology (Vincent, 1988; Heumans et al., 1992), after which observations align naturally with the latent field; blocks are then extended by including first-order neighbors. In the continuous case, observations are assigned to partitions through the projection matrix $\mathbf{A}_i$, with those linked only to separator nodes attached to the nearest partition; separator nodes are then distributed either by dilation or by observation assignments. As in the discrete case, extended blocks are obtained by

including first-order neighbors from $\mathbf{Q}_i$, ensuring correlations across partitions are properly captured.

This procedure is summarized schematically in Algorithm 3. Figure 8.1 illustrates two examples (grid and mesh) of the initial partitioning process together with their associated separators, which are later integrated into the partitions during the algorithm. The final step consists of assigning these separator nodes to specific partitions, ensuring that the resulting partitioning faithfully reproduces the precision matrix of the latent effect from which it was constructed.

Algorithm 3 Partitioning by separators (nested dissection)

Input: Precision matrix $\mathbf{Q}$ or mesh structure, latent field component $\mathbf{x}_i$.

STEP 0: Initialize the partitions and separator sets using the graph derived from the precision matrix or mesh.

while *desired number of partitions not reached* **do**

STEP 1: Identify separators of the induced graph from $\mathbf{Q}$ or mesh structure.
Optionally, apply recursively for hierarchical partitioning.

end

STEP 2: Construct initial blocks for the partitions and define the separator set.

STEP 3: Assign observations and separator nodes to the corresponding partitions derived from the initial blocks.

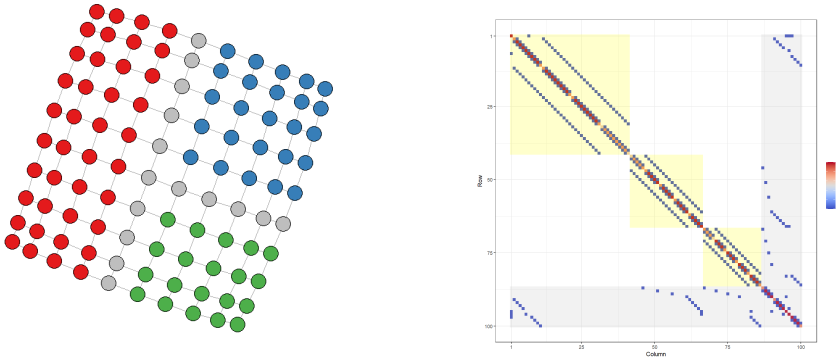
STEP 4: Build the block-diagonal structure of the precision matrix for the partitions.

STEP 5: Extend the blocks to incorporate neighboring correlations and finalize the partitions of $\mathbf{x}_i$.

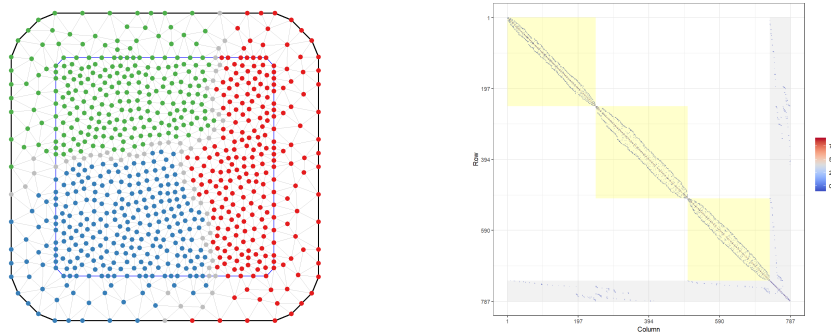
Output: Partitioned latent field and extended blocks.

8.5.2.2 Algorithm 2: Block-diagonal partitioning (bandwidth reduction)

The second algorithm generates a partition by exploiting the block-diagonal structure of the precision matrix through bandwidth reduction techniques. Unlike the previous approach, this method allows us to work directly with the precision matrix without explicitly constructing the graph or its associated Laplacian, although it can also be applied starting from the graph or the mesh in the SPDE-FEM setting.



(A) Graph (left) and Laplacian matrix (right) highlighting the partitions (red, green, and blue) and the set of separator nodes (grey) in a discrete spatial case.



(B) Graph (left), directly derived from the SPDE-FEM mesh, and its Laplacian matrix (right) in a continuous spatial case. Partitions are shown in different colors (red, green, and blue), while separator nodes are marked in grey.

FIGURE 8.2: Example of Algorithm 3 applied to build three partitions in two scenarios: (a) a discrete case, corresponding to a fixed-location network, and (b) a continuous case, based on the SPDE-FEM methodology. In the graphs, partitions are highlighted in colors and separators in grey, whereas in the Laplacian matrices partition blocks are marked with a yellow background and separator blocks with a grey background.

The algorithm consists of four main steps: (i) reordering the precision matrix, (ii) constructing diagonal blocks, (iii) assigning observations to the partitions derived from these blocks, and (iv) extending the blocks—and consequently the partitions. In addition, the procedure can be applied recursively to define a hierarchical partitioning algorithm based on bandwidth reduction.

The first step involves reordering the precision matrix using bandwidth reduction methods (Duff, 2017), such as the RCM algorithm (Liu and Sherman, 1976) or the GPS algorithm (Gibbs et al., 1976). Once reordered, the matrix can be split into diagonal blocks. The number and size of these blocks may be determined by the total number of nodes and their allocation per block, by the number of observations per block, or by other design or data-driven criteria that define the desired partitioning structure.

Once the blocks are defined, we can partition the latent field associated with the effect used for block construction. At this stage, the key task is assigning observations to partitions. In the discrete case, this assignment is straightforward, since each observation is uniquely linked to a single node of the latent field $\mathbf{x}_i$. In contrast, in the continuous case, some observations may be associated with nodes spanning multiple partitions. In such cases, observations can be distributed across partitions to ensure a balanced allocation of information.

As in the previous algorithm, the diagonal blocks and their induced partitions can be used directly to define the priors for each partition (skipping step iii), or they can be extended to include first-order neighbors. Without this extension, inter-block correlations would be lost and the original precision matrix could not be recovered. By extending the blocks, these correlations are preserved, although overlapping entries must be rescaled. This extension affects only the latent field related to the effect $\mathbf{x}_i$ driving the partition, without altering the assignment of observations to partitions.

Finally, the algorithm can be implemented hierarchically by applying it recursively. In this case, steps (i) and (ii) are repeated across levels until the desired number of blocks is reached, after which steps (iii) and (iv) are carried out. This recursive procedure yields a hierarchical partitioning structure based on bandwidth reduction.

Algorithm 4 Block-diagonal partitioning (bandwidth reduction)

Input: Precision matrix $\mathbf{Q}$ (or equivalently, graph/mesh structure in the SPDE-FEM setting)

STEP 0: Initialize permutation scheme for bandwidth reduction.

while *desired partitioning not reached* **do**

STEP 1: Reorder the precision matrix using a bandwidth reduction algorithm (e.g., Reverse Cuthill–McKee or Gibbs–Poole–Stockmeyer) to expose diagonal block structure.

STEP 2: Construct diagonal blocks by detecting contiguous sets of indices with weak or no coupling in the permuted $\mathbf{Q}$.

end

STEP 3: Assign observations to the partitions derived from these blocks, either by reordering $\mathbf{Q}$ or through the projection matrix $\mathbf{A}_i$.

STEP 4: Extend the blocks, and consequently the partitions, by including neighboring nodes to better capture correlations between original blocks.

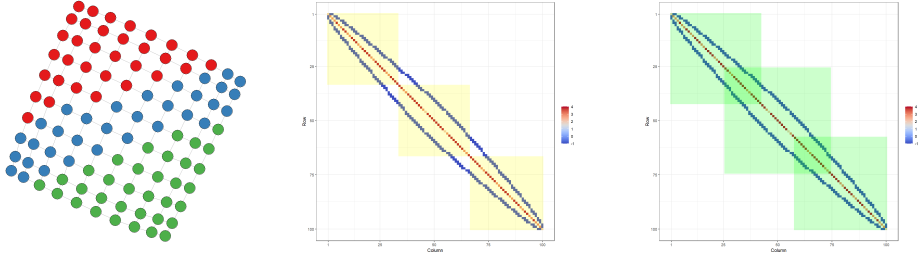
Output: Partitioned latent field and block-diagonal structure.

The procedure can be applied recursively, yielding a hierarchical partitioning algorithm based on bandwidth reduction.

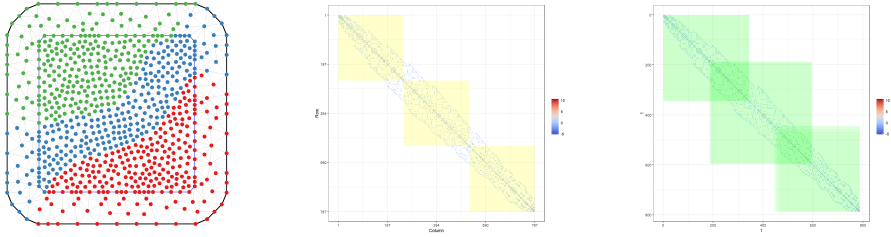
8.5.2.3 Algorithm 3: Spectral clustering for graph partitioning

The third algorithm relies on spectral clustering techniques applied either to the normalized Laplacian matrix associated with the graph induced by the precision matrix of the considered effect, or directly to the precision matrix itself, without explicitly constructing the Laplacian.

The algorithm proceeds in five steps: (i) spectral decomposition of the graph Laplacian, the normalized Laplacian, or the precision matrix; (ii) partitioning of nodes using either the ordered Fiedler vector or by applying K-means (or a similar method) to a subset of eigenvectors; (iii) identification of diagonal blocks in the precision matrix according to the node partitions; (iv) assignment of observations to the partitions defined by the node grouping; and (v) extension of the diagonal blocks—and consequently of the partitions of the latent field.



(A) Graph (left), precision matrix derived from the graph with block-diagonal partitions highlighted in yellow (middle), and precision matrix with extended block partitions highlighted in green (right), for a discrete case.



(B) Graph (left), precision matrix with block-diagonal partitions highlighted in yellow (middle), and precision matrix with extended block partitions highlighted in green (right), for a mesh and matrix derived from the SPDE-FEM approach.

FIGURE 8.3: Example of Algorithm 2 applied to construct three partitions in two settings: a discrete case with a fixed-location network, and a continuous case based on the SPDE-FEM methodology. The partitions in the graph are highlighted in different colors (blue, red, and green), while in the precision matrix the blocks corresponding to each partition are shaded in yellow, and the extended blocks are shaded in green.

As in the previous algorithm, steps (i) and (ii) can be applied recursively to obtain a hierarchical partitioning, which requires rebuilding the graphs (or selecting the corresponding sub-blocks of the precision matrix) at each stage.

In step (i), the spectral decomposition may be performed on the precision matrix, the normalized Laplacian of the equivalent graph of $\mathbf{Q}_i$, or the graph associated with the mesh in the SPDE-FEM framework. If the decomposition is carried out on the precision matrix, typically parameterized by a vector of hyperparameters $\mathbf{Q}_i(\boldsymbol{\theta}_i)$, specific values must be fixed. In some cases, however, it is possible to exploit the matrix structure to avoid fixing values explicitly, for example when hyperparameters affect all entries globally (e.g., marginal precision) or only the eigenvalue matrix.

In step (ii), let $\mathbf{M}$ denote the matrix on which the spectral decomposition is performed. Its decomposition is expressed as

$$\mathbf{M} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top,$$

where the columns of $\mathbf{U}$ are the eigenvectors and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues (assumed ordered from largest to smallest). Partitions can be defined either through the Fiedler vector—by sorting its entries and grouping nodes accordingly—or by applying clustering algorithms such as K-means or bisecting K-means to a subset of eigenvectors in $\mathbf{U}$.

In step (iii), the diagonal blocks corresponding to the node partitions are identified. Step (iv) assigns observations to these partitions. This assignment is straightforward in the discrete case, whereas in the continuous case, observations are allocated to the nearest partitions. Observations linked to nodes in multiple partitions via the projection matrix $\mathbf{A}_i$ are distributed among them according to a chosen rule, e.g., balancing the number of observations across partitions. Finally, in step (v), diagonal blocks are extended to include the neighbors of nodes within each partition.

To incorporate a hierarchical structure, after completing step (ii), steps (i)–(ii) are repeated on the induced subgraphs of each partition, recursively refining the decomposition. Once the last hierarchical level is reached, steps (iii)–(v) are carried out as described above.

This spectral clustering procedure is summarized in Algorithm 5. Figure 8.4 illustrates the outcome of applying the algorithm to a grid (discrete case) and to a mesh (continuous case). The figure displays the associated graphs, the precision matrices with block-diagonal partitions, and their extended blocks that capture correlations between original partitions. When overlaps occur between extended blocks, they must be rescaled by the number of partitions sharing those elements, as in the previous algorithms.

Algorithm 5 Spectral clustering for graph partitioning

Input: Graph Laplacian (standard or normalized) or a fixed precision matrix $\mathbf{Q}$.

STEP 0: Initialize number of partitions and choice of spectral method.

STEP 1: Perform spectral decomposition of $\mathbf{L}$ (graph Laplacian, normalized Laplacian, or $\mathbf{Q}$).

STEP 2: Partitioning of nodes using either the ordered Fiedler vector or clustering (e.g., K-means) on the first p eigenvectors.

STEP 3: Identify diagonal blocks in $\mathbf{Q}$ corresponding to the node partitions.

STEP 4: Assign observations to partitions defined by the node grouping (directly in discrete settings, or via projection matrices $\mathbf{A}_i$ if available).

STEP 5: Extend diagonal blocks by including neighboring nodes to capture local correlations, thereby refining the partitions of the latent field.

Output: Partitioned latent field with an extended block structure.

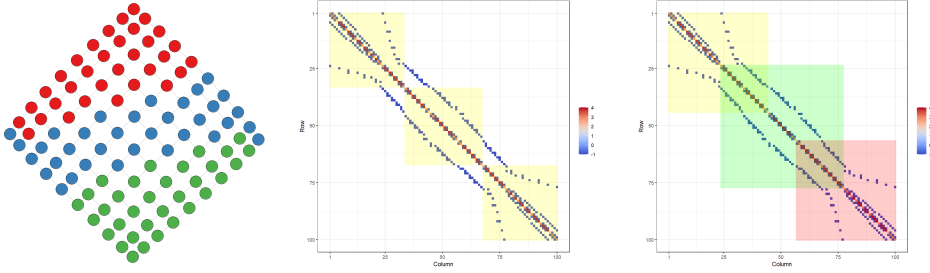
The procedure can be applied recursively, yielding a hierarchical partitioning algorithm based on bandwidth reduction.

8.5.2.4 Algorithm 4: Block partitioning via Voronoi tessellation

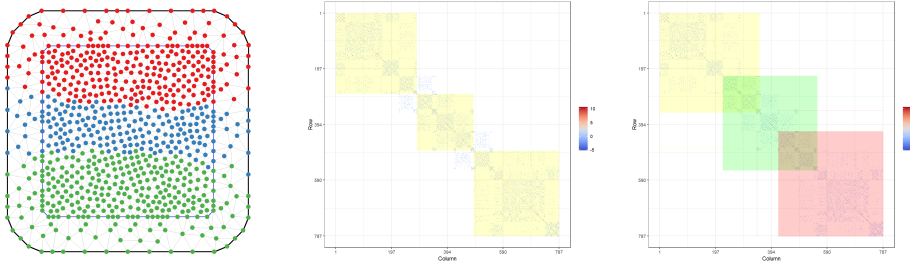
The final algorithm is primarily designed for cases where the effect to be partitioned is linked to a continuous space. Unlike the previous algorithms, this one begins by clustering the observations using a structure analogous to the latent field component to be partitioned—typically spatial, temporal, or spatio-temporal. Once the observations are clustered, a Voronoi diagram is constructed to assign the latent field nodes to each partition. This ensures a more balanced distribution of observations across the resulting partitions.

The algorithm proceeds in four steps: (i) partition the data using K-means, spectral clustering, a community detection algorithm, another clustering method, or a custom partition; (ii) construct a Voronoi diagram based on the centroids of the partitions derived from the observations; (iii) assign the latent field nodes to partitions according to the regions of the Voronoi diagram, which defines the block-diagonal structure in the precision matrix; and (iv) extend these blocks to capture correlations across neighboring partitions.

In the first step, clustering is applied directly to the observations using attributes relevant to the latent field component under consideration. For instance, when



(A) Graph (left), precision matrix derived from the graph with block-diagonal partitions highlighted in yellow (middle), and precision matrix with extended block partitions highlighted with different colors (right), for a discrete case.



(B) Graph (left), precision matrix with block-diagonal partitions highlighted in yellow (middle), and precision matrix with extended block partitions highlighted with different colors (right), for a mesh and its corresponding SPDE-FEM representation.

FIGURE 8.4: Example of Algorithm 3 applied to construct three partitions in two settings: a discrete case (network of fixed locations) and a continuous case based on the SPDE-FEM methodology. The graph partitions are shown in different colors (blue, red, and green), while in the precision matrix, blocks corresponding to each partition are highlighted with a yellow background, and extended blocks are highlighted with yellow, green and red backgrounds.

partitioning a spatio-temporal component, spatial and temporal coordinates of the observations can be used. Alternatively, spectral clustering may be applied to a distance matrix (e.g., spatial or spatio-temporal), potentially restricted to a fixed number of neighbors or a distance threshold, to define the neighborhood structure. This provides a partition of the observations from which a Voronoi tessellation can be constructed.

In the second step, the Voronoi diagram is built in the relevant space using the cluster centroids as seeds. The Voronoi tessellation restricts the approach in practice to spaces in $\mathbb{R}^1$, $\mathbb{R}^2$, or $\mathbb{R}^3$. In the third step, the latent field nodes are assigned to partitions according to their corresponding Voronoi region, thereby defining

the block-diagonal structure of the precision matrix. Finally, in the fourth step, the blocks are extended to include first-order neighbors, incorporating correlations between adjacent partitions.

A hierarchical structure can also be introduced in this framework by applying a hierarchical clustering algorithm in the first step, analogous to the recursive partitioning described in previous algorithms. The remaining steps then follow in the same way once the observation partitions are established.

The steps of this procedure are summarized in Algorithm 6, and its application to a set of observations on an SPDE-FEM mesh is illustrated in Figure 8.5. The figure shows how the algorithm partitions the spatial effect by assigning observation points within the continuous domain defined by the mesh. The block-diagonal partition retains only correlations within each partition, while the extended blocks incorporate neighboring elements across rows and columns, enabling the reconstruction of the original matrix from the extended structure of blocks.

Algorithm 6 Block partitioning via Voronoi tessellation

Input: Observations $\mathbf{y}$ with associated attributes (e.g., spatial, temporal, or spatio-temporal coordinates); latent field nodes $\mathbf{u}$; precision matrix $\mathbf{Q}$ (or mesh structure in the SPDE-FEM setting).

STEP 1: Partition the data. Apply K-means, spectral clustering, a community detection algorithm, another clustering method, or a custom partition to group the observations.

STEP 2: Construct Voronoi diagram. Use the centroids of the partitions obtained in Step 1 to build a Voronoi tessellation of the observation space (restricted to $\mathbb{R}^1$, $\mathbb{R}^2$, or $\mathbb{R}^3$).

STEP 3: Assign latent nodes. Map each node of the latent field $\mathbf{x}$ to the Voronoi region it belongs to. This induces the block-diagonal structure in the precision matrix $\mathbf{Q}$.

STEP 4: Extend the blocks. Enlarge each block by incorporating first-order neighboring nodes, thereby capturing correlations across adjacent partitions.

Output: Partitioned latent field with an extended block structure.

The procedure can be made hierarchical by applying a hierarchical clustering algorithm in Step 1, yielding nested partitions of the observations and corresponding refinements of the latent field partitioning.

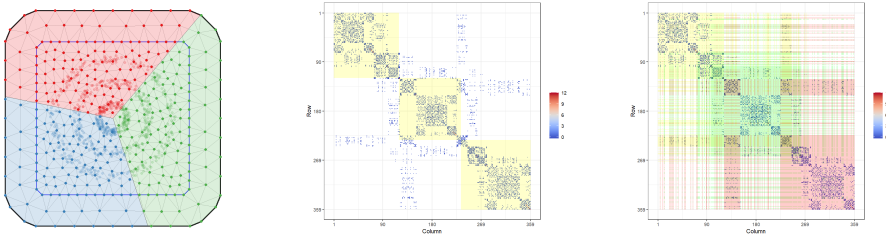


FIGURE 8.5: Example of Algorithm 4 applied to a continuous case using an SPDE-FEM mesh with observational data distributed in concentric circles. Graph partitions are highlighted in blue, red, and green; corresponding blocks in the precision matrix are shown in yellow, with their extended versions in yellow, green and red.

8.6 Implementing distributed and recursive inference with R-INLA

In this section, we present the algorithms used to implement the distributed and recursive inference methods described in Sections 8.3 and 8.4, leveraging the functionalities of the R-INLA package. The implementation of these algorithms can take advantage of the partitioning strategies introduced in Section 8.5, which allow the automatic construction of latent field partitions and help reduce computational costs.

A key aspect is that using R-INLA enables us to exploit the INLA methodology, which relies on highly efficient techniques for sparse linear algebra, numerical methods, and the automatic construction of statistical models in C++. However, this also constrains us to the set of functions currently implemented in the package, which may not always be optimal for directly applying distributed and recursive methodologies with maximum precision. In other words, achieving results as close as possible to simultaneous inference — particularly when integrating latent field partitioning of specific components, as outlined in Section 8.5 — would require developing the full procedure independently of R-INLA. Nevertheless, building upon this package significantly reduces the effort needed to implement the methods efficiently.

8.6.1 Distributed inference

The structure of the algorithm for implementing the distributed inference method described in Section 8.3 is presented in Algorithm 7. Once the partition has been defined, the algorithm can be summarized in the following steps:

1. identify the modal configuration of the marginal posterior distribution of the hyperparameters, leveraging Eq. (8.17);
2. construct the integration scheme and define the support points;
3. obtain, in a distributed manner, the conditional posterior distribution of the latent field and the marginal posterior distribution of the hyperparameters for each partition at each support point;
4. aggregate the conditional posterior distributions (after appropriate scaling) and reconstruct the joint marginal posterior distribution of the hyperparameters across partitions, for each support point;
5. compute the marginal posterior distribution at each support point.

In the R-INLA implementation, we assume that the partitioning of the data and the latent field has already been performed, based on one of the latent field components as described in Section 8.5, e.g., a spatial or spatio-temporal component. Thus, we work with the n partitions of the observations $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, the explanatory variables $\mathbf{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$, and the latent field $\mathbf{x} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$.

The partitioning of the latent field, and its role in distributed inference, can be carried out in different ways, as discussed in Section 8.3. One option is to appropriately rescale the precision matrix of the prior distribution of the latent field, such that $\pi(\mathbf{x} \mid \boldsymbol{\theta}_2) = \prod_{i=1}^n \pi(\mathbf{x}_i \mid \boldsymbol{\theta}_2)$, by properly defining the precision matrix for each $\pi(\mathbf{x}_i \mid \boldsymbol{\theta}_2)$. However, the approach we adopt is to perform the rescaling in the conditional posterior distribution of the latent field through an additional component $\pi(\mathbf{x}_0 \mid \boldsymbol{\theta}_2)$, so that

$$\pi(\mathbf{x} \mid \boldsymbol{\theta}_2) = \prod_{i=1}^n \pi(\mathbf{x}_i \mid \boldsymbol{\theta}_2) / \pi(\mathbf{x}_0 \mid \boldsymbol{\theta}_2).$$

While this adjustment does not affect the result under Gaussian likelihoods, it does alter the location of the modal configuration when using non-Gaussian likelihoods

using R-INLA, since the extra component cannot be integrated into the optimization routine and is therefore not used in locating the mode of the hyperparameters.

Another key aspect influencing the determination of the hyperparameter mode is the definition of their prior distributions. Here, the modal configuration is obtained as

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \{-\log \pi(\boldsymbol{\theta} \mid \mathbf{y})\}$$

and, following Eqs. (8.15) and (8.17), the prior in each partition must satisfy $\pi(\boldsymbol{\theta}) \propto \prod_{i=1}^n \pi_i(\boldsymbol{\theta})$. This can be implemented either by choosing from the available prior distributions or by specifying a user-defined prior through the `expression`, `table`, or `rprior` functionalities. Although the flexibility of prior specification varies across models, when possible, $\pi_i(\boldsymbol{\theta})$ can be approximated through an optimization procedure that identifies a function $f(\boldsymbol{\theta} \mid \boldsymbol{\alpha})$ such that

$$\boldsymbol{\alpha}^* = \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \{\mathcal{L}(\log f(\boldsymbol{\theta} \mid \boldsymbol{\alpha})), \log \pi(\boldsymbol{\theta})\}$$

where $\mathcal{L}$ is a loss function, e.g., Kullback–Leibler divergence, squared loss, etc.

Once the partitioning scheme and the model structure are defined, we can implement the first step described above on the servers or machines where the inference will be distributed. Each of the n partitions is assigned to a server or machine, where the function `inla(...)` is used. In this function, the `data` and `formula` arguments specify the data corresponding to that partition and the structure of the latent field, respectively, while the `family` argument defines the likelihood. The model specification must also include the necessary constraints to ensure identifiability within each partition. Moreover, since at this stage we are only interested in obtaining the modal configuration using the Gaussian approximation in the hyperparameter space, we explicitly set `only.hyperparam = TRUE` and `control.compute = list(config = TRUE)` in the function call.

```

1 # Modal configuration and Gaussian approximation at the mode of the hyperparameters
2 mod <- inla(data, family, formula, only.hyperparam = TRUE,
3           control.compute = list(config = TRUE))
4 # Mean and covariance matrix of the Gaussian approximation at the modal configuration of
   the hyperparameters
5 mu.GaussApprox <- mod$misc$theta.mode
6 cov.GaussApprox <- mod$misc$cov.intern

```

Distributing this operation across different servers allows us to obtain, for each partition, the modal configuration of the hyperparameters together with its Gaussian approximation. The mode and the covariance matrix at the modal configuration can be found in `mod$misc$theta.mode` and `mod$misc$cov.intern`, respectively,

both expressed in the internal parametrization used for the hyperparameters. Based on this approximation, we combine the information to derive an approximation of the Laplace approximation at the modal configuration of the hyperparameters for the complete dataset, as given by Eq. (8.22). Once this approximation is available, we can construct the integration scheme, i.e., the set of support points for the hyperparameters $\{\theta^k\}_{k=1}^K$. This scheme may be a CCD, which comes with its own design weights as described in Rue et al. (2009), or alternatively another integration design with corresponding weights can be used (Myers, 2016).

At this stage, we are in the third step of the algorithm: we already have the modal configuration of the hyperparameters θ^* and the integration scheme $\{\theta^k\}_{k=1}^K$. The next step is to compute the conditional posterior distribution of the latent field for each support point and for each partition. This can be parallelized across the servers or machines hosting the partitions. For this computation, two alternatives are available: (i) impose the identifiability constraints directly and later adjust the results by correcting the global intercept, or (ii) skip the constraints when computing the conditional posteriors, first combine the results across partitions for each support point, and then impose the constraints using *conditioning by kriging*. In practice, with R-INLA this is implemented by fixing the hyperparameter values (`theta.supp`) and computing only the conditional posterior distribution—without evaluating the marginals of the latent field—by setting `control.compute = list(return.marginal = FALSE)`.

```

1 # Conditional posterior for a fixed support point
2 cond <- inla(data, family, formula, control.mode = list(theta = theta.supp, fixed =
  TRUE),
3             control.compute = list(return.marginals = FALSE))
4 # Mean of the conditional posterior for the support point (theta.supp)
5 mu.cond <- cond$misc$configs$config[[1]]$improved.mean
6 # Fixing the upper triangular matrix to symmetric precision matrix of the conditional
  posterior
7 Q.cond <- fix.Q(cond$misc$configs$config[[1]]$Q)
```

The marginal posterior for each node of the latent field can be computed using Eq. (8.23), based on the conditional posterior distributions obtained at each support point and combined across partitions. Subsequently, the marginal posterior of the hyperparameters can be derived with the Gaussian approximation given by the Eq. (8.22) or by the *numerical integration free algorithm* leveraging Eqs.(8.27) and (8.28); the latter would need to compute the Hessian for the posterior of the hyperparameters at each support point which can be done leveraging the `inla(...)` function. With all marginals—both for the latent field and the hyperparameters—available, model fit criteria such as DIC, WAIC, and CPO can then be computed.

The marginal posterior for each node of the latent field can be computed using Eq.(8.23), based on the conditional posterior distributions obtained at each support point and subsequently combined across partitions. The marginal posterior of the hyperparameters can then be derived either through the Gaussian approximation given by Eq. (8.22) or using the *numerical integration free algorithm*, which relies on Eqs. (8.27) and (8.28). The latter requires computing the Hessian of the hyperparameter posterior at each support point, which can be obtained using the `inla(...)` function. Once all marginals—both for the latent field and for the hyperparameters—are available, model fit criteria such as DIC, WAIC, and CPO can be computed.

Algorithm 7 Distributed inference with R-INLA

Input: Partitioned data $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, explanatory variables $\mathbf{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$, and latent field $\mathbf{x} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The partition can be defined with respect to a component of the latent field.

for each partition do

STEP 1: Identify modal configuration. For each partition, use R-INLA to obtain the mode $\boldsymbol{\theta}^*$ of the marginal posterior of the hyperparameters, leveraging Eq. (8.17). Store the Gaussian approximation at the mode.

STEP 2: Construct the integration scheme. Combine the Gaussian approximations across partitions to approximate the Laplace approximation at $\boldsymbol{\theta}^*$ for the full dataset. Define the set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ (e.g., CCD or another design with associated weights).

for each support point do

STEP 3: Compute conditional posteriors. Compute the conditional posterior of the latent field in parallel across servers/machines. Constraints may be imposed directly, or added later through conditioning by kriging.

end

end

STEP 4: Aggregate across partitions. For each support point, combine the conditional posterior distributions (after appropriate scaling) and compute the joint marginal posterior of the hyperparameters.

STEP 5: Compute marginals. Using Eq. (8.23), the marginal posterior of each node of the latent field can be obtained. The marginal posteriors of the hyperparameters can then be computed either via the Gaussian approximation or through the numerical integration-free algorithm.

Output: Marginal posterior distributions of the latent field and hyperparameters.

8.6.2 Recursive inference

The structure of the algorithm for implementing the recursive inference method described in Section 8.4 is presented in Algorithm 8. In the case of recursive inference, the procedure follows a similar strategy to that of distributed inference, but with two key differences. First, the recursive procedure is inherently sequential and is typically executed on the same server or machine. Second, when defining the integration scheme or the support points, two approaches can be considered: (i) *a Gaussian update approach*, identical to that used in distributed inference, where a Gaussian approximation at the modal configuration is computed for each partition and combined to update the Gaussian approximation of the hyperparameters across all data; or (ii) *a high-density region update approach*, in which a wider region is constructed with an increased number of support points to track the evolution of the marginal posterior of the hyperparameters. This second approach starts from the Gaussian approximation at the mode of the first partition and adapts the integration scheme as the modal configuration drifts from its initial location, increasing the number of support points in certain directions when necessary.

Assuming that the data and latent field have already been partitioned, the recursive algorithm can be summarized as follows:

1. identify the modal configuration of the marginal posterior of the hyperparameters for each partition, or define an extended region to allocate the support points;
2. construct the integration scheme and determine the support points;
3. compute, recursively, the conditional posterior of the latent field and the marginal posterior of the hyperparameters for each partition at each support point, reusing information from the corresponding posteriors of previous partitions;
4. aggregate the conditional posteriors (after appropriate scaling if required) and reconstruct the joint marginal posterior of the hyperparameters across partitions for each support point;
5. evaluate the marginal posterior distribution at each support point.

The first step of the recursive algorithm is to select the approach for defining the support points: (i) the *Gaussian update approach* or (ii) the *high-density region*

update approach. The Gaussian update is identical to that of the distributed method: the modal configuration and its Gaussian approximation are computed for each partition, which allows reusing this Gaussian approximation throughout the recursive algorithm. However, the conditional posterior of the latent field and the marginal posterior of the hyperparameters must be recalculated at each step. In contrast, the high-density region approach defines a wider region starting from the Gaussian approximation at the mode of the first partition, allocates a larger number of support points, and monitors whether the support point associated with the mode drifts away from its initial location. This makes it possible to update the conditional and marginal posteriors at the support points without revising the integration scheme until the mode has shifted beyond a predefined threshold.

Once the integration scheme has been established, the conditional posterior of the latent field and the marginal posterior of the hyperparameters are computed by reusing information from previous partitions within the recursive framework. For the recursive computation of the conditional posterior using R-INLA, the model formula is reconstructed at each support point, including only the GMRF corresponding to the conditional posterior obtained from the analysis of previous partitions, while keeping the likelihood hyperparameters fixed. In this context, one may either use the conditional posterior without constraints and impose them later when computing the marginals, or alternatively impose the constraints directly during the computation. For the marginal posterior, the marginal likelihood provided by the `inla(...)` output is used. When evaluated at each support point—i.e., with all hyperparameters fixed—this likelihood is equivalent to the marginal posterior of the hyperparameters under the method employed by R-INLA, and it is subsequently corrected in the successive steps of the recursive algorithm. A simplified version of this model update at each support point is illustrated in the following code snippet:

```

1 # Analysis of the first partition
2 rec1 <- inla(data, family, formula = form_rec1,
3             control.mode = list(theta = theta.supp, fixed = TRUE),
4             control.compute = list(return.marginals = FALSE))
5 marg.hyper <- marg.hyper + rec1$mlik[1]
6 # Fixing the upper triangular precision matrix Q
7 Q1 <- fix.Q(rec1$misc$configs$config[[1]]$Q)
8 mul <- r1$misc$configs$config[[1]]$improved.mean
9 # Extracting constraints
10 A.constr <- rec1$misc$configs$constr$A
11 e.constr <- rec1$misc$configs$constr$e
12 # Modify the formula for the second step of the recursive algorithm
13 form_rec <- y ~ offset(mul) +
14             f(idx.latent_GMRF, model = "generic0", Cmatrix = Q1,
15             hyper = list(prec = list(initial = 0, fixed = TRUE)),
16             extra.constr = list(A = A.constr, e = e.constr))
17 rec2 <- inla(data, family, formula = form_rec2,
18             control.mode = list(theta = theta.supp, fixed = TRUE),
19             control.compute = list(return.marginals = FALSE))
20 # Correcting the density from the "generic" model
21 marg.hyper <- marg.hyper + rec2$mlik[1] + 0.5*determinant(Q1)$modulus

```

Finally, as in the distributed case, once the conditional posterior distributions of the latent field and the marginal posterior distributions of the hyperparameters have been obtained, it is possible to compute the marginal distributions for each node of the latent field and for each hyperparameter, as well as the model fit criteria.

Algorithm 8 Recursive inference with R-INLA

Input: Partitioned data $\mathbf{y} = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, explanatory variables $\mathbf{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$, and latent field $\mathbf{x} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The partition can be defined with respect to a component of the latent field.

if *Gaussian update approach* **then**

while *i-th partition* $\leq$ *number partitions* **do**

STEP 1: Identify modal configuration. For the *i*-th partition, use R-INLA to obtain the mode $\boldsymbol{\theta}^*$ of the marginal posterior of the hyperparameters and store the Gaussian approximation at the mode.

STEP 2: Construct the integration scheme. Combine the Gaussian approximations across partitions to approximate the Gaussian approximation at $\boldsymbol{\theta}^*$. Define the set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ (e.g., CCD or another design with associated weights).

for each support point do

STEP 3: Compute conditional posteriors. Compute the conditional posterior of the latent field in parallel across support points, reusing the conditional posterior from the $(i - 1)$ -th step as the prior for the latent field. Constraints may be imposed directly, or added later through conditioning by kriging.

end

end

end

else if *high-density region update approach* **then**

while *i-th partition* $\leq$ *number partitions* **do**

STEP 1: Identify high density region. For the first partition, use R-INLA to obtain the Gaussian approximation at the mode $\boldsymbol{\theta}^*$ of the marginal posterior of the hyperparameters and define a high density region. This region is updated as needed in subsequent steps.

STEP 2: Construct the integration scheme. Define the set of support points $\{\boldsymbol{\theta}^k\}_{k=1}^K$ to track changes in the marginal posterior of the hyperparameters, computed by Eq. (8.39).

for each support point do

STEP 3: Compute conditional posteriors. Compute the conditional posterior of the latent field in parallel across support points, reusing the conditional posterior from the $(i - 1)$ -th step as the prior for the latent field. Constraints may be imposed directly, or added later through conditioning by kriging.

end

end

end

STEP 4: Compute marginals. Using Eq. (8.40), compute the marginal posterior of each node of the latent field. Then, apply the interpolation strategy, the Gaussian approximation or the *numerical integration free algorithm* to obtain the marginal posterior of the hyperparameters.

Output: Marginal posterior distributions of the latent field and hyperparameters.

8.7 Examples

In this section, we illustrate the methodology described in Sections 8.3 and 8.4, which present the foundations of the distributed and recursive approaches, as well as Section 8.6, which details their implementation using the R-INLA software. We also demonstrate the implementation of the algorithms introduced in Section 8.5 for generating partitions based on components of the latent field. In particular, three implementation examples are presented: the first is a simple simulated spatio-temporal case with Gaussian likelihood, used to illustrate the application of the distributed and recursive inferential methods. The second is a real-data example for PM_{2.5} pollutant levels in the United States, employing a more complex spatio-temporal model with space–time interaction and a non-Gaussian likelihood. Finally, the third example demonstrates the implementation of the distributed and recursive procedures for wildfire data in Spain, modeled through a log-Gaussian Cox process (LGCP).

8.7.1 Distributed and recursive inference for a simple spatio-temporal model

In this first example, we simulate a spatio-temporal dataset to illustrate, in a controlled and simple setting, the implementation of the previously described procedures for distributed and recursive inference.

The model used for data simulation is defined as follows:

$$\begin{aligned} \mathbf{y} &\sim \text{MVN}(\boldsymbol{\mu}, \tau \mathbf{I}), \\ \boldsymbol{\mu} &= \beta_0 \mathbf{1} + \beta_{cov} \mathbf{x}_{cov} + \boldsymbol{\beta}_{cat} \mathbf{X}_{cat} + \mathbf{u}_s + \mathbf{u}_t, \end{aligned} \tag{8.41}$$

where β_0 denotes the intercept, β_{cov} is the coefficient related to the $\mathbf{x}_{cov}$ covariate values vector, $\boldsymbol{\beta}_{cat} \mathbf{u}_s$ is a spatial component simulated using the SPDE–FEM approach, and $\mathbf{u}_t$ is a temporal component simulated from a first-order random walk.

In the simulation, 300 spatial locations were randomly distributed within the study region across 60 temporal nodes (years). For the simulation, the values used for the different parameters are as follow: $\beta_0 = 2$, $\beta_{cov} = 1.5$ and $\boldsymbol{\beta}_{cat} = (-1.92, 2.28, -0.36)$ are the linear coefficients for the intercept, covariate and the

levels of the categorical variable. $\tau_G = 20$ is the precision of the Gaussian likelihood, $\tau_{rw} = 20$ is the precision of the random walk, and $\sigma_{sp} = 2$ and $\rho_{sp} = 0.4$ are the marginal standard deviation and spatial range of the spatial effect (SPDE-FEM), respectively. Figure 8.6 shows the spatial effect, the temporal effect, and the spatial patterns of both the covariate and the categorical variable.

The inference was carried out using the same model as in the simulation. The distributed and recursive approaches were implemented by dividing the data into two partitions, obtained by partitioning the temporal precision matrix of the prior distribution using Algorithm 4. The CCD integration scheme derived for both approaches is presented in Fig. 8.7, where the results are shown to be in close agreement with those obtained from the standard full-data analysis.

The marginal distributions for the fixed effect and the covariate are displayed in Fig. 8.8, while Fig. 8.9 reports the corresponding results for the levels of the categorical variable. The temporal trend, together with its 95% credible intervals, is presented in Fig. 8.10. In turn, Fig. 8.11 illustrates the posterior summaries of the spatial effect, including the posterior mean, standard deviation, and the 0.025 and 0.975 quantiles.

Finally, the marginal posteriors of the hyperparameters for the distributed and recursive approaches were computed using the Gaussian approximation at the mode. These results are compared in Fig. 8.12 with those obtained from the standard full-data analysis, where the marginals were calculated using the *numerical integration free algorithm*. It is important to note, however, that applying the *numerical integration free algorithm* directly through the `inla(...)` function is neither straightforward nor computationally efficient.

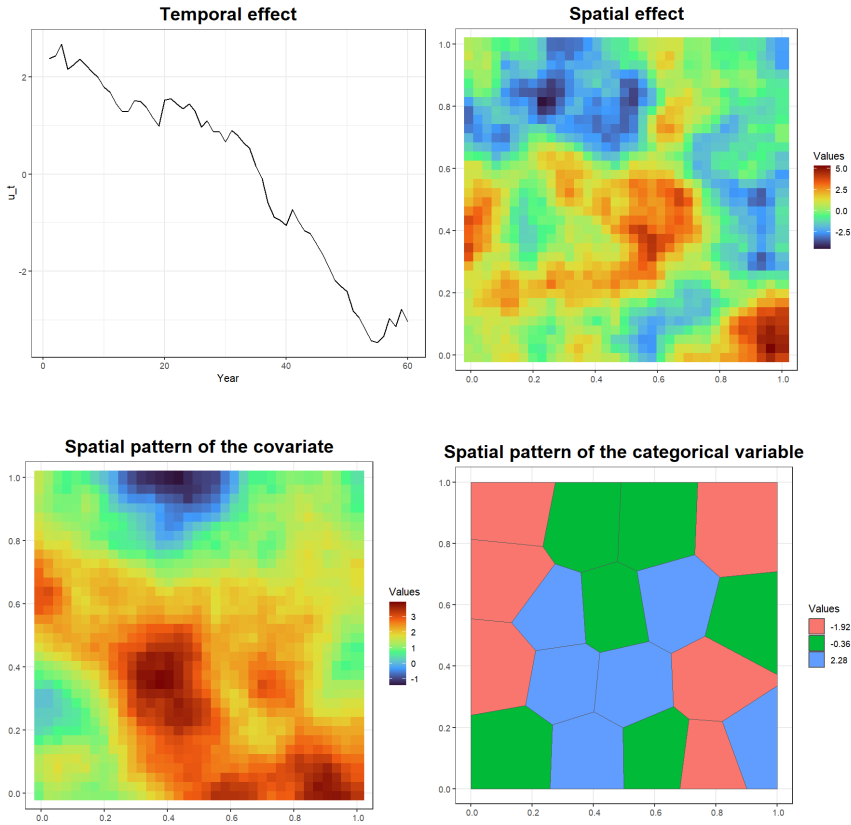


FIGURE 8.6: Plots of the temporal and spatial effects, together with the spatial patterns of the covariate and the categorical variable.

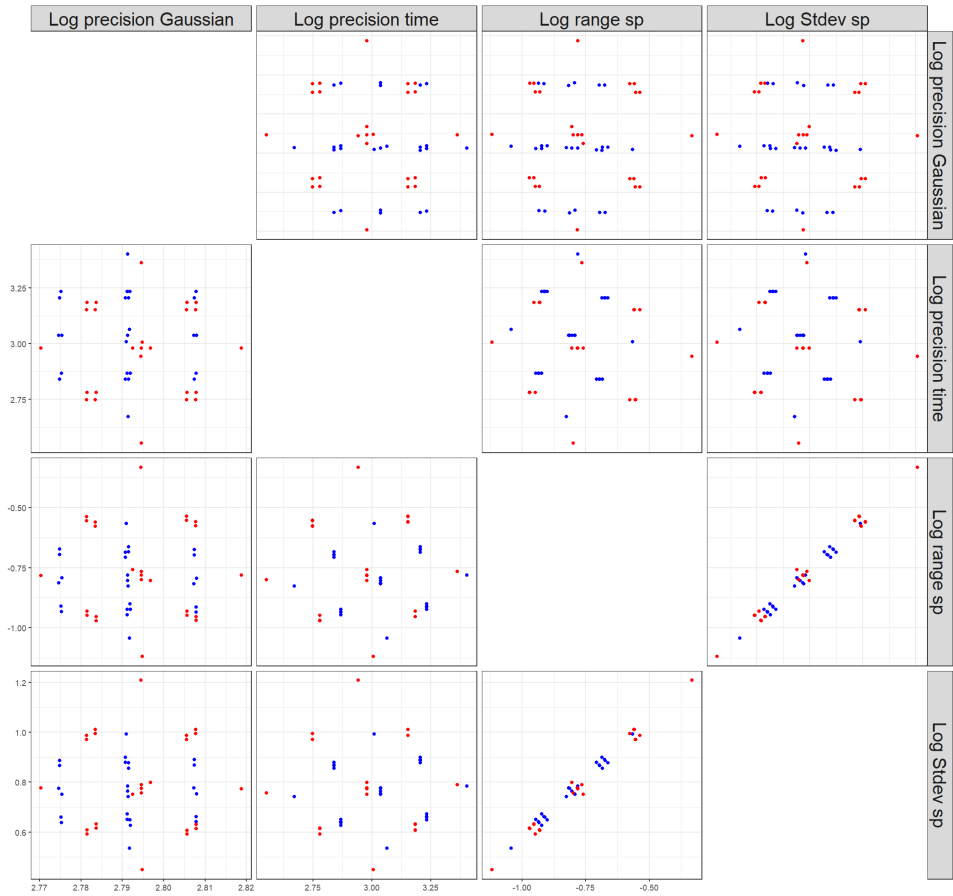


FIGURE 8.7: CCD for the full data inference (red) and from the combination of the distributed inference (blue).

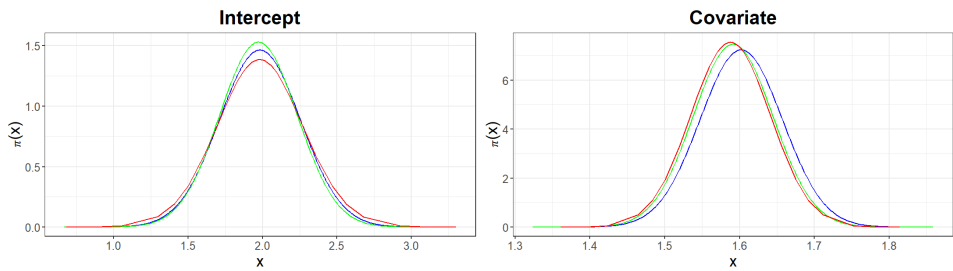


FIGURE 8.8: Posterior distributions of the intercept and covariate parameters from the full-data analysis (red), distributed approach (blue), and recursive approach (green).

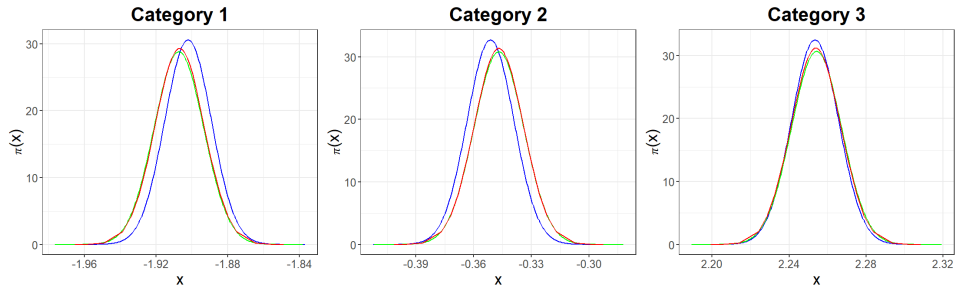


FIGURE 8.9: Posterior distributions of the categorical variable parameters from the full-data analysis (red), distributed approach (blue), and recursive approach (green).

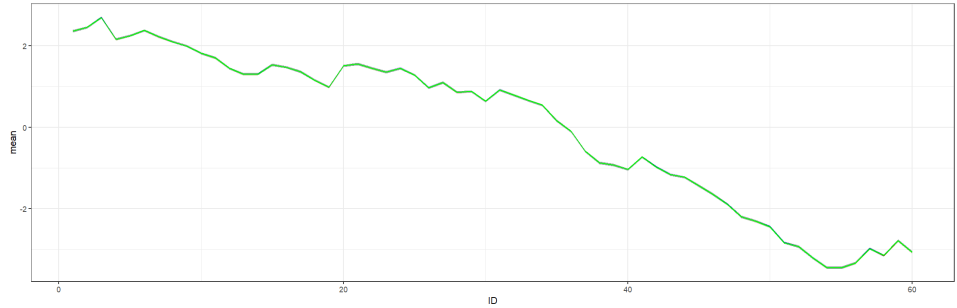


FIGURE 8.10: Posterior mean and 95% credible intervals of the temporal effect under the full-data (red), distributed (blue), and recursive (green) analyses.

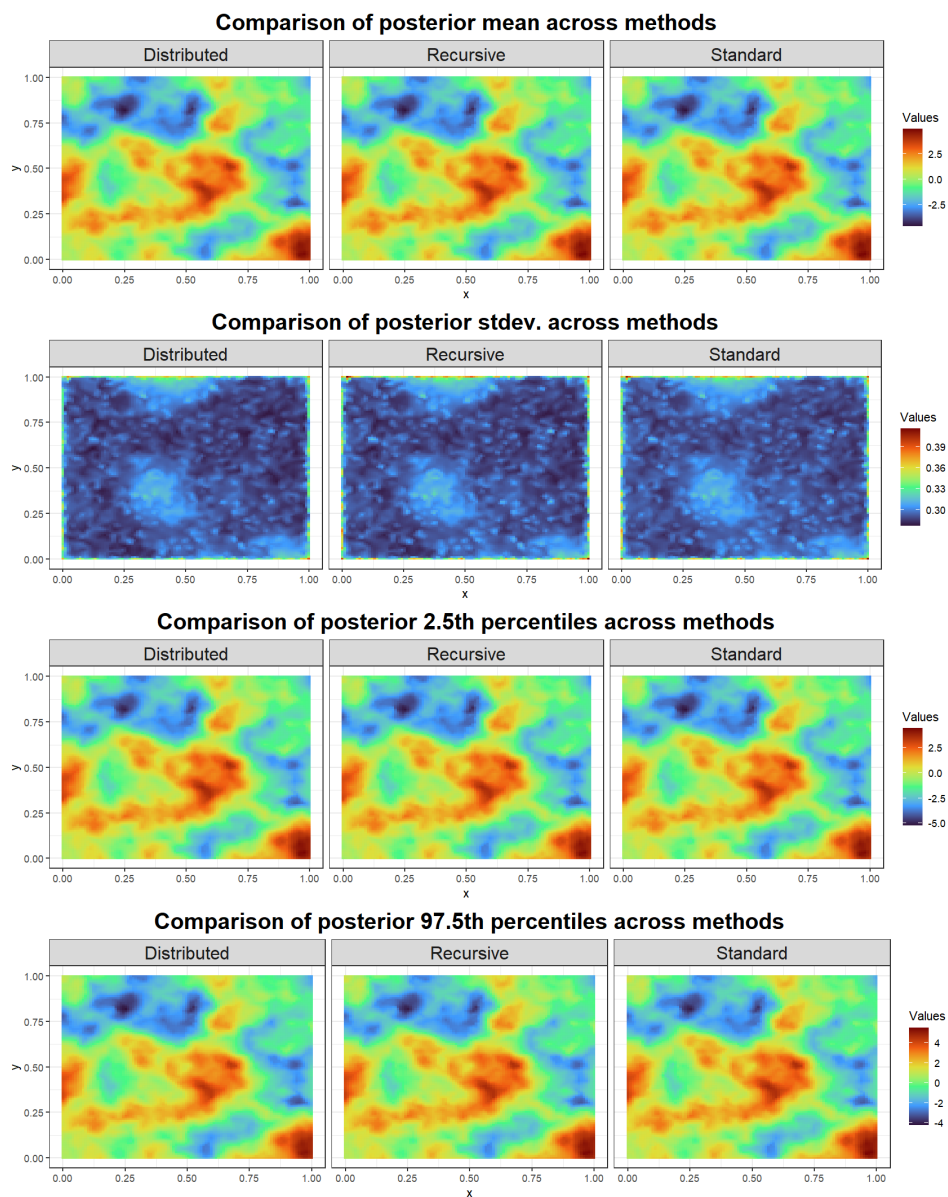


FIGURE 8.11: Comparison of the posterior mean, standard deviation, and 2.5th and 97.5th percentiles of the spatial effect across distributed, recursive, and standard full-data analyses.

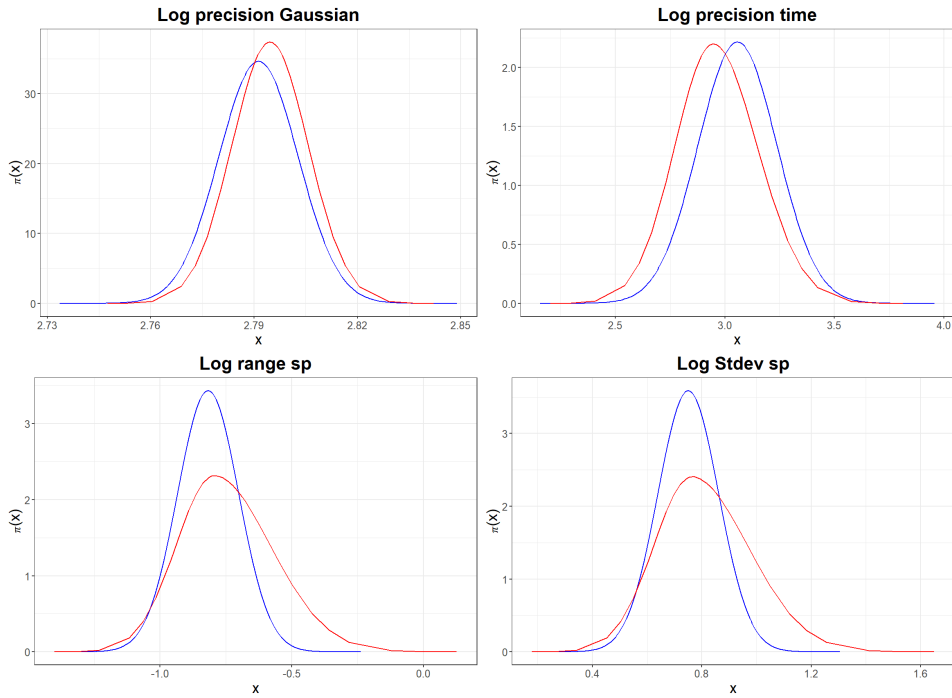


FIGURE 8.12: Posterior distributions of the hyperparameters on the internal scale, obtained from the full-data analysis (red) and from the distributed and recursive approaches (blue).

8.7.2 Top-down distributed for a complex spatio-temporal model

In this example, we analyze a real dataset of $\text{PM}_{2.5}$ concentrations across the conterminous United States. Fine particulate matter ($\text{PM}_{2.5}$) is one of the six criteria pollutants regulated by the U.S. Environmental Protection Agency (EPA). Data on $\text{PM}_{2.5}$ mass concentrations and chemical speciation were obtained from the EPA's Air Quality System (AQS) database (https://aq5.epa.gov/aq5web/airdata/download_files.html), which compiles measurements from both the CSN and IMPROVE networks. For the spatio-temporal analysis, the dataset was cleaned by treating near-zero negative values as negligible (10^{-4}) and removing extreme outliers, both negative and positive. After this preprocessing, the dataset contained 873,762 annual average observations of $\text{PM}_{2.5}$, expressed in micrograms per cubic meter, collected between 1997 and 2024.

In Fig. 8.13, the complete dataset after preprocessing is displayed across the study period. The figure clearly shows the presence of spatial patterns in $\text{PM}_{2.5}$ concentrations as well as a temporal trend indicating a decrease in pollutant levels over time. To analyze these data, we employed a model with a Gamma likelihood, as the observations are strictly positive. The latent field consists of a global intercept, a purely temporal component (to capture the clear global temporal trend in the data), and a spatio-temporal interaction component that accounts for variations in the spatial pattern over time:

$$\begin{aligned} y_i &\sim \text{Gamma}(\mu_i, \phi), \\ \mu_i &= \beta_0 + u_{ti} + u_{sti}, \end{aligned} \tag{8.42}$$

where β_0 is the global intercept, u_{ti} is the purely temporal component modeled as a first-order random walk, and u_{sti} is the spatio-temporal interaction component. The latter is defined as

$$\mathbf{u}_{st} = \text{GMRF}(\mathbf{0}, \mathbf{Q}_{st}),$$

with precision matrix $\mathbf{Q}_{st} = \mathbf{Q}_t \otimes \mathbf{Q}_s$. That is, a separable spatio-temporal interaction represented by the Kronecker product of the precision matrix of the temporal structure (here based on a first-order autoregressive process) and the precision matrix of the spatial structure (defined using the SPDE-FEM approach). Thus,

$$\mathbf{Q}_{st} = \mathbf{Q}_{st}(\sigma_{st}, \rho_{st}, \phi_{st}),$$

where this separable spatio-temporal component is characterized by three hyperparameters: the marginal standard deviation σ_{st} , the spatial correlation range ρ_{st} , and the temporal autocorrelation parameter ϕ_{st} . This spatio-temporal interaction component, based on a mesh with 1,211 spatial nodes and 28 temporal nodes, results in a precision matrix $\mathbf{Q}_{st}$ of size $33,908 \times 33,908$, which can be demanding in terms of memory usage.

In this case, the partitioning of both the data and the latent field was also carried out using the structure of the temporal precision matrix $\mathbf{Q}_t$ through Algorithm 4. This approach reduces the dimension of the spatio-temporal interaction component to $1211 \times (14 + 1)$, where extending to the immediate neighbors of the temporal component is relatively straightforward to handle in the subsequent computations.

The results of the distributed computation and the standard full-data analysis for the CCD integration scheme are presented in Fig. 8.14. Substantial differences can be observed, which lead to clear discrepancies in the marginal distributions of the hyperparameters. However, these differences do not propagate to the marginals of the latent field, which remain largely consistent across methods. The divergence can arise partly from heterogeneity in the hyperparameters—for instance, the spatial range may vary across years—and is further amplified by the non-Gaussian nature of both the hyperparameter posterior and the likelihood. Together, these factors accentuate the gap between the posterior obtained with the standard inferential method and that of the distributed approach.

Figure 8.15 displays the mean and standard deviation of the spatio-temporal effect for a subset of years, comparing the results obtained with the distributed method and the standard full-data analysis. The figure shows that both methods yield very similar outcomes. Figure 8.16 presents the mean and the 95% credible interval of the temporal effect for both methods, again indicating close agreement between the two approaches. Finally, Figure 8.17 shows the marginal distributions of the hyperparameters on their internal scale, as well as the intercept. While the hyperparameters exhibit noticeable differences in some cases, as previously discussed, the marginal distribution of the intercept remains quite similar across both methods.

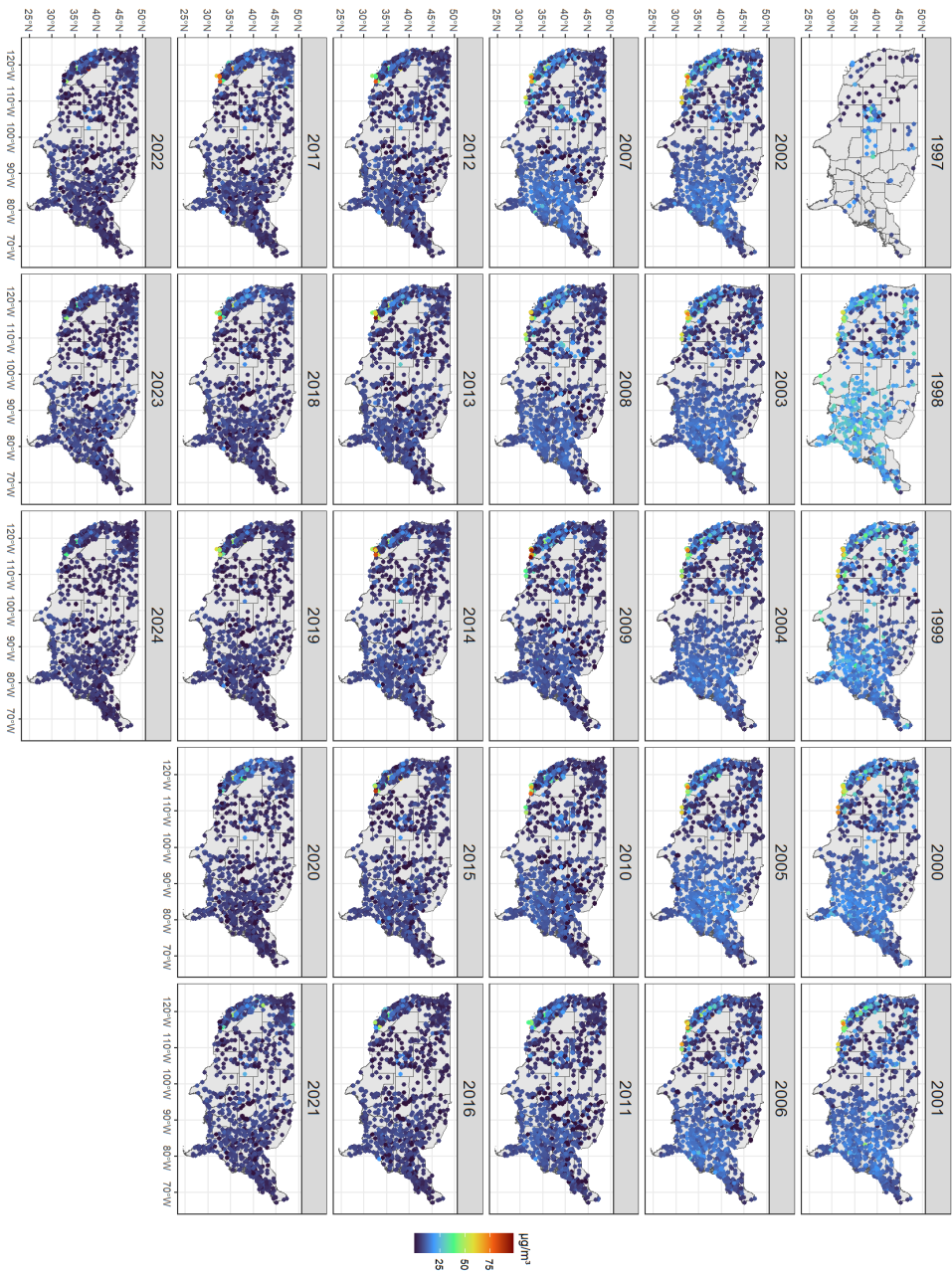


FIGURE 8.13: PM_{2.5} data from monitoring stations across the contiguous United States, measured between 1997 and 2024. Values are reported in micrograms per cubic meter [$\mu\text{g}/\text{m}^3$].

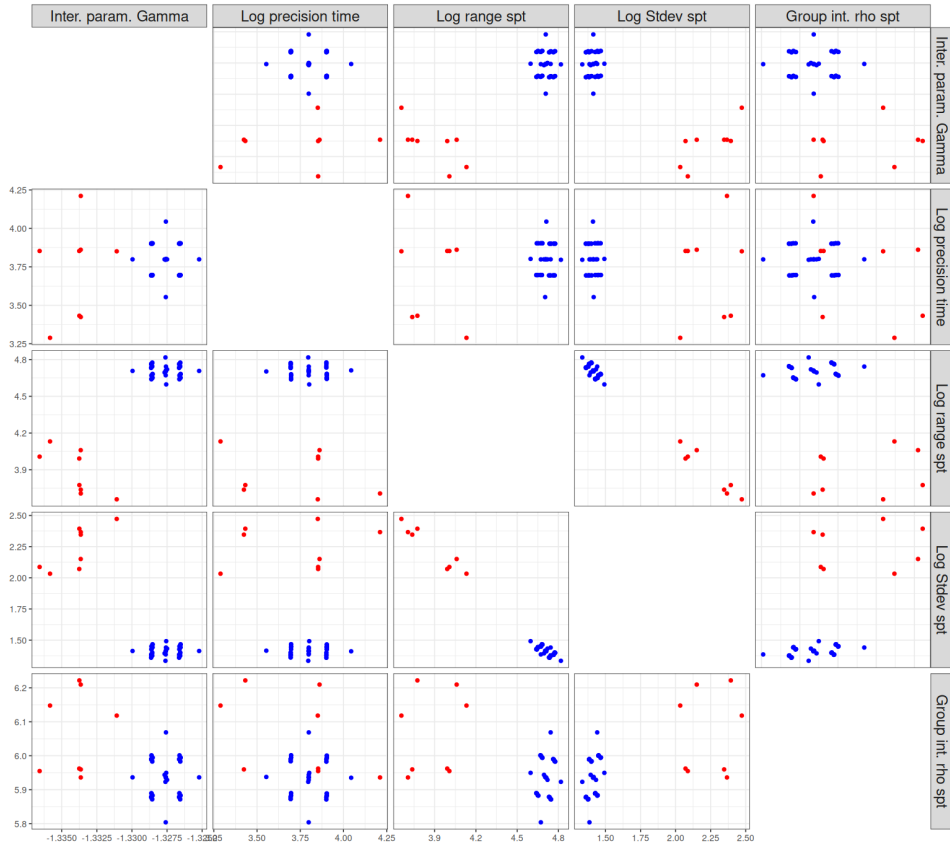
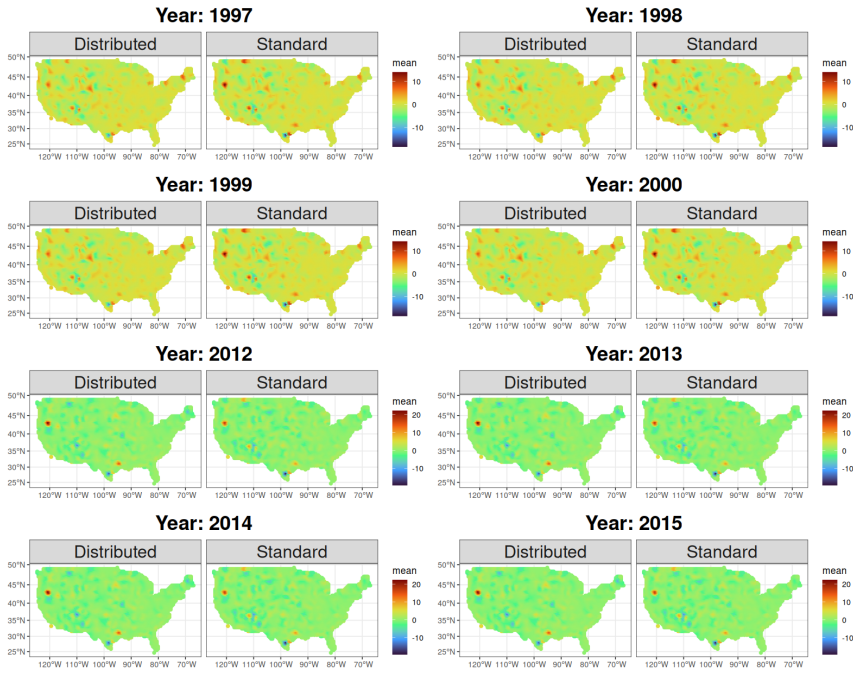
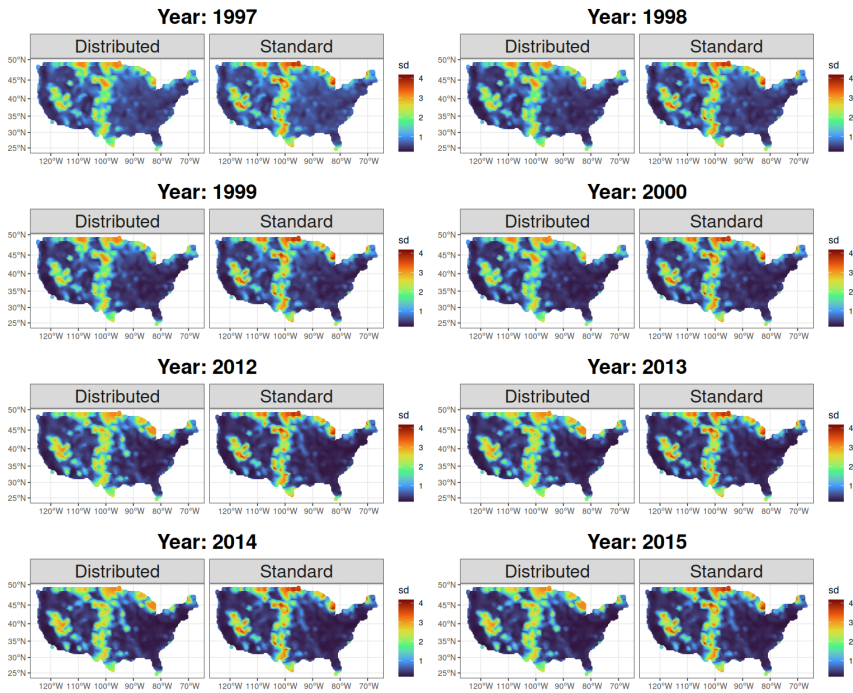


FIGURE 8.14: CCD plot of the integration scheme in the hyperparameter space for full data inference (red) and combined distributed inference (blue). Hyperparameters are shown on the internal scale and include: the internal precision parameter of the Gamma likelihood, the log-precision of the temporal component, the log-precision and log-range of the spatio-temporal component, and the internal parameter for temporal autocorrelation in the spatio-temporal interaction.



(A) Estimated posterior mean of the spatio-temporal effect.



(B) Estimated posterior standard deviation of the spatio-temporal effect.

FIGURE 8.15: Comparison of posterior mean and standard deviation of the spatial effect for selected years between distributed and standard full data analyses.

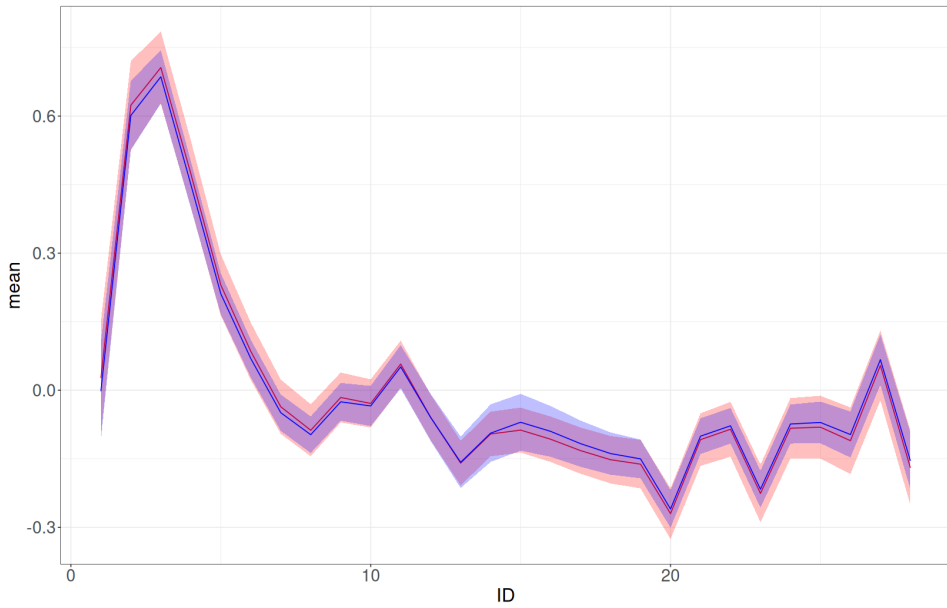


FIGURE 8.16: Posterior mean and 95% credible intervals of the temporal effect under the full-data (red) and distributed (blue) analyses.

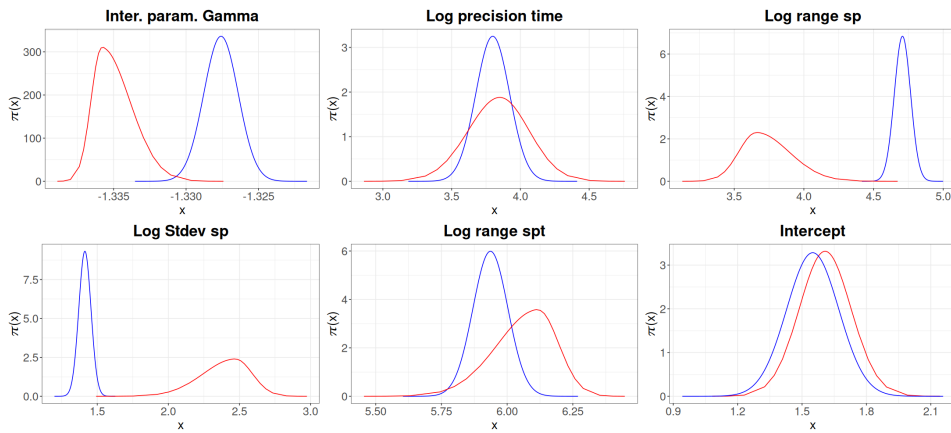


FIGURE 8.17: Posterior distributions of the hyperparameters on the internal scale and the posterior of the intercept, obtained from the standard full-data analysis (red) and the distributed method (blue).

8.7.3 Distributed inference for Log-Gaussian Cox processes

In this final example, we apply the distributed approach using a spatial partition of wildfire data from Spain between 2010 and 2025. The analysis focuses on mainland Spain, where the centroids of the available polygons are used as the raw data, representing the origins of the fires. These data were provided by the European Forest Fire Information System (EFFIS) of the European Commission's Joint Research Centre (<https://forest-fire.emergency.copernicus.eu>).

The partitioning of the wildfire data was carried out using Algorithm 6, which applies a Voronoi tessellation based on centroids obtained from K-means clustering of the mesh nodes. Figure 8.18 displays the resulting spatial partition, the data across years, and the integration areas employed in the point process analysis. The statistical model used to analyze these data was a Log-Gaussian Cox Process (LGCP) with distinct temporal and spatial components, but without explicit spatio-temporal interaction.

In the LGCP framework, the observed locations $\mathbf{s} = \{s_1, \dots, s_n\}$ are considered a realization of a non-homogeneous Poisson process, where the intensity function $\lambda(s)$ itself is modeled as a random field with a log-Gaussian distribution. More formally, an LGCP is defined over a bounded region $\Omega \subset \mathbb{R}^2$ of area $|\Omega|$. For any subregion $D \subset \Omega$, the number of events in D follows a Poisson distribution with mean $\Lambda(D) = \int_D \lambda(s) ds$, where $\lambda(s)$ encodes the spatial structure of the process.

The likelihood of the LGCP, given the intensity function and the observed point pattern $\mathbf{s} = s_1, \dots, s_n$, takes the form

$$\pi(y(\mathbf{s}) \mid \lambda) = \exp \left[|\Omega| - \int_{\Omega} \lambda(s) ds \right] \prod_{s_i \in \mathbf{s}} \lambda(s_i). \quad (8.43)$$

This likelihood can be approximated using numerical integration methods (Illian et al., 2012; Diggle et al., 2013; Simpson et al., 2016; Sørbye et al., 2019).

Numerical integration leads to the following approximation of the LGCP log-likelihood:

$$\log \pi(y(\mathbf{s}) \mid \lambda) \approx |\Omega| - \sum_{l=1}^L a_l \lambda(s_l) + \sum_{i=1}^n \log \lambda(s_i), \quad (8.44)$$

where the integral is approximated as $\int_{\Omega} \lambda(s) ds \approx \sum_{l=1}^L a_l \lambda(s_l)$ using L integration points with weights a_l . These weights correspond to the areas associated with the

Voronoi cells defined by the integration points. In this formulation, the term $\sum_{i=1}^n \log \lambda(s_i)$ plays the role of the Poisson contribution ($y_i = 1$) to the likelihood. The constant $|\Omega|$ does not influence the optimization of the hyperparameters and latent field, and can thus be omitted from the inferential procedure.

For model specification, the log-intensity $\log \lambda(s)$ is expressed as

$$\log \lambda(s_i) = \beta_0 + u_{t,i} + u_{s,i}, \quad (8.45)$$

where β_0 is the intercept, $u_{t,i}$ is a purely temporal component, and $u_{s,i}$ is a purely spatial component. The same formulation applies when evaluating the intensity at the integration points s_l .

The construction of the CCD for the hyperparameters, obtained using both the distributed method and the standard full-data analysis, is shown in Fig. 8.19. This figure illustrates that both integration schemes are quite similar, which in turn results in comparable marginal distributions for the hyperparameters. The marginals of the hyperparameters, together with that of the intercept, are displayed in Fig. 8.20. Finally, Figs. 8.21 and 8.22 present the marginals of the temporal and spatial effects, respectively. These figures display the posterior mean and the 95% credible interval for the temporal effect, and the posterior mean and standard deviation for the spatial effect, under both the distributed approach and the standard full-data analysis. Overall, these results confirm that the distributed approach is able to closely reproduce the outcomes of the standard full-data analysis, both for the hyperparameters and for the latent field components, thereby supporting its reliability and practical usefulness.

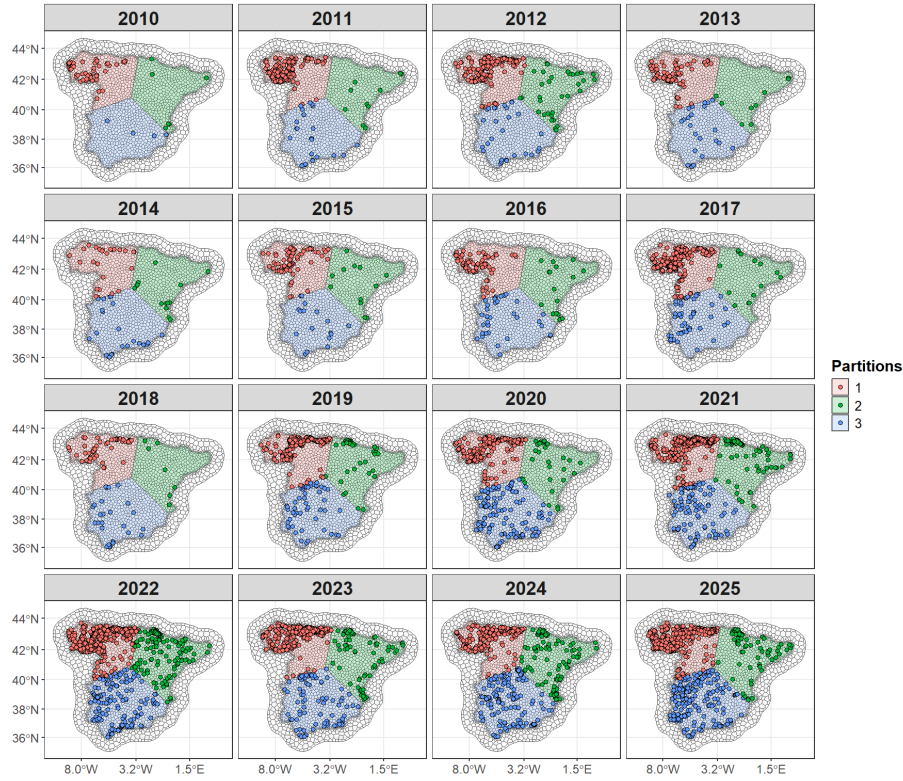


FIGURE 8.18: Spatial partitions of the data with integration points for the point process analysis across years.

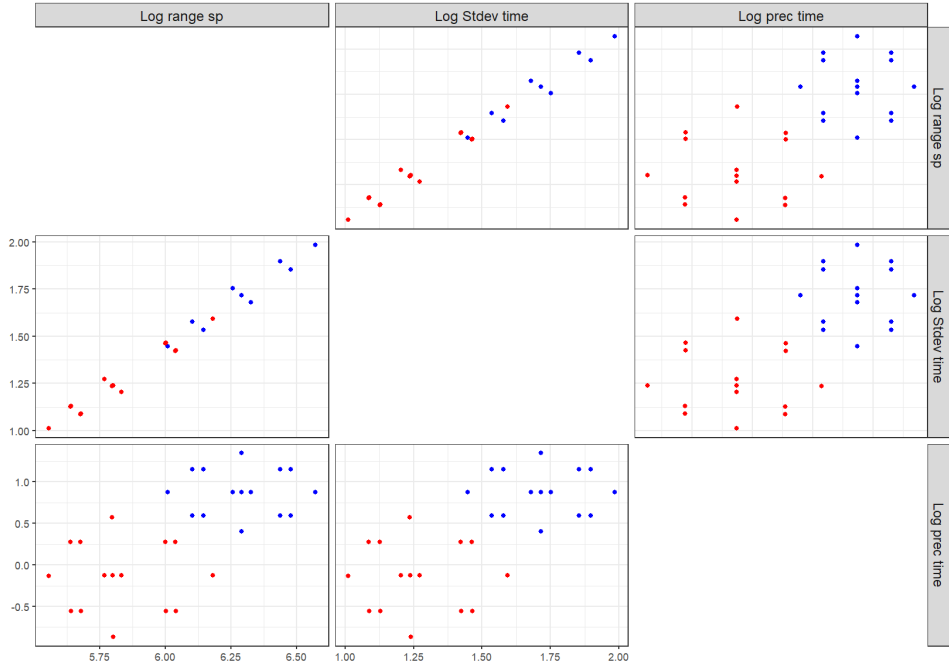


FIGURE 8.19: CCD plot of the integration scheme in the hyperparameter space for full-data inference (red) and combined distributed inference (blue). Hyperparameters are shown on the internal scale.

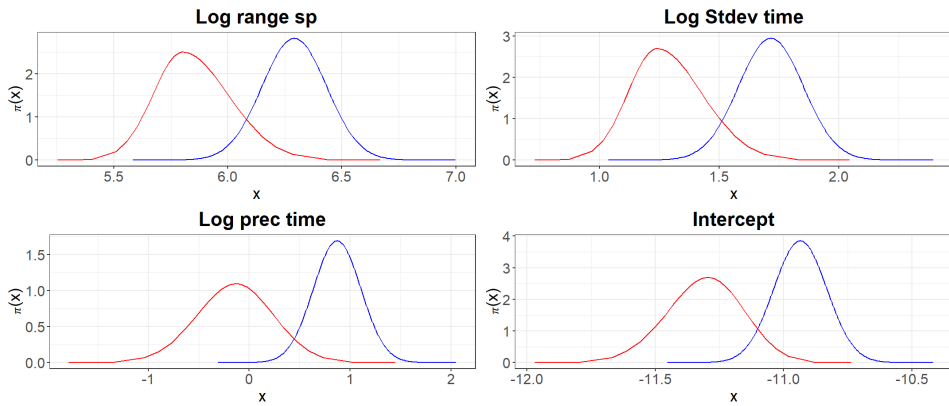


FIGURE 8.20: Posterior distributions of the hyperparameters on the internal scale and the posterior of the intercept, obtained from the standard full-data analysis (red) and the distributed method (blue).

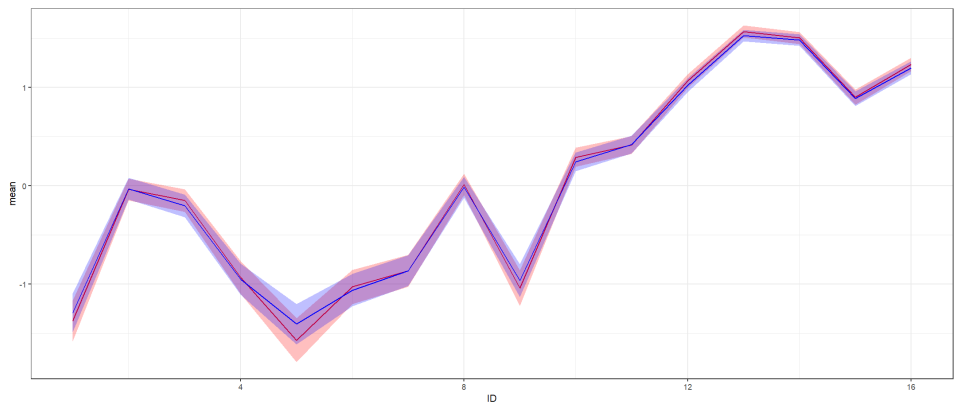


FIGURE 8.21: Posterior mean and 95% credible intervals of the temporal effect under the full-data (red) and distributed (blue) analyses.

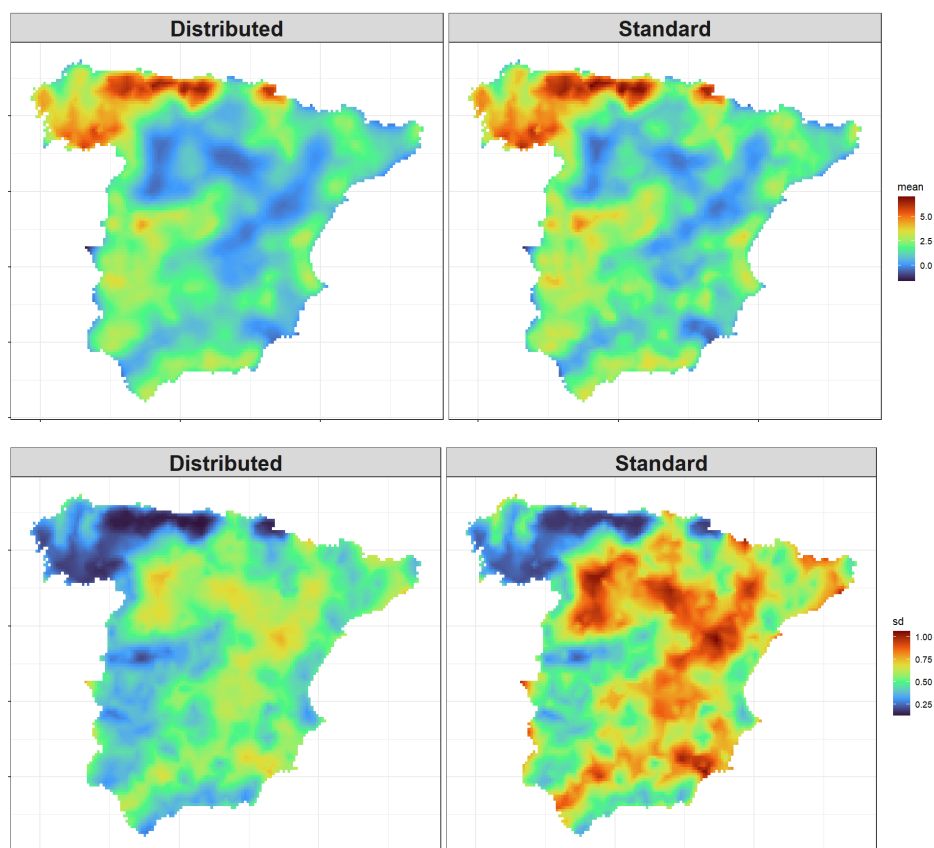


FIGURE 8.22: Comparison of posterior mean (top) and standard deviation (bottom) of the spatial effect for selected years between distributed and standard full-data analyses.

8.8 Conclusions

This work has presented a methodology for performing distributed and recursive inference by leveraging the INLA framework. In particular, we have provided a detailed exposition of the different procedures that can be followed to conduct inference under these two paradigms. Special attention has been given to the strategies available for partitioning both the data and the latent field, as well as their implementation within distributed and recursive inferential procedures. The implementation in the **R-INLA** software has been described to facilitate the application of these methods without the need to program and design every step from scratch, thus making full use of the **R-INLA** package. Finally, the methodology has been illustrated through its application to both a simulated dataset and a real dataset.

Throughout the development, we have highlighted the advantages of the proposed methods, particularly their ability to partition the latent field and the data jointly, thereby reducing computational costs in terms of both memory and processing time. At the same time, we have discussed the limitations of the approach, which become apparent in the second example when identifying support points and marginal posterior distributions. These limitations stem from the geometry of the joint marginal posterior distribution of the hyperparameters. In cases where hyperparameters could themselves vary along the analyzed data, assuming a single posterior distribution for the entire dataset may introduce divergences. This issue is further compounded by the non-Gaussian nature of the posterior, which amplifies discrepancies when information is transferred through Gaussian approximations around the modal configurations.

Nevertheless, the methodology proves capable of recovering the structure of the latent field, even when it has been partitioned. This constitutes one of its main advantages, as it enables a more flexible combination of information from diverse sources, even when these sources do not share the same latent structure, when certain components of the latent field are not common, or when the hyperparameters differ. The extension of the latent field, together with the modal information of the hyperparameters encoded through a Gaussian approximation, allows the mean vector and precision matrix to be expanded as described in various sections of this article. In other words, these methodologies open the door to addressing problems such as meta-analysis, inference under data-privacy constraints, the integration of

heterogeneous information sources (including expert elicitation), and, importantly, big data contexts requiring scalable Bayesian inference.

References

- Alon, N., Seymour, P., and Thomas, R. (1994). “Planar Separators.” *SIAM Journal on Discrete Mathematics*, 7(2): 184–193.
- Altman, M., Wood, A., O’Brien, D. R., and Gasser, U. (2018). “Practical approaches to big data privacy over time.” *International Data Privacy Law*, 8(1): 29–51.
- Amestoy, P. R., Davis, T. A., and Duff, I. S. (1996). “An Approximate Minimum Degree Ordering Algorithm.” *SIAM Journal on Matrix Analysis and Applications*, 17(4): 886–905. Publisher: Society for Industrial & Applied Mathematics (SIAM).
- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D., and Lindgren, F. (2018). “Spatial modeling with R-INLA: A review.” *WIREs Computational Statistics*, 10(6): e1443.
- Bishop, C. M. (2019). *Pattern recognition and machine learning*. Information Science and Statistics. New York, NY: Springer Science+Business Media, LLC.
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). “Fast unfolding of communities in large networks.” *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10): P10008.
- Bochkina, N. (2023). “Bernstein - von Mises theorem and misspecified models: a review.” In *Foundations of Modern Statistics*, volume 425, 355–380. Springer International Publishing AG.
- Bolla, M. (2013). *Spectral clustering and biclustering: learning large graphs and contingency tables*. Chichester, West Sussex, United Kingdom: John Wiley & Sons Inc.
- Broderick, T., Boyd, N., Wibisono, A., Wilson, A. C., and Jordan, M. I. (2013). “Streaming Variational Bayes.” In Burges, C. J., Bottou, L., Welling, M., Ghahramani, Z., and Weinberger, K. Q. (eds.), *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc.

- Brown, P. T., Joshi, C., Joe, S., and Rue, H. (2021). “A novel method of marginalisation using low discrepancy sequences for integrated nested Laplace approximations.” *Computational Statistics & Data Analysis*, 157: 107147.
- Carvalho, C. M., Johannes, M. S., Lopes, H. F., and Polson, N. G. (2010). “Particle Learning and Smoothing.” *Statistical Science*, 25(1).
- Castillo, I. and Nickl, R. (2014). “On the Bernstein–von Mises phenomenon for nonparametric Bayes procedures.” *The Annals of Statistics*, 42(5).
- Chung, F. R. K. (1997). *Spectral graph theory*. Number 92 in CBMS regional conference series in mathematics. Providence (R. I.): American mathematical society.
- Diggle, P. J., Moraga, P., Rowlingson, B., and Taylor, B. M. (2013). “Spatial and Spatio-Temporal Log-Gaussian Cox Processes: Extending the Geostatistical Paradigm.” *Statistical Science*, 28(4): 542 – 563.
- Duff, I. S. (2017). *Direct Methods for Sparse Matrices*. Numerical Mathematics and Scientific Computation Ser. Oxford: Oxford University Press USA - OSO, 2nd ed edition.
- Ezugwu, A. E., Ikotun, A. M., Oyelade, O. O., Abualigah, L., Agushaka, J. O., Eke, C. I., and Akinyelu, A. A. (2022). “A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects.” *Engineering Applications of Artificial Intelligence*, 110: 104743. Publisher: Elsevier BV.
- Farley, S. S., Dawson, A., Goring, S. J., and Williams, J. W. (2018). “Situating Ecology as a Big-Data Science: Current Advances, Challenges, and Solutions.” *BioScience*, 68(8): 563–576.
- Fattah, E. A. and Rue, H. (2022). “Approximate Bayesian Inference for the Interaction Types 1, 2, 3 and 4 with Application in Disease Mapping.” *arXiv*.
- Fattah, E. A., Van Niekerk, J., and Rue, H. (2022). “Smart Gradient - An adaptive technique for improving gradient estimation.” *Foundations of Data Science*, 4(1): 123.
- Fiedler, M. (1973). “Algebraic connectivity of graphs.” *Czechoslovak Mathematical Journal*, 23(2): 298–305.

- Fortunato, S. (2010). “Community detection in graphs.” *Physics Reports*, 486(3-5): 75–174. ArXiv:0906.0612 [physics].
- Fortunato, S. and Hric, D. (2016). “Community detection in networks: A user guide.” *Physics Reports*, 659: 1–44.
- Freedman, D. (1999). “Wald Lecture: On the Bernstein-von Mises theorem with infinite-dimensional parameters.” *The Annals of Statistics*, 27(4).
- Gao, Y., Liu, W., Wang, H., Wang, X., Yan, Y., and Zhang, R. (2022). “A review of distributed statistical inference.” *Statistical Theory and Related Fields*, 6(2): 89–99.
- George, A. (1973). “Nested Dissection of a Regular Finite Element Mesh.” *SIAM Journal on Numerical Analysis*, 10(2): 345–363. Publisher: Society for Industrial & Applied Mathematics (SIAM).
- Gibbs, N. E., Poole, W. G., and Stockmeyer, P. K. (1976). “An Algorithm for Reducing the Bandwidth and Profile of a Sparse Matrix.” *SIAM Journal on Numerical Analysis*, 13(2): 236–250. Publisher: Society for Industrial and Applied Mathematics.
- Gilbert, J. R. and Tarjan, R. E. (1986). “The analysis of a nested dissection algorithm.” *Numerische Mathematik*, 50(4): 377–404. Publisher: Springer Science and Business Media LLC.
- Goodrich, M. (1995). “Planar Separators and Parallel Polygon Triangulation.” *Journal of Computer and System Sciences*, 51(3): 374–389.
- Gómez-Rubio, V. (2020). *Bayesian Inference with INLA*. Chapman & Hall/CRC Press.
- Har-Peled, S. (2011). *Geometric approximation algorithms*. Number Volume 173 in Mathematical surveys and monographs. Providence (R.I.): American mathematical society.
- Heumans, H., Nacken, P., Toet, A., and Vincent, L. (1992). “Graph morphology.” *Journal of Visual Communication and Image Representation*, 3(1): 24–38.
- Hoffman, M. D., Blei, D. M., Wang, C., and Paisley, J. (2013). “Stochastic variational inference.” *Journal of Machine Learning Research*.
- Hooten, M. B., Johnson, D. S., and Brost, B. M. (2021). “Making Recursive Bayesian Inference Accessible.” *The American Statistician*, 75(2): 185–194.

- Huang, Z. and Gelman, A. (2005). “Sampling for Bayesian Computation with Large Datasets.” Technical report, Columbia University.
- Illian, J. B., Sørbye, S. H., and Rue, H. (2012). “A toolbox for fitting complex spatial point process models using integrated nested Laplace approximation (INLA).” *The Annals of Applied Statistics*, 6(4): 1499 – 1530.
- Jain, A. K. (2010). “Data clustering: 50 years beyond K-means.” *Pattern Recognition Letters*, 31(8): 651–666.
- Jianbo Shi and Malik, J. (2000). “Normalized cuts and image segmentation.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(8): 888–905.
- Johnson, S. C. (1967). “Hierarchical Clustering Schemes.” *Psychometrika*, 32(3): 241–254. Publisher: Cambridge University Press (CUP).
- Karypis, G. and Kumar, V. (1998). *METIS: A Software Package for Partitioning Unstructured Graphs, Partitioning Meshes, and Computing Fill-Reducing Orderings of Sparse Matrices*.
- Kaufman, L. and Rousseeuw, P. J. (1990). *Finding Groups in Data: An Introduction to Cluster Analysis*. Wiley Series in Probability and Statistics. Wiley, 1 edition. ISSN: 1940-6347.
- Kessler, S., Cobb, A., Rudner, T. G. J., Zohren, S., and Roberts, S. J. (2023). “On Sequential Bayesian Inference for Continual Learning.” *Entropy*, 25(6): 884.
- Knorr-Held, L. (2000). “Bayesian modelling of inseparable space-time variation in disease risk.” *Statistics in Medicine*, 19(17-18): 2555–2567.
- Koers, G., Szabó, B., and Vaart, A. v. d. (2025). “Misspecified Bernstein-Von Mises theorem for hierarchical models.” ArXiv:2308.07803 [math.ST].
- Lindgren, F. and Rue, H. (2015). “Bayesian Spatial Modelling with R-INLA.” *Journal of Statistical Software*, 63(19): 1–25.
- Lipton, R. J. and Tarjan, R. E. (1980). “Applications of a Planar Separator Theorem.” *SIAM Journal on Computing*, 9(3): 615–627.
- Liu, J. W. H. (1985). “Modification of the minimum-degree algorithm by multiple elimination.” *ACM Transactions on Mathematical Software*, 11(2): 141–153. Publisher: Association for Computing Machinery (ACM).

- Liu, W.-H. and Sherman, A. H. (1976). “Comparative Analysis of the Cuthill-McKee and the Reverse Cuthill-McKee Ordering Algorithms for Sparse Matrices.” *SIAM Journal on Numerical Analysis*, 13(2): 198–213. Publisher: Society for Industrial and Applied Mathematics.
- Lunn, D., Barrett, J., Sweeting, M., and Thompson, S. (2013). “Fully Bayesian Hierarchical Modelling in Two Stages, with Application to Meta-Analysis.” *Journal of the Royal Statistical Society Series C: Applied Statistics*, 62(4): 551–572.
- Luo, J.-C. (1992). “Algorithms for reducing the bandwidth and profile of a sparse matrix.” *Computers & Structures*, 44(3): 535–548.
- Martins, T. G., Simpson, D., Lindgren, F., and Rue, H. (2013). “Bayesian computing with INLA: New features.” *Computational Statistics & Data Analysis*, 67: 68–83.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., and Arcas, B. A. y. (2017). “Communication-Efficient Learning of Deep Networks from Decentralized Data.” In Singh, A. and Zhu, J. (eds.), *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, 1273–1282. PMLR.
- Monien, B. and Schulze, J. (1997). “Parallel sparse Cholesky factorization.” In *Lecture Notes in Computer Science*, 255–272. Berlin, Heidelberg: Springer Berlin Heidelberg. ISSN: 0302-9743, 1611-3349.
- Murtagh, F. and Contreras, P. (2012). “Algorithms for hierarchical clustering: an overview.” *WIREs Data Mining and Knowledge Discovery*, 2(1): 86–97. Publisher: Wiley.
- Myers, R. H. (2016). *Response Surface Methodology: Process and Product Optimization Using Designed Experiments*. New York Academy of Sciences Series. Newark: John Wiley & Sons, Incorporated, 1st ed edition.
- Naumann, U. and Schenk, O. (2012). *Combinatorial scientific computing*. Boca Raton, Fla.: CRC Press. OCLC: 1300505893.
- Orozco-Acosta, E., Adin, A., and Ugarte, M. D. (2023). “Big problems in spatio-temporal disease mapping: Methods and software.” *Computer Methods and Programs in Biomedicine*, 231: 107403.
- Rendell, L. J., Johansen, A. M., Lee, A., and Whiteley, N. (2021). “Global Consensus Monte Carlo.” *Journal of Computational and Graphical Statistics*, 30(2): 249–259.

- Rue, H. and Held, L. (2005). *Gaussian Markov Random Fields*. Chapman and Hall/CRC.
- Rue, H., Martino, S., and Chopin, N. (2009). “Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations.” *Journal of the Royal Statistical Society. Series B: Statistical Methodology*, 71(2): 319–392.
- Rue, H. and Martino, S. (2007). “Approximate Bayesian inference for hierarchical Gaussian Markov random field models.” *Journal of Statistical Planning and Inference*, 137(10): 3177–3192.
- Sarwate, A. D. and Chaudhuri, K. (2013). “Signal Processing and Machine Learning with Differential Privacy: Algorithms and Challenges for Continuous Data.” *IEEE Signal Processing Magazine*, 30(5): 86–94.
- Savarese, S. M. and Boley, D. L. (2001). “On the performance of bisecting K-means and PDDP.” In *Proceedings of the 2001 SIAM International Conference on Data Mining*, 1–14. Philadelphia, PA: Society for Industrial and Applied Mathematics.
- Scott, S. L., Blocker, A. W., Bonassi, F. V., Chipman, H. A., George, E. I., and McCulloch, R. E. (2016). “Bayes and big data: the consensus Monte Carlo algorithm.” *International Journal of Management Science and Engineering Management*, 11(2): 78–88.
- Sebestyén, V., Czvetkó, T., and Abonyi, J. (2021). “The Applicability of Big Data in Climate Change Research: The Importance of System of Systems Thinking.” *Frontiers in Environmental Science*, 9: 619092.
- Simpson, D., Illian, J. B., Lindgren, F., Sørbye, S. H., and Rue, H. (2016). “Going off grid: computationally efficient inference for log-Gaussian Cox processes.” *Biometrika*, 103(1): 49–70.
- Singh, P., Singh, M. K., Singh, R., and Singh, N. (2022). “Federated Learning: Challenges, Methods, and Future Directions.” In Yadav, S. P., Bhati, B. S., Mahato, D. P., and Kumar, S. (eds.), *Federated Learning for IoT Applications*, 199–214. Cham: Springer International Publishing. Series Title: EAI/Springer Innovations in Communication and Computing.
- Smith, A., Thakurta, A., and Upadhyay, J. (2017). “Is Interaction Necessary for Distributed Private Learning?” In *2017 IEEE Symposium on Security and Privacy (SP)*, 58–77. San Jose, CA, USA: IEEE.

- Sørbye, S. H., Illian, J. B., Simpson, D. P., Burslem, D., and Rue, H. (2019). “Careful Prior Specification Avoids Incautious Inference for Log-Gaussian Cox Point Processes.” *Journal of the Royal Statistical Society Series C: Applied Statistics*, 68(3): 543–564.
- Spokoiny, V. (2014). “Bernstein - von Mises Theorem for growing parameter dimension.” ArXiv:1302.3430 [math.ST].
- Srivastava, S., Li, C., and Dunson, D. B. (2018). “Scalable Bayes via Barycenter in Wasserstein Space.” *Journal of Machine Learning Research*, 19(8): 1–35.
- Vaart, A. W. v. d. (2007). *Asymptotic statistics*. Number 3 in Cambridge series on statistical and probabilistic mathematics. Cambridge New York: Cambridge University Press, first paperback edition, 8. printing edition.
- Van Niekerk, J. and Rue, H. (2024). “Low-rank Variational Bayes correction to the Laplace method.” *Journal of Machine Learning Research*, 25(62): 1–25.
- Van Niekerk, J., Krainski, E., Rustand, D., and Rue, H. (2023). “A new avenue for Bayesian inference with INLA.” *Computational Statistics & Data Analysis*, 181: 107692.
- Varian, H. R. (2014). “Big Data: New Tricks for Econometrics.” *Journal of Economic Perspectives*, 28(2): 3–28.
- Vehtari, A., Gelman, A., Sivula, T., Jylänki, P., Tran, D., Sahai, S., Blomstedt, P., Cunningham, J. P., Schiminovich, D., and Robert, C. P. (2020). “Expectation propagation as a way of life: a framework for Bayesian inference on partitioned data.” *J. Mach. Learn. Res.*, 21(1). Publisher: JMLR.org.
- Vincent, L. (1988). “Mathematical Morphology On Graphs.” 95. Cambridge, MA.
- Von Luxburg, U. (2007). “A tutorial on spectral clustering.” *Statistics and Computing*, 17(4): 395–416.
- Wang, X. and Dunson, D. B. (2013). “Parallelizing MCMC via Weierstrass Sampler.” ArXiv:1312.4605 [stat.CO].
- Xu, R. and WunschII, D. (2005). “Survey of Clustering Algorithms.” *IEEE Transactions on Neural Networks*, 16(3): 645–678.
- Yannakakis, M. (1981). “Computing the Minimum Fill-In is NP-Complete.” *SIAM Journal on Algebraic Discrete Methods*, 2(1): 77–79. Publisher: Society for Industrial & Applied Mathematics (SIAM).

- Zhou, L., Gong, Z., and Xiang, P. (2024). “Distributed Computing and Inference for Big Data.” *Annual Review of Statistics and Its Application*, 11(1): 533–551.

Concluding Remarks and Future Directions



The final oath has been honored



Throughout the thesis, several methods have been developed for transferring information across inferential procedures, whether through *sequential* (also known as *sequential learning* in the machine learning community), or through *distributed inference* (also referred to as *distributed learning*, with *federated learning* as a special case). This transfer of information is not limited to models partitioned along a specific structure (temporal, spatial, or otherwise), but also extends to models that share certain components without being fully overlapping. In other words, the goal is to combine sources of information related to connected or interdependent phenomena, allowing learning to take place jointly from both. However, these procedures can also be useful in cases where unified or complete data sources are available, but analyzing all the data simultaneously may be unreliable. In such settings, sequential or distributed inference can also be applied to handle models that are computationally demanding, either due to the sheer volume of data or because of the structural complexity of the model.

This theme is central to four articles corresponding to Chapters 3, 5, 7, and 8, which present: a Bayesian feedback procedure (Chapter 3), two sequential consensus algorithms (Chapter 5), and recursive (Chapters 7 and 8) as well as distributed (Chapter 8) procedures. In addition, two other articles, corresponding to Chapters 4 and 6, indirectly address this theme through applications. Chapter 4 introduces a Shiny application that implements data analysis functionality together with the Bayesian feedback procedure. Chapter 6, motivated by the European project LAMASUS (Land Management for Sustainability), applies the sequential consensus

approach (Chapter 5) to large-scale compositional datasets. Specifically, it addresses a case involving vast amounts of data with a spatial support structure that changes over time (across three different sets of spatial areal supports), where a downscaling model is employed.

In overall, along this PhD, we have proposed different approaches and algorithms to deal with combination of data, combination of models, transferring information along models, splitting and distributing inference, which allows to perform analysis in complex, real-time and vast data or model settings. However, there are many other elements that, due to time constraints, could not be appropriately structured within the main body of the thesis. These refer to ongoing work, future lines of research, or developments that build upon the directions already presented in this manuscript. We now briefly review these research lines.

9.1 Minor research lines

The following sketches and ideas are among the minor research directions under development:

1. This line of work within spatial statistics aims to develop an automatic procedure for analyzing data that naturally lie on non-trivial surfaces, while accounting for the underlying “terrain” topology (e.g., temperature data). The procedure, which is already very advanced, builds upon the SPDE-FEM ([Lindgren et al., 2011](#)) approach and leverages the `INLA` and `fmesh` packages. The key idea is to model the topological information of elevation—or more generally, the distance with respect to a planar or spherical reference mesh. Under certain assumptions, it is then possible to construct a non-planar modified mesh, adapted to the actual topology by means of a surrogate model for this distance. This approach can be further generalized to incorporate cost-based distances ([Muñoz, 2012](#)) into spatial and spatio-temporal models. The resulting models remain homogeneous and isotropic, but with respect to the new spatial metric induced by the modified topology of the mesh.
2. Modification of the sequential consensus algorithm to better handle hyperparameter information. The main idea is to define a joint prior on the internal parameterization of the hyperparameters. This joint prior may be based either on the Gaussian approximation at the modal configuration or on a

multivariate skew-normal distribution that captures marginal skewness at the mode. For the Gaussian field, different strategies will be explored, such as transferring the Gaussian approximation at the modal configuration, or redefining the prior using a block-diagonal structure to reconstruct priors with hyperparameters that mimic the original prior structure. These will then be compared with the current approach, which combines sequential transfer of marginals with either multivariate (of the conditional posterior at the mode) consensus or univariate (of the marginal posterior) consensus within an ensemble of posterior information. By incorporating these elements into the sequential consensus algorithm, we aim to examine their impact on data analysis and imputation, as well as robustness in the presence of missing data, misalignment, or misspecification.

3. Modification of the `dirinla` package to incorporate any of the random effects predefined in the `INLA` package, in collaboration with Joaquín Martínez-Minaya and Finn Lindgren. The package functions have already been modified, but it is still necessary to verify that the algorithms implemented internally in `dirinla` converge properly, and to consider adjustments or changes if the benchmarks do not yield positive results. Some preliminary tests in simple cases have already been carried out, and the results have been positive.
4. Development of graph-geometry-driven matrix storage formats. This involves the design of specialized storage formats for matrices derived from intrinsic properties of graph geometry. Specifically, the goal is to exploit local symmetries of graphs to define compact representations. A trivial example arises in matrices associated with regular grids or “line graphs” (RW1, RW2, AR1, ...), where repeated structural patterns can be encoded explicitly. By identifying such redundancies in the sparsity pattern, either at the global or subgraph level, it becomes possible to reduce memory requirements and improve scalability for very large graphs. Unlike standard universal sparse matrix formats (CSC, CSR, COO), these geometry-aware formats would encode graph-specific patterns and therefore require a redefinition of fundamental linear algebra operations¹ in accordance with the adopted structure. Conceptually, this approach bridges matrix-free methods (which avoid explicit storage) and hierarchical matrix methods (which exploit block-structured low-rank properties), yielding a hybrid hierarchical/matrix-free framework tailored for large-scale graphical models.

¹Huge troubles!

9.2 Distributed inference

This is a line of research that can be further exploited, improving upon the results obtained by the methodology proposed in Chapter 8, while still building on it.

As a first step, it remains to develop an R package that enables *distributed inference* in the most automated way possible, minimizing the burden on the user. Ideally, this would include everything from the automatic construction of partitions to the splitting of the latent field and its subsequent reassembly with the appropriate weights, while providing functions that support the implementation of different distributed procedures.

9.2.1 Improving hyperparameter exploration

Within the distributed approach methodology, there are also fertile directions for exploration, the most relevant of which concerns the treatment of hyperparameters. This also opens the possibility of improving procedures for transferring hyperparameter information across partitions in recursive inference. As indicated in Chapter 8, the posterior distribution of the joint hyperparameters can be expressed as:

$$\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}) = \prod_{i=1}^n \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i) = \prod_{i=1}^n \frac{\pi(\mathbf{y}_i \mid \mathbf{x}_i, \boldsymbol{\theta}) \pi^*(\mathbf{x}_i \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta})^{w_i}}{|\mathbf{Q}_{\mathbf{x}_i \mid \mathbf{y}_i, \boldsymbol{\theta}}|} \Bigg|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})}, \quad (9.1)$$

where $\sum_i w_i = 1$ and $(\forall i) w_i > 0$. In this case, by denoting $\mathbf{x}_i$ for each partition, we are assuming that, in general, the latent field may itself be partitioned. As described in the section on data and latent field partitioning in Chapter 8, it is assumed that the latent field can be split across partitions in such a way that the product of the partition-specific priors reconstructs the original prior for the latent field, with the appropriate rescaling applied to shared elements. Accordingly, in this equation we denote the prior of the latent field associated with each partition as $\pi^*(\mathbf{x}_i \mid \boldsymbol{\theta}, \mathbf{y}_i)$. Based on this, we can propose two possible avenues for improving the procedure presented in Chapter 8.

Distributed modal approximation

The first avenue concerns the fact that the marginal posterior of the hyperparameters may lose substantial information about its geometry when using a Gaussian

approximation. A natural generalization would therefore be to approximate each $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$ by a Generalized Skew-Normal (GSN) distribution via a skew-modal (SM) approximation, theoretically justified by the skewed Bernstein–von Mises theorem established by [Durante et al. \(2024\)](#). To approximate $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$ with a skew-normal distribution, one may employ a matching method such as the derivative matching (DM) scheme ([Zhou et al., 2024](#)). This approach enables the construction of a GSN approximation at the modal configuration of $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$, such that:

$$\tilde{\pi}_{GSN}(\boldsymbol{\theta} \mid \mathbf{y}_i) = \text{GSN}(\boldsymbol{\theta}; \boldsymbol{\mu}, \boldsymbol{\Sigma}, \pi) \approx \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i). \quad (9.2)$$

In this SM approximation approach, although the procedure may not be computationally cheap, it only needs to be performed once per partition, and each partition may reside on a separate server or machine. Once these GSN distributions are obtained, we can compute the modal configuration approximation of $\tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y})$, denoted as $\tilde{\pi}_d^*(\boldsymbol{\theta} \mid \mathbf{y})$. We then define this modal configuration approximation as:

$$\tilde{\pi}_d^*(\boldsymbol{\theta} \mid \mathbf{y}) = \tilde{\pi}_{GSN}^*(\boldsymbol{\theta} \mid \mathbf{y}) \approx \prod_{i=1}^n \tilde{\pi}_{GSN}(\boldsymbol{\theta} \mid \mathbf{y}_i). \quad (9.3)$$

On top of this approximation, we may apply a Gaussian approximation at the modal configuration of $\tilde{\pi}_d^*(\boldsymbol{\theta})$ to construct the CCD. Alternatively, we may compute a new skew-modal approximation of $\tilde{\pi}_d^*(\boldsymbol{\theta})$ via matching, and then calculate the High Density Regions (HDRs), defined as

$$R_c = \{\boldsymbol{\theta} : \tilde{\pi}_{GSN}^*(\boldsymbol{\theta} \mid \mathbf{y}) \geq c\}$$

where the threshold c is chosen such that $\mathbb{P}(R_c) = 1 - \alpha$. Based on this, we could design an integration scheme centered at the modal location together with a set of additional locations covering the hypersurfaces associated with the HDRs. This procedure could also be applied in the recursive framework, at least for the information transfer component related to the hyperparameters.

The computation of these HDRs is non-trivial, but several strategies can be proposed. One option is (a) to numerically determine the threshold c via Monte Carlo simulation of the GSN, since GSNs are simulable (and, as long as the dimensionality of the hyperparameters is not too large, this method remains computationally feasible). Another option is (b) to approximate the HDRs by a modified Gaussian

ellipsoid, corrected in each direction according to the skewness, in a manner conceptually analogous to the correction applied to the CCD in **R-INLA**. Additional strategies may also be developed to efficiently compute the HDRs and to place support points along their associated surfaces.

Distributed modal optimization

In a second approach, we note that Eq. (9.4) has been used for computing the weights in the integration scheme for the marginal distributions of each latent field node. However, it has not been directly used to determine the integration scheme itself. Therefore, the methodological improvement would be to employ an optimization procedure directly over Eq. (9.4), treating it as the target distribution within the optimization process. This implies that, on each server or machine, we have

$$\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i) = \frac{(\log \pi(\mathbf{y}_i \mid \mathbf{x}_i, \boldsymbol{\theta}) + \log \pi^*(\mathbf{x}_i \mid \boldsymbol{\theta}) + w_i \log \pi(\boldsymbol{\theta}) - \log |\mathbf{Q}_{\mathbf{x}_i \mid \mathbf{y}_i, \boldsymbol{\theta}}|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})})}{\sum_{i=1}^n (\log \pi(\mathbf{y}_i \mid \mathbf{x}_i, \boldsymbol{\theta}) + \log \pi^*(\mathbf{x}_i \mid \boldsymbol{\theta}) + w_i \log \pi(\boldsymbol{\theta}) - \log |\mathbf{Q}_{\mathbf{x}_i \mid \mathbf{y}_i, \boldsymbol{\theta}}|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})})}, \quad (9.4)$$

and the objective is to minimize $-\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y})$ using finite-difference differentiation on each block $\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i)$ across the servers or machines.

Information from these servers or machines can then be aggregated as

$$\nabla_{\boldsymbol{\theta}} \log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}) = \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y}_i), \quad (9.5)$$

thus yielding an optimization procedure for this function. An analogous approach can be followed to compute the Hessian at the modal configuration obtained through the minimization of $-\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y})$.

This allows us to solve the optimization problem by locating the mode $\boldsymbol{\theta}^*$ of the marginal posterior under the distributed framework:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \{-\log \tilde{\pi}_d(\boldsymbol{\theta} \mid \mathbf{y})\}$$

and subsequently construct the CCD or another integration scheme, leveraging either a Gaussian approximation or an SM approximation at the mode $\boldsymbol{\theta}^*$.

It is fairly straightforward, at least from a formal standpoint, to extend this framework to cases where not all hyperparameters are shared across blocks in the

distributed inference. The optimization procedure would still progress according to Eq. (9.5), while the negative Hessian would be computed as

$$(\mathbf{H}_{\tilde{\pi}_d(\boldsymbol{\theta}|\mathbf{y})})_{jk} = -\frac{\partial^2 \log \tilde{\pi}_d(\boldsymbol{\theta} | \mathbf{y})}{\partial \theta_j \partial \theta_k} = -\sum_{i=1}^n \frac{\partial^2 \log \tilde{\pi}_d(\boldsymbol{\Theta}_i | \mathbf{y}_i)}{\partial \theta_j \partial \theta_k} \Big|_{\mathbf{x}=\boldsymbol{\mu}_i(\boldsymbol{\theta})}, \quad (9.6)$$

where $\boldsymbol{\Theta}_i$ is used to distinguish the subset of hyperparameters associated with each partition from the full hyperparameter vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_M)$.

Let us consider, for example, a case with two partitions, $\mathbf{y} = \{\mathbf{y}_1, \mathbf{y}_2\}$, and three hyperparameters in total, $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$. The first partition depends on all three hyperparameters, $\boldsymbol{\Theta}_1 = (\theta_1, \theta_2, \theta_3)$, whereas the second involves only two, $\boldsymbol{\Theta}_2 = (\theta_1, \theta_2)$. In this setting, the negative Hessian is computed as

$$\begin{aligned} \mathbf{H}_{\tilde{\pi}_d(\boldsymbol{\theta}|\mathbf{y})} = & - \left(\begin{array}{ccc} \frac{\partial^2}{\partial \theta_1 \partial \theta_1} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_1 \partial \theta_2} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_1 \partial \theta_3} \log \tilde{\pi}_d^1 \\ \frac{\partial^2}{\partial \theta_2 \partial \theta_1} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_2 \partial \theta_2} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_2 \partial \theta_3} \log \tilde{\pi}_d^1 \\ \frac{\partial^2}{\partial \theta_3 \partial \theta_1} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_3 \partial \theta_2} \log \tilde{\pi}_d^1 & \frac{\partial^2}{\partial \theta_3 \partial \theta_3} \log \tilde{\pi}_d^1 \end{array} \right) \Big|_{\mathbf{x}=\boldsymbol{\mu}_1(\boldsymbol{\theta})} \\ & - \left(\begin{array}{ccc} \frac{\partial^2}{\partial \theta_1 \partial \theta_1} \log \tilde{\pi}_d^2 & \frac{\partial^2}{\partial \theta_1 \partial \theta_2} \log \tilde{\pi}_d^2 & 0 \\ \frac{\partial^2}{\partial \theta_2 \partial \theta_1} \log \tilde{\pi}_d^2 & \frac{\partial^2}{\partial \theta_2 \partial \theta_2} \log \tilde{\pi}_d^2 & 0 \\ 0 & 0 & 0 \end{array} \right) \Big|_{\mathbf{x}=\boldsymbol{\mu}_2(\boldsymbol{\theta})} \end{aligned}, \quad (9.7)$$

where $\log \tilde{\pi}_d^1 = \log \tilde{\pi}_d(\boldsymbol{\Theta}_1 | \mathbf{y}_1)$ and $\log \tilde{\pi}_d^2 = \log \tilde{\pi}_d(\boldsymbol{\Theta}_2 | \mathbf{y}_2)$. This situation is equivalent to having two datasets and two models, where the models share part of their structure (θ_1, θ_2) while another part (θ_3) is specific to only one of them.

9.2.2 Workflow designs for distributed inference

In the previous section, we discussed in detail two procedures for improving the estimation of the hyperparameters: (a) a distributed modal approximation approach, and (b) a distributed modal optimization approach, designed to locate the modal configuration through distributed optimization.

Figure 9.1 illustrates the workflow for computing the marginal posterior of the hyperparameters under the distributed modal approximation approach. In this setup, we assume n servers (S) or machines, with each data partition assigned to one server. On each server, the steps for computing $\tilde{\pi}_{GSN}(\boldsymbol{\theta} | \mathbf{y}_i)$ are carried out. Once the marginal posterior for each partition has been obtained, the results are sent to a

master server (MS), where the ensemble of the posteriors is performed; the MS can also be one of the n servers (or machines). As shown, the communication required between servers is minimal, and all information exchanged preserves the privacy of the data associated with each partition $\mathbf{y}_i$, making the procedure privacy-safe.

Finally, Figure 9.2 illustrates the workflow for computing $\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{y})$ by the distributed modal optimization approach. The optimization algorithm implemented may vary; classical methods such as gradient descent, conjugate gradient, stochastic gradient descent, or their mini-batch variants are all valid options. The underlying assumption in this setup remains the same as in the previous case. In this setting, communication between the servers (S) and the master server (MS), as well as vice versa, is intensive. However, since the exchanged information is low-dimensional, the loss in computational performance due to transfer is expected to be negligible. Moreover, as in the previous approach, the method preserves the privacy of the data associated with each partition, as the transferred information consists only of gradients, Hessians, updated exploration points, and the modal location obtained upon convergence of the optimization algorithm.

References

- Durante, D., Pozza, F., and Szabo, B. (2024). “Skewed Bernstein–von Mises theorem and skew-modal approximations.” *The Annals of Statistics*, 52(6).
- Lindgren, F., Rue, H., and Lindström, J. (2011). “An explicit link between gaussian fields and gaussian markov random fields: The stochastic partial differential equation approach.” *Journal of the Royal Statistical Society. Series B: Statistical Methodology*, 73(4).
- Muñoz, F. (2012). “Geoestadística en regiones heterogéneas con distancia basada en el coste.” Ph.D. thesis, University of Valencia. Supervisor: Antonio López-Quílez.
- Zhou, J., Grazian, C., and Ormerod, J. T. (2024). “Tractable skew-normal approximations via matching.” *Journal of Statistical Computation and Simulation*, 94(5): 1016–1034.

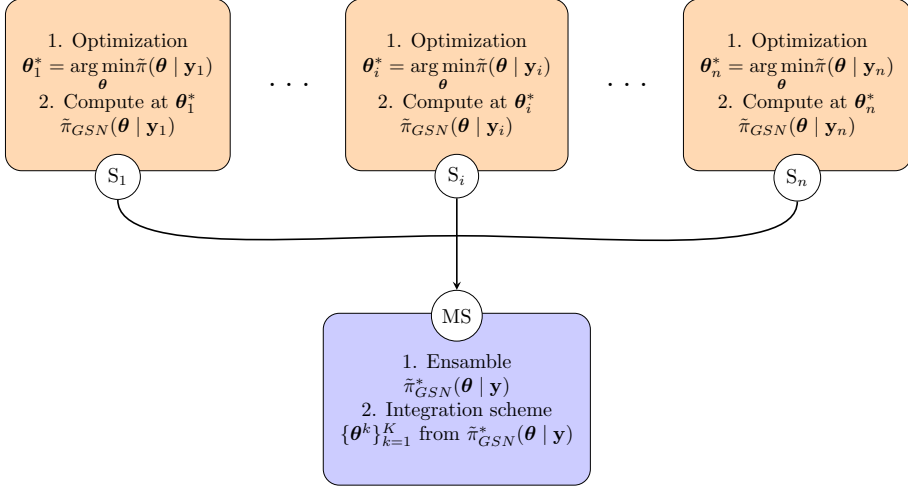


FIGURE 9.1: Workflow for the distributed modal approximation approach.

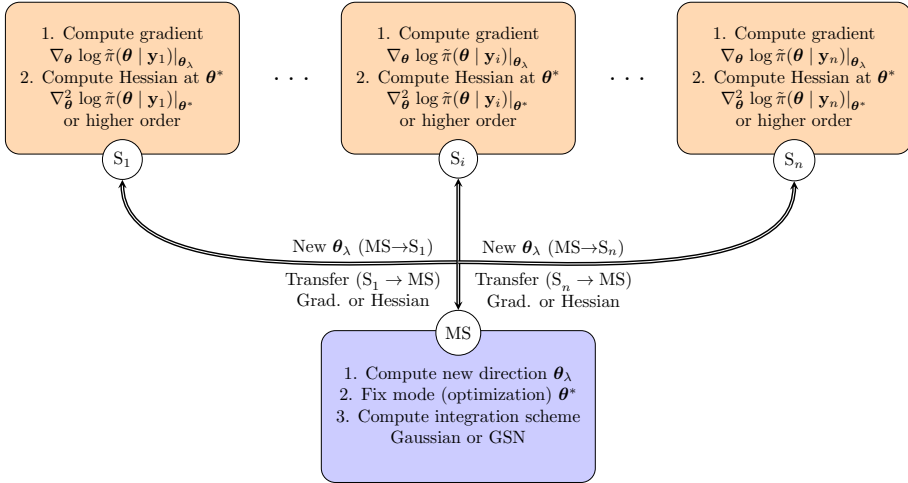


FIGURE 9.2: Workflow for the distributed modal optimization approach.

General references

- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D., and Lindgren, F. (2018). “Spatial modeling with R-INLA: A review.” *WIREs Computational Statistics*, 10(6): e1443.
- Banerjee, S., Carlin, B. P., and Gelfand, A. E. (2015). *Hierarchical modeling and analysis for spatial data*. Number 135 in Monographs on statistics and applied probability. Boca Raton: CRC Press, Taylor & Francis Group, second edition edition.
- Cressie, N. (1993). *Statistics for Spatial Data*. Wiley-Interscience, revised edition edition.
- Cressie, N., Sainsbury-Dale, M., and Zammit-Mangion, A. (2022). “Basis-Function Models in Spatial Statistics.” *Annual Review of Statistics and Its Application*, 9(1): 373–400.
- Diggle, P. J., Menezes, R., and Su, T. I. (2010). “Geostatistical inference under preferential sampling.” *Journal of the Royal Statistical Society. Series C: Applied Statistics*, 59(2).
- Diggle, P. J., Moraga, P., Rowlingson, B., and Taylor, B. M. (2013). “Spatial and Spatio-Temporal Log-Gaussian Cox Processes: Extending the Geostatistical Paradigm.” *Statistical Science*, 28(4): 542 – 563.
- Gelfand, A. E. (2001). “On the change of support problem for spatio-temporal data.” *Biostatistics*, 2(1): 31–45.
- Gelfand, A. E. and Schliep, E. M. (2018). *Bayesian Inference and Computing for Spatial Point Patterns*. NSF-CBMS Regional Conference Series in Probability and Statistics. Beachwood, Ohio; Alexandria, Virginia: Institute of Mathematical Statistics and American Statistical Association.

- Gelfand, A. E., Schmidt, A. M., Banerjee, S., and Sirmans, C. F. (2004). “Nonstationary multivariate process modeling through spatially varying coregionalization.” *Test*, 13(2): 263–312.
- Gelfand, A. E., Diggle, P., Guttorp, P., and Fuentes, M. (eds.) (2010). *Handbook of Spatial Statistics*. CRC Press, 0 edition.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC, 0 edition.
- Gonçalves, F. B. and Gamerman, D. (2018). “Exact Bayesian Inference in Spatiotemporal Cox Processes Driven by Multivariate Gaussian Processes.” *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 80(1): 157–175.
- Gómez-Rubio, V. (2020). *Bayesian Inference with INLA*. Chapman & Hall/CRC Press.
- Lindgren, F. and Rue, H. (2015). “Bayesian Spatial Modelling with R-INLA.” *Journal of Statistical Software*, 63(19): 1–25.
- Lindgren, F., Rue, H., and Lindström, J. (2011). “An explicit link between gaussian fields and gaussian markov random fields: The stochastic partial differential equation approach.” *Journal of the Royal Statistical Society. Series B: Statistical Methodology*, 73(4).
- Lindgren, F., Bolin, D., and Rue, H. (2022). “The SPDE approach for Gaussian and non-Gaussian fields: 10 years and still running.” *Spatial Statistics*, 50: 100599.
- Lindgren, F., Bakka, H., Bolin, D., Krainski, E., and Rue, H. (2024). “A diffusion-based spatio-temporal extension of Gaussian Matérn fields.” *SORT-Statistics and Operations Research Transactions*, 48(1): 3–66.
- MacNab, Y. C. (2011). “On Gaussian Markov random fields and Bayesian disease mapping.” *Statistical Methods in Medical Research*, 20(1): 49–68.
- (2022). “Bayesian disease mapping: Past, present, and future.” *Spatial Statistics*, 50: 100593.
- Martínez-Beneito, M. A. and Botella-Rocamora, P. (2019). *Disease Mapping*. Chapman and Hall/CRC.
- Rue, H. and Held, L. (2005). *Gaussian Markov Random Fields*. Chapman and Hall/CRC.

- Rue, H., Martino, S., and Chopin, N. (2009). “Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(2): 319–392.
- Simpson, D., Lindgren, F., and Rue, H. (2012). “In order to make spatial statistics computationally feasible, we need to forget about the covariance function.” *Environmetrics*, 23(1): 65–74.
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., and Van Der Linde, A. (2002). “Bayesian measures of model complexity and fit.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(4): 583–639.
- Van Niekerk, J. and Rue, H. (2024). “Low-rank Variational Bayes correction to the Laplace method.” *Journal of Machine Learning Research*, 25(62): 1–25.
- Van Niekerk, J., Krainski, E., Rustand, D., and Rue, H. (2023). “A new avenue for Bayesian inference with INLA.” *Computational Statistics & Data Analysis*, 181: 107692.
- Watanabe, S. (2013). “A widely applicable Bayesian information criterion.” *J. Mach. Learn. Res.*, 14(1): 867–897.